Andrology & Sexual Medicine Rapid Revision

Campbell-Walsh-Wein · Chapters 66–76

For MCh Urology theory & viva · NEET-SS / competitive MCQs

How to use (total ≈ 6h 12min even at a slow, careful pace)

- Each chapter reads top-to-bottom in the order you would write an essay answer: definition → pathology → diagnosis → management → complications → follow-up. The same material serves theory and viva.
- Boxed **ALGORITHM** and shaded **KEY POINT** panels carry the reasoning examiners ask about ("why do we…?").
- **MCQ one-liners** end each chapter: re-read these the day before the paper.
- The last section puts the big treatment choices side by side for essay questions.

Conventions: **underlined bold** = number to memorise · MCQ = exam trivia only · → *Ch 72* = cross-reference · † = not in the source chapter notes (added because it is essential in practice). Abbreviations are spelled out at first use in each chapter.

Chapter	Topic	Slow-reader time
66	Surgical, Radiographic and Endoscopic Anatomy of the Male Reproductive System	28 min
67	Male Reproductive Physiology	33 min
68	Integrated Men's Health: Androgen Deficiency, Cardiovascular Risk and Metabolic Syndrome	20 min
69	Male Infertility	34 min
70	Surgical Management of Male Infertility	42 min
71	Physiology of Penile Erection and Pathophysiology of ED	47 min
72	Evaluation and Management of Erectile Dysfunction	44 min
73	Priapism	27 min
74	Disorders of Male Orgasm and Ejaculation	33 min
75	Surgery for Erectile Dysfunction	27 min
76	Diagnosis and Management of Peyronie Disease	27 min
★	Putting it together: andrology decisions	10 min
	Total (26059 words, at 70 words/min)	**6h 12min**

66 Surgical, Radiographic and Endoscopic Anatomy of the Male Reproductive System

Reading time ≈ 28 min (slow pace)

1 · SCROTUM

Testis and cord are invested in layers carried down from the abdominal wall during descent; each layer is a surgical plane. The median raphe marks the septum between the two compartments.

Layer (superficial → deep)	Continuous with	Note
Tunica dartos	Scarpa, Colles, penile dartos	Smooth muscle
External spermatic fascia	External oblique	—
Cremasteric fascia	**Internal oblique**	**Striated** cremaster; thermoregulation
Internal spermatic fascia	Transversalis fascia	—
Tunica vaginalis (parietal, visceral)	Peritoneum	Hydrocele space; visceral layer fused to albuginea

- **Anterior wall**: external pudendal vessels; ilioinguinal and genitofemoral nerves. **Posterior wall**: perineal vessels and nerve, posterior femoral cutaneous nerve.
- Vessels and nerves run parallel to the rugae; nothing crosses the raphe. Lymph → **ipsilateral superficial inguinal** nodes.

> **KEY POINT** Scrotal lymph goes to the groin, testicular lymph to the retroperitoneum — †so scrotal violation in testis cancer adds inguinal nodes to the at-risk field.

2 · TESTIS

Structure

- Volume **15–25 mL**, length 4.5–5.1 cm. Capsule: visceral tunica vaginalis → **tunica albuginea** (contains smooth muscle — possible contractility) → tunica vasculosa.
- **Mediastinum testis**: posterior echogenic band on US. Appendix testis (upper pole) = **müllerian** remnant.
- ~250 lobules; **600–1200 seminiferous tubules** (280–400 m) ending in rete testis. Interstitium **20–30%** of volume — **Leydig cells** (testosterone). **Sertoli cells** line tubules; their tight junctions form the blood–testis barrier.
- Outflow: tubule → rete → 6–12 ductuli efferentes → single epididymal duct (~6 m) → vas.

Arterial supply — three sources

Artery	Origin	Point
Testicular (internal spermatic)	Aorta below renal artery	Main supply; larger than the other two combined
Deferential	Superior (or inferior) vesical	Vas, cauda; key collateral — lost after vasectomy
Cremasteric	Inferior epigastric	Tunica vaginalis; anastomoses at mediastinum

- Branches usually within the inguinal canal. At the mediastinum it coils, then runs inferiorly along the **posterior** surface within the parenchyma and fans anteriorly over both poles → centrifugal septal arteries → intertubular and peritubular capillaries.
- Collaterals may preserve the testis after testicular artery ligation, but atrophy/azoospermia do occur.

> **KEY POINT** **Surgical rules**: no traction suture through the lower pole; biopsy the **medial or lateral upper pole**; the midportion has the fewest vessels. After prior vasectomy the deferential collateral may be gone → preserve the testicular artery in later cord surgery and counsel about devascularisation.

Veins — pampiniform plexus

Intratesticular veins do not follow arteries. They join deferential veins as the plexus, which surrounds the artery and cools it by **countercurrent heat exchange** to **2–4 °C below rectal** temperature; varicocele disrupts this and warms the testis.

Group	Accompanies	Drains to
Anterior	Testicular artery	Inferior vena cava (IVC) (right) / left renal vein (left)
Middle	Vas	Pelvic veins
Posterior	Posterior cord	Epigastric and pudendal veins

- Right vein enters IVC obliquely (valve-like); left enters left renal vein → **~90% of varicoceles left**.
- Middle/posterior groups give collateral return after repair; pudendal, cremasteric and vasal anastomoses explain recurrence after embolisation.
- **High (retroperitoneal) ligation** takes testicular artery and vein above the exit of deferential/external vessels — collaterals suffice. **Inguinal/subinguinal** ligation takes veins only; inadvertent arterial ligation here carries higher devascularisation risk.

Lymphatics and nerves

- Lymph ascends in the cord → **para-aortic and interaortocaval** nodes. Lymphatic division at varicocelectomy → **hydrocele**; microsurgery preserves lymphatics.
- Autonomic supply from renal/aortic plexuses (and pelvic plexus via vas); **no somatic innervation**; seminiferous epithelium is nerve-free. Genital branch of genitofemoral nerve supplies tunica vaginalis.
- **"Trifecta" nerve complex** in chronic orchialgia: perivasal, posterior periarterial/lipomatous, intracremasteric — targets of microsurgical spermatic cord denervation.
- Some fibres cross to the contralateral pelvic plexus — may explain contralateral effects of unilateral varicocele or tumour.

Blood–testis barrier (anatomy)

Primary level: Sertoli–Sertoli tight junctions dividing basal from adluminal compartments; additional levels at capillary endothelium and peritubular myoid cells. Evidence: tubular fluid differs markedly from plasma and lymph → *Ch 67*.

Imaging

Modality	Role
Ultrasonography (US), 7.5–10 MHz + Doppler	**First line**: pain, mass, swelling, epididymo-orchitis, recurrent varicocele
Computed tomography (CT)	Not primary; cord/inguinal disease; air in Fournier gangrene
Magnetic resonance imaging (MRI)	Second line: indeterminate US, intra- vs extratesticular; nonpalpable testis (MR angiography)

- Normal: scrotal wall 3–4 mm; small physiological fluid between tunica layers (more = hydrocele); homogeneous medium echoes; intratesticular arteries low impedance with high diastolic flow.
- **Suspected torsion → explore**; Doppler must not delay surgery.
- Varicocele is a **clinical diagnosis**; US (erect, Valsalva) only when examination is equivocal or limited.

3 · EPIDIDYMIS

- Caput, corpus, cauda on the posterolateral testis. Pseudostratified epithelium: **principal** cells (tight junctions = blood–epididymis barrier), basal cells, **clear** cells (acidify lumen → quiescent sperm storage). Cilia only in efferent ducts.
- Arteries: testicular branch → caput/corpus; deferential → cauda; cremasteric contributes — collaterals after testicular artery ligation.
- Lymph: caput/corpus → preaortic; **cauda → external iliac** (with vas).
- Nerves and smooth muscle increase toward the cauda (storage → propulsion).
- US: head ~9 mm, body 3.5 mm, tail 4.8 mm; highly sensitive for cysts and dilation. Inverted epididymis (caput at lower pole) in a small minority. MRI T1 signal may predict need for vasoepididymostomy rather than vasovasostomy.

Wolffian (mesonephric) duct	Müllerian (paramesonephric) duct
Epididymis, vas, seminal vesicle, ejaculatory duct, **appendix epididymis** (efferent ductules from mesonephric tubules)	**Appendix testis**, prostatic utricle

4 · VAS DEFERENS

- Course: cauda → convoluted 2–3 cm → straight, **posterior** to cord vessels → inguinal canal → pelvic side wall → crosses **anterior to the ureter** → behind bladder → ampulla → joins seminal vesicle.
- Length **30–35 cm**; lumen **0.2–0.7 mm**; **three muscle layers** (outer longitudinal, middle circular, inner longitudinal).
- Deferential artery (superior vesical). Lymph → external and internal iliac.
- ~Half the cord nerves surround the vas: sympathetic drives emission peristalsis; sensory fibres cause chronic scrotal pain.
- **Vasogram**: hemivasotomy → 24 G angiocatheter → indigo carmine (seen endoscopically) or contrast (fluoroscopy). It can cause obstruction, so do it **only at the time of reconstruction**.
- US confirms congenital absence of the vas when examination is equivocal.

5 · SEMINAL VESICLES AND EJACULATORY DUCTS

- Behind bladder base, above prostate; join the ampulla → ejaculatory duct → verumontanum at **5 and 7 o'clock**.
- Bud from mesonephric duct at 12 weeks, developmentally linked to the ureteric bud → **renal anomalies coexist**.
- Arteries: vesiculodeferential and inferior vesical; lymph → internal iliac. Noradrenergic > cholinergic → contraction at emission.
- Fluid ~70% of semen volume; alkaline; **fructose**, prostaglandins, **semenogelin** (coagulum, liquefied by prostate-specific antigen (PSA)).
- **transrectal ultrasound (TRUS)**: post-ejaculation anteroposterior (AP) diameter **>16 mm** or volume **>6 mL** = obstruction; guides aspiration and vesiculography. **MRI** is modality of choice; CT for stones. Vesiculoscopy (4.5–6.9 Fr) before endoscopic treatment.

> **KEY POINT** A normal vesicle empties at ejaculation; one that stays large is the imaging signature of ejaculatory duct obstruction.

6 · PROSTATE

Zone (McNeal)	Share	Significance
Peripheral	**70%**	Most cancers; hyperechoic on TRUS
Central	25%	Surrounds ejaculatory ducts

Zone (McNeal)	Share	Significance
Transition	~5%	Benign prostatic hyperplasia (BPH) → outlet obstruction
Anterior fibromuscular stroma	Nonglandular	—

- Normal **~25 mL**; enlarges with age, fastest in transition zone. Internal (preprostatic) sphincter — circular smooth muscle thickest at bladder neck — continence at rest.
- Puboprostatic ligaments suspend the apex; parietal and visceral endopelvic fascia meet at the arcus tendineus. Superficial branch of dorsal venous complex lies in retropubic fat outside the prostatic fascia; Denonvilliers fascia lies posteriorly.
- "Capsule" 0.5–2 mm, best posterolaterally; deficient at apex (blends with sphincter).
- **Cavernous nerves run in the lateral prostatic fascia** (posterolateral neurovascular bundle) → preserve for erectile function at radical prostatectomy.
- Prostate-specific antigen (PSA) normally stays luminal; disrupted architecture (inflammation, cancer) lets it reach serum. Cancer = **loss of basal cell layer**.
- Arteries: superior pedicle (urethral group → median lobe, proximal urethra); inferior pedicle (capsular group → apex, peripheral zone); variable origin; extraprostatic anastomoses matter for embolisation.
- Veins: periprostatic plexus + deep dorsal vein = **dorsal venous complex**, ligated at radical prostatectomy.
- Lymph: **obturator and internal iliac**, plus presacral and external iliac.
- Parasympathetic → secretion; sympathetic (**α1**) → expulsion at emission.
- TRUS volume = L × W × H × **0.52** (underestimates). **Multiparametric MRI**: T2, diffusion-weighted, dynamic contrast; Prostate Imaging Reporting and Data System (PI-RADS) v2.1; targeted biopsy reduces insignificant-cancer detection.

7 · URETHRA

Segment	Features
Prostatic	Verumontanum, utricle (müllerian) flanked by ejaculatory ducts
Membranous (**2–2.5 cm**)	Striated sphincter; narrowing on retrograde urethrogram (RUG)
Bulbar	Widest; Cowper glands at 3 and 9 o'clock
Penile (~15 cm)	Glands of Littré
Fossa navicularis	Lacuna magna on roof; squamous epithelium

- **Perineal membrane** divides posterior (prostatic, membranous) from anterior urethra. Normal diameter 8–9 mm.
- Epithelium: urothelium (prostatic) → pseudostratified columnar → stratified squamous in the glans.
- Striated sphincter: pudendal nerve, S2–S3, motor neurons in **Onuf nucleus**. Arteries: internal pudendal → bulbourethral.
- Imaging: sonourethrography (anterior strictures), RUG (stricture, trauma), MRI (malignancy), endoscopy.

8 · PENIS

Structure

- Paired **corpora cavernosa** (crura on pubic arch) + **corpus spongiosum** (bulb, urethra, glans).
- Septum incomplete distally → corpora act as **one unit** (†one-sided injection fills both).
- **Tunica albuginea** 2–3 mm: inner **circular** layer (contains sinusoids, gives intracavernosal struts); outer **longitudinal** layer (glans → inferior pubic ramus). Thin ventrally in the spongiosal groove.
- **Buck fascia** encloses the corpora and splits around the spongiosum; contains deep dorsal vein, dorsal arteries, dorsal nerves (midline → lateral: **vein → arteries → nerves**).

- Fundiform (superficial, rectus sheath) and suspensory (deep, pubis) ligaments.
- Sinusoidal smooth muscle **~45%** of volume; endothelial nitric oxide synthase sustains erection.

Arteries

Superficial: external pudendal (skin). Deep: internal pudendal → Alcock canal → **common penile artery**:

Branch	Supplies
Bulbourethral	Spongiosum, urethra, glans
Cavernosal	Enters at hilum, runs centrally → helicine arteries → sinusoids (erection)
Dorsal	Under Buck fascia → glans; circumflex branches

Variants common; accessory pudendal arteries may supplement or replace.

Veins, lymph, nerves

- Superficial dorsal vein (outside Buck) drains skin. **Deep dorsal vein** (under Buck) → periprostatic plexus.
- Distal two-thirds: sinusoids → subtunical plexus → **emissary veins** (oblique through tunica) → circumflex → deep dorsal vein. Proximal third: cavernous and crural veins → internal pudendal.
- Shaft skin lymphatics converge dorsally → **bilateral** superficial inguinal nodes; glans → superficial and deep inguinal bilaterally.
- Shaft skin mobile (dartos loosely attached to Buck fascia); glans skin fixed to the albuginea.
- **Dorsal nerve** (pudendal, **S2–S4**) — sensation. **Cavernous nerves** (pelvic plexus, autonomic) — erection.

> **KEY POINT** Erection is veno-occlusive: filled sinusoids compress subtunical and oblique emissary veins against the inelastic tunica; †tunical or smooth-muscle damage defeats this → venous leak.

Penile duplex (after intracavernosal injection)

Parameter	Threshold	Meaning
Peak systolic velocity	**<30 cm/s**	Arterial insufficiency
End-diastolic velocity	**>5 cm/s**	Veno-occlusive dysfunction
Resistive index	**<0.80**	Veno-occlusive dysfunction

Systolic values judge inflow, diastolic values outflow. Duplex also shows Peyronie plaques; MRI shows fracture, prosthesis malposition and tumour.

9 · MCQ ONE-LINERS

1. Cremaster: striated, from internal oblique.
2. Internal spermatic fascia continues transversalis fascia.
3. Scrotal lymph → superficial inguinal; testicular lymph → para-aortic/interaortocaval.
4. Testicular artery arises from aorta below renal artery.
5. Cremasteric artery from inferior epigastric; deferential from superior vesical.
6. Safest testis biopsy site: medial or lateral upper pole.
7. Countercurrent exchange: testicular arterial blood 2–4 °C below rectal.
8. ~90% of varicoceles are left-sided.
9. Hydrocele after varicocelectomy = lymphatic ligation.
10. Appendix testis müllerian; appendix epididymis wolffian.
11. Epididymal cauda lymph → external iliac nodes.
12. Vas crosses anterior to the ureter; lumen 0.2–0.7 mm.
13. Vasography only at reconstruction.
14. Seminal vesicle >16 mm or >6 mL post-ejaculation suggests obstruction.
15. Ejaculatory duct orifices: 5 and 7 o'clock.
16. Peripheral zone: 70% of gland, most cancers; BPH in transition zone.
17. TRUS ellipsoid coefficient 0.52.
18. Cavernous nerves travel in lateral prostatic fascia.
19. Striated sphincter motor neurons: Onuf nucleus.
20. Cowper glands open into bulbar urethra.

21. Common penile artery: bulbourethral, cavernosal, dorsal branches.
22. Deep dorsal vein lies beneath Buck fascia.
23. Glans is the distal expansion of corpus spongiosum.
24. Corporal smooth muscle ≈ 45% of volume.
25. Membranous urethra 2–2.5 cm long.
26. Penile duplex: peak systolic velocity (PSV) <30 cm/s arterial; end-diastolic velocity (EDV) >5 cm/s venous leak.

67 Male Reproductive Physiology

Reading time ≈ 33 min (slow pace)

1 · HYPOTHALAMIC–PITUITARY–GONADAL AXIS

The hypothalamic–pituitary–gonadal (HPG) axis governs phenotypic sex development, puberty, testosterone production (endocrine testis) and sperm production (exocrine testis). It is read by its **rhythm** as much as its level: pulse frequency and negative feedback set downstream responsiveness.

	Peptide hormones	**Steroid hormones**
Examples	Luteinising hormone (LH), follicle-stimulating hormone (FSH)	Testosterone, oestradiol
Storage	Secretory granules	Not stored — secretion = production
Receptor	Cell surface → second messenger (cyclic adenosine monophosphate, cAMP)	Intracellular → gene transcription; carried bound in plasma

Hypothalamus

- Gonadotrophin-releasing hormone (GnRH): 10-amino-acid peptide from preoptic and arcuate nuclei, reaching the anterior pituitary via the portal system; sole known function is LH/FSH release. Half-life **5–7 min**.
- Rhythms: pulsatile (every **90–120 min**), circadian (testosterone highest in early morning), seasonal.
- **Pulsatility is essential**: continuous GnRH agonist (leuprolide) converts cyclic to constant exposure → desensitisation → castrate testosterone.
- **Kallmann syndrome**: GnRH neurones fail to migrate → hypogonadotrophic hypogonadism (delayed puberty, infertility). †Associated anosmia, since the neurones migrate with the olfactory placode.

GnRH modulation	**Factors**
Negative	Opioids (β-endorphin), FSH, LH, testosterone
Positive	Prostaglandin E2, insulin, **kisspeptin** (puberty), **leptin**
Variable	Dopamine (catecholamines)

Anterior pituitary

- LH and FSH are glycoproteins: common **α subunit** (shared with thyroid-stimulating hormone (TSH), hCG) + unique **β subunit** conferring specificity; both needed. Act via adenylate cyclase → cAMP.
- **Prolactin**: may raise Leydig LH receptors and support libido. **Hyperprolactinaemia abolishes gonadotrophin pulsatility** by disrupting episodic GnRH release → low testosterone. Low prolactin is not pathological.

	LH	**FSH**
Pulses	8–16/24 h; closely follow GnRH	Smaller; partly GnRH-independent (inhibin, activin)
Target	**Leydig cell**	**Sertoli cell** (and spermatogonia)
Action	Steroidogenesis: cholesterol → pregnenolone → testosterone	Tubule growth; **initiates spermatogenesis at puberty**; maintains quantitatively normal output
Main feedback	Testosterone (hypothalamus)	Oestradiol (pituitary) + **inhibin B**

Testicular hormones and feedback

- Testosterone ~**5 mg/day**, pulsatile. Metabolites: **dihydrotestosterone (DHT)** via 5α-reductase — more potent, needed in most peripheral tissues (not testis or muscle); **oestradiol** via aromatase —

feedback especially on FSH.
- FSH induces Sertoli products: androgen-binding protein (ABP), transferrin, lactate, inhibin and growth factors.
- **Inhibin** (Sertoli, 32 kDa): FSH stimulates it; it suppresses FSH. **Activin** (TGF-β family): stimulates FSH.

> **KEY POINT** The axis runs two loops: LH ↔ Leydig governed by testosterone, and FSH ↔ Sertoli governed by inhibin and oestradiol. Hence **LH reports Leydig function; FSH and inhibin B report the tubules** — raised FSH with low inhibin B signals spermatogenic failure.

Development and ageing

- **SRY** (Yp) encodes a high-mobility-group (HMG) box protein that bends DNA → testis formation. **DAX1** and **WNT4** are "antitestis" genes that can actively induce female development even with SRY — the female pathway is not a passive default.

Fetal cell	Product	Effect
Leydig	Testosterone	Wolffian duct → internal genitalia
Leydig	Insulin-like peptide 3 (INSL3)	Transabdominal testicular descent
Target tissue	DHT	External genitalia
Sertoli	Müllerian-inhibiting substance (MIS/AMH)	Müllerian regression; keeps germ cells out of meiosis

- Neonatal gonadotrophin surge → childhood quiescence → puberty with GnRH pulsing (~12 years), triggered by kisspeptin, leptin (fat-store signal) and melatonin.
- **Ageing** (partial androgen deficiency of the aging male): testosterone ~one-third lower by the seventh decade; FSH rises ~threefold; fewer Leydig cells, more binding protein, lost diurnal rhythm, blunted feedback and irregular GnRH pulses — both peripheral and central failure.

2 · TESTIS

Architecture and blood flow

- Volume 15–25 mL; **600–1200 tubules**, ~250 m; interstitium 20–30%; ~**700 million Leydig cells** in young adults.
- Arteries: testicular (aorta), deferential (richest collateral), cremasteric. Cord holds one artery in ~half, two in ~30%, three in ~20% — relevant to microsurgical varicocelectomy. With a single artery, ligation can cause atrophy.
- **Countercurrent exchange** in the pampiniform plexus cools arterial blood **2–4 °C** below core and recycles testosterone from vein to artery. Loss of this gradient underlies dysfunction in varicocele and cryptorchidism.
- Intratesticular arteries run on the posterior surface and fan over the poles; midsection has fewer vessels — matters for biopsy, sperm retrieval and orchidopexy.
- Cord lymphatic obstruction → hydrocele after varicocelectomy or herniorrhaphy.
- Tubular fluid is reabsorbed in the rete and efferent ductules under oestrogen control; its composition differs from plasma — the basis of the blood–testis barrier.

Leydig cells and testosterone synthesis

- Differentiate by week 7 under hCG/LH; second (neonatal) wave ~2–3 months imprints hypothalamus, liver and prostate.
- **Rate-limiting step**: LH → cAMP → **steroidogenic acute regulatory (StAR) protein** moves cholesterol into the inner mitochondrial membrane → side-chain cleavage → pregnenolone → smooth endoplasmic reticulum (3β-hydroxysteroid dehydrogenase, 17α-hydroxylase/17,20-lyase, 17β-hydroxysteroid dehydrogenase) → testosterone.
- Enzyme mutations → rare disorders of androgen biosynthesis in 46,XY.

- Modifiers: FSH and prolactin enhance LH response; oestrogens and androgens directly inhibit.
- †Circulating testosterone: ~44% bound to sex hormone-binding globulin (SHBG, tightly), ~54% to albumin (loosely), ~2% free; free + albumin-bound = bioavailable.

Testosterone peak	Timing	Role
Fetal	12–18 weeks	Builds reproductive tract
Neonatal	~2 months	Imprints target tissues
Pubertal	12–17 years	Masculinisation
Adult	Peak 2nd–3rd decade → decline	Maintains androgen-dependent organs

Sertoli cells

- Rest on basement membrane, extend to lumen; "nurse cells" for germ cells. Low mitotic index.
- **Sertoli–Sertoli tight junctions** (strongest in the body) → basal compartment (spermatogonia, preleptotene spermatocytes; plasma-like) and adluminal compartment (later spermatocytes, spermatids). Junctions open and close to let spermatocytes pass.
- Functions: unique adluminal milieu; ABP keeps intratubular androgen ~**50-fold** above serum; **inhibin B** (marker of Sertoli function in infertility); phagocytosis of residual bodies; fluid secretion; c-kit/kit ligand signalling with germ cells; spermiation.
- Peritubular myoid cells (contractile) secrete matrix and paracrine P-Mod-S that modulates Sertoli function.

Blood–testis barrier

Anatomical	Functional (immunological)
Mainly Sertoli tight junctions; also myoid cells and endothelium. Excludes large molecules and lymphocytes	Small continuous antigen leak in rete, efferent ducts and epididymis → T-suppressor tolerance; Sertoli immunosuppressants (galectin-1, TGF-β, androgens). Large antigen load → true immune response

- Forms at spermarche — postmeiotic antigens appear after immune self-tolerance is established. Hence prepubertal biopsy, torsion or trauma do not induce antisperm antibodies; after puberty, immunological infertility is a risk.
- May shelter tumour cells from chemotherapy → isolated testicular relapse.
- TGF-β3 and TNF-α reversibly open it (reduce occludin, ZO-1).

3 · SPERMATOGENESIS

- Duration classically **~64 days** (in vivo labelling 42–76 days). Output ~**123 million/day** (~1000 per heartbeat).
- Optimal at 2–4 °C below body temperature and needs very high intratesticular testosterone. †Hence exogenous testosterone suppresses LH, intratesticular testosterone and spermatogenesis.
- Three phases: **proliferative** (spermatogonia) → **meiotic** (spermatocytes → haploid spermatids) → **spermiogenesis** (spermatid → spermatozoon).
- Sequence: Ad (dark, reserve) → Ap (pale, divide every 16 days) → B spermatogonia (committed) → primary spermatocytes (preleptotene → leptotene → zygotene → pachytene) → secondary spermatocytes → spermatids Sa → Sb → Sc → Sd1 → Sd2.
- **Cycle** = succession of stages at one point; **stage** = cell association (**six** in humans); **wave** = helical arrangement along the tubule → continuous, not pulsatile, production.
- Daughter cells stay joined by cytoplasmic bridges → synchronised clones.

Stem cells

- Primordial germ cells migrate to the gonadal ridge → gonocytes → spermatogonia. Failed migration → extragonadal germ cell tumour, Sertoli cell–only azoospermia.
- Self-renewal via kit ligand/c-kit and glial cell line–derived neurotrophic factor (GDNF). Adult spermatogonial stem cells can be reprogrammed to pluripotent cells.

- Lifelong renewal → "selfish spermatogonia" with FGFR2/FGFR3 mutations → paternal-age disorders and spermatocytic seminoma.

Meiosis

	Mitosis	Meiosis
Divisions / daughters	1 / 2 identical	2 / 4 varied, haploid
Synapsis, crossover	None	Prophase I (synaptonemal complex, chiasmata)
Centromere division	Anaphase	**Anaphase II**

- Primary spermatocytes undergo the first (reduction) division, crossing into the adluminal compartment; second division is equational.
- **Defective recombination** found in about half of men with maturation arrest; altered crossover → aneuploidy — explains higher chromosomal abnormalities in offspring after in vitro fertilisation (IVF) and intracytoplasmic sperm injection (ICSI).

Spermiogenesis

No division. Golgi → **acrosome**; centriole → **flagellum**; nucleus condenses to ~10%; mitochondria gather in the midpiece; cytoplasm shed as residual body. Sertoli cytoplasm retracts to release sperm (spermiation).

Genetics

Y microdeletions (azoospermia factor, AZF): **15%** of azoospermic and 6–8% of severely oligospermic men — the commonest defined molecular cause.

Deletion	Histology	Sperm retrieval
Complete AZFa	Sertoli cell–only	Highly unlikely
Complete AZFb	Maturation arrest	Highly unlikely
AZFc	Hypospermatogenesis / focal spermatogenesis	**Often found**
Partial AZFa/b	—	Ejaculated sperm possible

X-linked candidates: SOX3, TEX11. **Paternal age**: ~23 spermatogonial divisions per year after puberty → more de novo dominant mutations (achondroplasia, Apert, Marfan), neurodevelopmental disorders (autism, schizophrenia), methylation drift, sex-chromosome aneuploidy and rising **DNA fragmentation** (oxidative stress) → poor embryo development and miscarriage.

4 · EPIDIDYMIS

- 3–4 m duct: caput (8–12 efferent ductules) → corpus → cauda. Testicular sperm are **immotile and cannot fertilise** (except by ICSI).
- Epithelium: principal cells (absorptive, secretory; stereocilia = microvilli), basal cells (macrophage-derived); ciliated cells only in efferent ductules. Blood–epididymis barrier less tight than testis.
- Muscle and sympathetic nerves increase toward the cauda (three layers there) → active transport and controllable storage.
- Lymph: caput/corpus → preaortic; cauda → external iliac.

Functions

- **Transport**: **2–12 days**; driven mainly by spontaneous rhythmic duct contraction (plus fluid resorption and efferent cilia). Recent ejaculation shortens only cauda transit.
- **Storage**: cauda holds about half of 155–209 million sperm per side. Prolonged storage → more DNA fragmentation; unejaculated sperm removed by spermiophages after vasectomy.
- **Maturation**: progressive motility (low-amplitude, high-frequency beat), chemotaxis, zona binding and penetration. Biochemically: negative surface charge, disulfide cross-linking (rigidity), membrane lipid/glycoprotein remodelling, increased glycolysis. **Fertility is reached mostly by late corpus/early cauda**.

- Epididymosomes deliver small RNAs that sperm carry to the embryo.

> **KEY POINT** Pregnancy occurs even after vasoepididymostomy at the efferent ductules (obstruction may let sperm mature proximally), but **the more distal the anastomosis, the better the fertility**. After vasectomy >15 years, motility falls after reversal — acquired epididymal dysfunction.

- **Androgen dependent**, needing higher levels than other glands, mainly via DHT → explains impaired semen with 5α-reductase inhibitors.
- **Heat** abolishes storage (varicocele, cryptorchidism); **denervation** impairs transport (spinal cord injury, diabetes).

5 · VAS DEFERENS, SEMINAL VESICLES, EJACULATORY DUCTS

- Vas: wolffian, 30–35 cm, five regions; three muscle layers; **muscle:lumen ~10:1** — most muscular hollow viscus. Unobstructed lumen 200–700 µm.
- Rich adrenergic (hypogastric → presacral) innervation with purinergic cotransmission → strong peristalsis just before emission. Testicular-side segment has fewer nerves at vasovasostomy.
- Vasal sperm reserve ~130 million. Androgen dependent; converts testosterone to DHT.
- **Seminal vesicle**: 5–7 cm coiled tubule; supplies ~**80%** of seminal volume in the late ejaculate fractions; alkaline, **fructose**, prostaglandins; **semenogelin I** forms the coagulum. Semen also provides antioxidant protection (glutathione peroxidase, superoxide dismutase, catalase).
- **Ejaculatory duct**: extraprostatic → intraprostatic → verumontanum; no sphincter — reflux prevented by the **acute insertion angle**.
- Seminal vesicle–ejaculatory duct unit behaves like bladder–urethra: obstruction may be **physical** (blocked duct) or **functional** (vesicle myopathy, e.g. diabetes, drugs); static transrectal ultrasound (TRUS) may not distinguish.
- **Ejaculate pH <7.2 (acidic), low volume, no fructose** → absent or blocked seminal vesicles (CBAVD, ejaculatory duct obstruction).

6 · SPERMATOZOON, CAPACITATION AND FERTILISATION

Part	Contents
Head (4.5 × 3 µm)	Compacted nucleus; acrosome (enzymes)
Neck	Proximal centriole; axoneme starts
Midpiece	~75 mitochondria in helix, outer dense fibres; ends at annulus
Principal piece / endpiece	Axoneme with fibrous sheath

- Total length ~60 µm. Axoneme **"9 + 2"**; **dynein** (Mg^{2+}-ATPase) slides doublets → bending. Inner dynein arm mutants: no motility; outer arm: reduced. Nexin links keep shape.
- Sperm mtDNA (16,569 bp, circular, no histones/introns) mutates 40–100× faster; **paternal mitochondria are degraded** in the egg (ubiquitin) → maternal inheritance.
- Within the female tract sperm undergo **capacitation** — acquiring hyperactivated motility and readiness for the **acrosome reaction**. Semenogelin coagulum prevents premature capacitation.
- Head proteins bind zona glycoprotein **ZP3** → acrosome reaction → zona penetration; **fertilin (PH30)** mediates sperm–oolemma fusion; sperm **phospholipase C zeta (PLCζ)** triggers calcium oscillations = oocyte activation.

Sperm defects

- Morphology graded by strict (Kruger) criteria; most head defects do not reflect DNA quality, except **pinhead** and **multiple-tailed** sperm.
- **Globozoospermia** (round head, no acrosome): failed fertilisation even with ICSI — PLCζ deficiency.

	Ciliary dyskinesia (immotile cilia)	MMAF (fibrous sheath dysplasia)
Motility	<10%	Severely reduced
Morphology	Normal on light microscopy; axonemal defect on electron microscopy	Short, bent, coiled, absent flagella
Cause	Recessive genetic; with respiratory disease, dextrocardia (Kartagener)	Genetic, but also varicocele, reactive oxygen species
Treatment	IVF-ICSI	Correct cause; else IVF-ICSI

7 · EJACULATION PHYSIOLOGY †

ALGORITHM

Emission (sympathetic, T10–L2 via hypogastric nerve; α1): vas, seminal vesicle and prostate contract → semen into posterior urethra; **bladder neck closes** → **Expulsion** (somatic, S2–S4 pudendal): rhythmic bulbospongiosus and pelvic floor contractions with relaxed external sphincter → antegrade ejaculate.

- Order of fractions: sperm-rich epididymal/vasal and prostatic first, seminal vesicle last (source).
- †Failed bladder neck closure (α-blocker, diabetes, bladder neck surgery, retroperitoneal lymph node dissection injury) → retrograde ejaculation; loss of emission → anejaculation.

8 · MCQ ONE-LINERS

1. GnRH: decapeptide, half-life 5–7 min, pulses every 90–120 min.
2. Continuous GnRH agonist suppresses testosterone by abolishing pulsatility.
3. Kallmann syndrome: failed GnRH neurone migration.
4. Kisspeptin and leptin stimulate GnRH; opioids inhibit.
5. LH and FSH share the α subunit; β confers specificity.
6. Hyperprolactinaemia abolishes gonadotrophin pulsatility.
7. Testosterone production ~5 mg/day.
8. Testosterone feeds back mainly at hypothalamus; oestradiol at pituitary.
9. Inhibin B: Sertoli-derived, suppresses FSH.
0. SRY encodes an HMG-box protein; DAX1, WNT4 are antitestis genes.
1. INSL3 mediates transabdominal testicular descent.
2. StAR protein: rate-limiting cholesterol transport into mitochondria.
3. Young adult testis: ~700 million Leydig cells.
4. ABP keeps intratubular androgen ~50× serum.
5. Blood–testis barrier: Sertoli–Sertoli tight junctions.
16. Prepubertal testicular insult does not induce antisperm antibodies.
17. Spermatogenesis ~64 days; six stages in humans.
18. Helical waves make sperm production continuous.
19. Ap spermatogonia divide every 16 days.
20. Meiotic centromeres divide at anaphase II.
21. Acrosome from Golgi; flagellum from centriole.
22. AZFc deletion: sperm retrieval often possible; AZFa/b complete: unlikely.
23. Selfish spermatogonia (FGFR2/3) → spermatocytic seminoma.
24. Epididymal transit 2–12 days.
25. Fertilising ability acquired at late corpus/early cauda.
26. Vas: muscle-to-lumen ratio ~10:1.
27. Seminal vesicles supply ~80% of ejaculate; fructose source.
28. Ejaculate pH <7.2 suggests absent or obstructed seminal vesicles.
29. Semenogelin I forms the semen coagulum.
30. Paternal mtDNA is degraded after fertilisation.
31. Globozoospermia: PLCζ deficiency, ICSI may fail.
32. Inner dynein arm mutants: complete immotility.

68 Integrated Men's Health: Androgen Deficiency, Cardiovascular Risk and Metabolic Syndrome

Reading time ≈ 20 min (slow pace)

1 · THE MEN'S HEALTH CENTRE — CONCEPT

- Multidisciplinary centre (urology + primary care, cardiology, gastroenterology): improves preventive care and acts as the **front door** into urology practice.
- Why the urologist leads: men avoid preventive care but do seek care for problems "below the belt" — erectile dysfunction (ED), infertility, voiding — so the urologist is the entry point who must triage the whole man.
- Advanced practice providers (APPs) and telemedicine extend reach, especially in large, sparsely populated regions.
- Personalised care rests on genetics, phenotype and the patient's own goals; shared decision making replaces "one size fits all".

Age band	Dominant need	Key partners
18–39	Sexual, reproductive, testicular complaints ("rule out hypogonadism" a common visit); almost wholly preventive	Sports medicine, physiotherapy, mental health
40–64	Prostate and colon cancer screening begin; active disease management	Gastroenterology, cardiology, endocrinology, bariatrics, pulmonology
65+	Falls, neurodegeneration, hearing/vision, nutrition; coordination	Geriatrics, neurology, audiology, ophthalmology

In-house versus referral provision is decided by volume.

2 · THE LONGEVITY GAP

- Life expectancy at birth: males **76.1** vs females 81.1 years (2016).
- Males have higher mortality in **12 of the 15** leading causes of death; largest gap = **suicide (>3-fold)**.
- More coronary disease and myocardial infarction (MI) before 75; type 2 diabetes mellitus (T2DM) rising faster in men.
- Prostate cancer: the only US malignancy with rising stage IV at diagnosis → discuss screening with every men's health patient.
- Causes poorly understood — hormonal (testosterone harm, oestrogen cardioprotection), behavioural and risk-taking factors — hence multidisciplinary care.

3 · UROLOGIC SCREENING

- Sexual complaints often coexist with voiding complaints — screen both.
- Prostate-specific antigen (PSA) and urinalysis per guidelines for prostate, bladder and upper-tract tumours.

Questionnaire	Screens
Sexual Health Inventory for Men (SHIM)	Erectile function
Androgen Deficiency in the Aging Male (ADAM)	Androgen-deficiency symptoms
International Prostate Symptom Score (IPSS)	Lower urinary tract symptoms (LUTS)

4 · UROLOGIC DISORDERS OF FOCUS

Testosterone deficiency (TD)

- Testosterone (T) falls with age from both testicular and hypothalamic–pituitary defects; symptoms overlap with age-related comorbidity.
- **Diagnosis**: low T confirmed on **more than one occasion** + well-documented symptoms. † American Urological Association (AUA): total T **<300 ng/dL** on **2 early-morning samples**.
- Before testosterone replacement therapy (TRT): exclude secondary causes. † Measure luteinising hormone (LH) to separate primary (↑ LH) from secondary (↓/normal LH) hypogonadism; prolactin if LH low.
- Chronic opioid use causes TD (~5 million US men) → Endocrine Society: screen and consider TRT.
- **Randomised evidence for TRT**: improves obesity, metabolic syndrome, diabetes, muscle function, bone. Mood, cognition and sexual-function benefits are observational (possible placebo).
- T and depression share causes (stress, obesity, diabetes, inactivity) and symptoms (low libido, fatigue, poor concentration, irritability) → strong association, but **TRT is not a treatment for depression**; likeliest responders: clearly deficient, <60 y, milder depression.

> **KEY POINT † Fertility caution**: exogenous T suppresses LH/follicle-stimulating hormone (FSH) and intratesticular T → oligo/azoospermia. Men wanting children: avoid TRT; use clomiphene, human chorionic gonadotropin (hCG) or aromatase inhibitors instead → *Ch 69*.

† TRT	Points
Contraindications	Desire for near-term fertility; breast cancer; untreated prostate cancer; haematocrit >50%; untreated severe obstructive sleep apnoea (OSA); uncontrolled heart failure; MI/stroke within 3–6 months
Target	Mid-normal T (≈450–600 ng/dL)
Monitoring	T, haematocrit and PSA at baseline, 3–6 months, then annually. Haematocrit **>54%** → hold/reduce dose, phlebotomy. PSA rise or abnormal digital rectal examination → urology evaluation

Multiple formulations (gels, injections, pellets), each with its own monitoring plan.

Erectile dysfunction

- >50% of men aged 40–70; multifactorial → full medical, surgical, psychosocial and exposure history.
- **Every man with ED, of any age, is at risk of cardiovascular disease (CVD)**, even without symptoms. † Rationale: penile arteries are small, so endothelial dysfunction shows as ED ~2–5 years before coronary events → assess cardiovascular risk factors.
- Sequence: lifestyle counselling → oral phosphodiesterase type 5 (PDE5) inhibitors (not with nitrates) → intracavernosal injection, intraurethral suppository, vacuum device → penile prosthesis (highest satisfaction; counsel irreversibility, infection, malfunction).
- Platelet-rich plasma, low-intensity shock wave therapy: only within trials.

Benign prostatic hyperplasia (BPH) and infertility

- BPH: treatment targets symptom relief; mostly medical (primary care) → refer when medication fails, is not tolerated or not wanted; transurethral resection/ablation remains a mainstay.
- Infertility: no pregnancy after 1 year; male factor in about half; semen analysis is the cornerstone → *Ch 69*.

5 · CARDIOVASCULAR PREVENTION

- ACC/AHA (American College of Cardiology/American Heart Association) atherosclerotic cardiovascular disease (ASCVD) risk calculator: blood pressure, lipids, smoking, diabetes, statin/aspirin → 10-year risk. Limitations: no family history, ignores diabetes type/control.

- Only **low-density lipoprotein cholesterol (LDL-C) lowering** benefits primary prevention; raising high-density lipoprotein (HDL) or lowering triglycerides does not. Lipid screening from age 20.

Statin decision (ACC/AHA)	Action
LDL-C ≥190 mg/dL	High-intensity statin ± nonstatin
LDL-C 70–189, 10-year risk ≥**7.5%**	Moderate–high intensity statin
Risk 5–7.5%	Moderate intensity may be started
Statin-intolerant, risk >20%	Ezetimibe or proprotein convertase subtilisin/kexin type 9 (PCSK9) inhibitor

- Moderate intensity preferred in primary prevention: no trial shows high is better, small extra LDL fall, better tolerated. US Preventive Services Task Force (USPSTF)/National Institute for Health and Care Excellence (NICE)/European Society of Cardiology (ESC) start at ≥10% risk.
- **Coronary artery calcification (CAC) score** (computed tomography): best in asymptomatic men >40, most useful at **intermediate risk**; zero score → <1% likelihood of atherosclerosis (strong negative predictor; less reliable in diabetics).
- Lipoprotein(a): near-linear risk, unaffected by statins.

BP (ACC/AHA 2017)	Systolic	Diastolic
Normal	<120	and <80
Elevated	120–129	and <80
Stage 1	130–139	or 80–89
Stage 2	≥140	or ≥90

Hypertension = most important modifiable risk factor. Elevated/stage 1 → lifestyle (salt restriction, weight loss, Dietary Approaches to Stop Hypertension (DASH) diet, exercise, less alcohol). Drugs: stage 2, or stage 1 + age ≥65, T2DM, CVD, chronic kidney disease or 10-year risk ≥10%.

6 · OBESITY AND METABOLIC SYNDROME

- Body mass index (BMI): overweight 25–29.9; obesity I 30–34.9, II 35–39.9, III ≥40 kg/m^2.
- Weight gain ≥5 kg after age 20 raises coronary and T2DM risk regardless of BMI.
- BMI ≥30, or 25–29.9 with CVD risk/OSA/osteoarthritis → active weight loss; glucagon-like peptide-1 (GLP-1) agonists (semaglutide) US Food and Drug Administration (FDA)-approved for weight management.
- † Metabolic syndrome (ATP III): ≥3 of waist >102 cm, triglycerides ≥150 mg/dL, HDL <40 mg/dL, BP ≥130/85, fasting glucose ≥100 mg/dL. Low T and metabolic syndrome are bidirectionally linked (adipose aromatase ↑ oestradiol → ↓ LH).

7 · GASTROINTESTINAL, SKIN AND MENTAL HEALTH

- **Constipation**: evaluate in every man with urinary tract infection (UTI) or urinary retention; anticholinergics cause it. Alarm features (weight loss, bleeding, iron deficiency anaemia, family history, positive faecal test) → colonoscopy.
- Iron deficiency anaemia: >30-fold risk of gastrointestinal (GI) malignancy. Colorectal screening **45–75** years.
- Reflux: lifestyle + proton pump inhibitor; long-term use → check for Barrett oesophagus.
- Skin: men photoage faster, use less sunscreen. Basal cell (commonest, rarely metastasises), squamous cell, melanoma (ABCDE). Psoriasis linked to MI. Androgenetic alopecia: 5α-reductase inhibitors + minoxidil.
- Suicide much higher in men; substance use, autism, attention-deficit/hyperactivity disorder male-predominant.

8 · MCQ ONE-LINERS

1. Largest male:female mortality gap among leading causes — suicide (>3-fold).
2. Males have higher mortality in 12 of 15 leading causes of death.
3. Only US cancer with rising stage IV at diagnosis — prostate cancer.
4. ADAM questionnaire screens androgen deficiency; SHIM screens erectile function.
5. TD diagnosis — low T on more than one occasion plus symptoms.
6. † AUA total T cut-off — <300 ng/dL, two early-morning samples.
7. † Haematocrit on TRT — AUA withholds at >50%; Endocrine Society at >54%.
8. † TRT in man wanting fertility — contraindicated; suppresses spermatogenesis.
9. Chronic opioids — screen for testosterone deficiency (Endocrine Society).
0. TRT for major depression — not recommended.
1. Randomised TRT benefits — obesity, metabolic syndrome, diabetes, muscle, bone.
2. Any man with ED — considered at risk of CVD.
13. Highest-satisfaction ED treatment — penile prosthesis.
14. PDE5 inhibitor absolute contraindication — nitrates.
15. Only lipid target with primary-prevention benefit — LDL-C.
16. LDL-C ≥190 mg/dL — high-intensity statin.
17. ACC/AHA statin threshold — 10-year ASCVD risk ≥7.5%.
18. CAC most useful at — intermediate risk; zero score = <1% atherosclerosis.
19. Lipoprotein(a) — not lowered by statins.
20. Stage 1 hypertension (2017) — 130–139/80–89 mm Hg.
21. Weight gain ≥5 kg after 20 — raises coronary and T2DM risk.
22. Iron deficiency anaemia — >30-fold GI malignancy risk.
23. Colorectal screening starts at — 45 years.
24. UTI or retention in older man — evaluate for constipation.
25. Commonest skin cancer — basal cell carcinoma.
26. Psoriasis associated with — MI/CVD and psoriatic arthritis.

69 Male Infertility

Reading time ≈ 34 min (slow pace)

1 · DEFINITION AND OVERVIEW

- Infertility: no pregnancy after **1 year** of unprotected intercourse; ~15% of couples. Males and females contribute equally — always evaluate the man.
- Most conceptions occur in the first 3–6 cycles; nearly all by 12. Reasonable to start evaluation after 6 months.
- Reproduction needs two patients and is **probabilistic**: history and tests modify a probability of conception rather than give a verdict.
- Clinic diagnoses: idiopathic (largest) and **varicocele (commonest identifiable cause)**, then obstruction; cryptorchidism, immunological, ejaculatory, endocrine, drugs/radiation, genetic causes are uncommon.

† Classification	Examples
Pre-testicular (hypothalamic–pituitary / endocrine)	Hypogonadotropic hypogonadism (HH), Kallmann, prolactinoma, exogenous androgens, opioids, obesity, congenital adrenal hyperplasia (CAH)
Testicular	Klinefelter, Y microdeletion, cryptorchidism, torsion, orchitis, varicocele, chemotherapy/radiation, Leydig failure
Post-testicular	Obstruction (epididymal, vasal, ejaculatory duct), congenital bilateral absence of vas deferens (CBAVD), ejaculatory dysfunction, antisperm antibodies, structural sperm defects

2 · HISTORY

- **Female partner's age is the most important determinant** of a couple's reproductive potential; fecundity falls sharply after 35.
- Advanced paternal age (≥40): counsel on ↑ offspring risk of bipolar disorder, schizophrenia, autism, attention deficit (ASRM/AUA [American Society for Reproductive Medicine / American Urological Association] 2021).
- Prior paternity eliminates many congenital causes.
- Intercourse: semen parameters peak after **1–2 days' abstinence** then decline — "saving up" worsens quality. Advise daily intercourse around ovulation (timed from 2–3 cycles of basal temperature/urinary kits).
- Lubricants: K-Y, Astroglide, Replens, saliva impair sperm; **Pre-Seed** (isotonic) is safe.
- Mnemonic **TICS**: Toxins, Infection, Childhood history, Sexual history.
- Diabetes, multiple sclerosis, spinal cord injury (SCI) → ejaculatory failure. Transurethral resection of prostate (TURP) and retroperitoneal lymph node dissection → retrograde ejaculation.
- Cancer impairs spermatogenesis even before chemotherapy; azoospermia may reveal testis cancer.
- Herniorrhaphy (especially mesh, including adolescent repair) can entrap the vas.

Gonadotoxins and drugs

> KEY POINT **Exogenous testosterone (T) is a contraceptive**: it is aromatised to oestradiol (E2) and feeds back on the pituitary → ↓ luteinising hormone (LH) → ↓ intratesticular T (which must be far higher than serum) → ↓ spermatogenesis. Recovery usually ≥4 months after stopping (≤2 years' use); after longer use, unknown.

- Endocrine modulators: antiandrogens, spironolactone, corticosteroids, oestrogens, anabolic steroids, some antiretrovirals; 5α-reductase inhibitors (limited effect).
- Cannabis: dose-dependent ↓ T and count. Alcohol: ↓ androgens, ↑ aromatisation. Smoking: oxidative stress, ↓ DNA quality, abnormal protamine ratio; additive with varicocele. Maternal

smoking → sons with smaller testes.
- Opioids, antipsychotics, environmental toxicants suppress the pituitary → remove agent.

Infection and childhood conditions

- Infection injures by **anatomical** (stricture, epididymal/vasal scarring; tuberculosis) or **functional** routes (direct sperm damage, leucocytes, reactive oxygen species [ROS], cytokines). Mumps orchitis history is reliable.
- Torsion: contralateral biopsy usually abnormal; about half have later spermatogenic effects; blood–testis barrier disruption → antisperm antibodies.

Cryptorchidism (after orchidopexy)	Unilateral	Bilateral
Paternity	Near normal	Markedly reduced
Inhibin B	~⅔ of controls	~⅓ of controls

- **Laterality is the key prognostic factor.** The lesion is germ-cell/Sertoli more than Leydig, begins in fetal life and worsens with time.
- Orchidopexy within **18 months** (AUA); germ cells are lost progressively from 15–18 months. If missed, before 10 years improves later surgical sperm retrieval; age at orchidopexy is second only to T in predicting retrieval.

3 · PHYSICAL EXAMINATION

- Androgenisation: body hair, voice, gynaecomastia (↑ E2). Eunuchoid habitus: arm span >5 cm over height, lower segment >5 cm over upper.
- Klinefelter classic habitus often absent — cannot exclude it clinically.
- Examine standing (vasovagal risk); interview the man privately.
- Easily palpable/engorged epididymis → obstruction. Testis size reflects spermatogenic mass.

Testis size	Abnormal
Seager caliper (long axis)	**≤4.6 cm**
Prader orchidometer (volume)	≤20 mL
Scrotal ultrasound	Most accurate; added value unclear

- **Vas**: if the presumed vas slips from the fingers three times, it is absent (Meacham). Unilateral absence → wolffian anomaly → renal ultrasound. Bilateral absence → cystic fibrosis transmembrane conductance regulator (CFTR) testing + renal ultrasound (renal agenesis ~1 in 10).
- Palpable seminal vesicles → ejaculatory duct obstruction (EDO). Digital rectal examination otherwise adds little.

Varicocele grade	Definition
Subclinical	Only on Doppler — **do not treat**
I	Palpable only with Valsalva, standing
II	Palpable without Valsalva, not visible
III	Visible through scrotal skin

4 · ENDOCRINE EVALUATION

ALGORITHM
Morning sample: total T + sex hormone–binding globulin (SHBG) + albumin → calculated bioavailable T → LH, follicle-stimulating hormone (FSH) → E2; prolactin only if indicated

- Adequate androgenisation: total T ≥**280–300 ng/dL**. Low T is common even with density >20 million/mL → assess every infertile man (full endocrine work-up unnecessary above ~66 million/mL).

- Bioavailable T = free (0.5–3%) + albumin-bound (54–68%); SHBG-bound (30–44%) is unavailable. Calculate by Vermeulen formula.
- SHBG ↓ in obesity, hypothyroidism, diabetes, steroids → total T under-reads androgen status; SHBG ↑ in ageing, cirrhosis, hyperthyroidism, HIV (human immunodeficiency virus), anticonvulsants → total T over-reads.
- Draw in the morning (diurnal peak, especially in young men). Borderline values: 3 samples 20 minutes apart, averaged (pulsatility).
- **LH localises**: ↑ LH + ↓ T = testicular (Leydig) failure; ↓/normal LH + ↓ T = pituitary/hypothalamic.
- **FSH indexes germ-cell mass**: germ-cell loss → ↓ inhibin B (Sertoli) → ↑ FSH. When sperm present, abnormal concentration odds rise from FSH 4.5 IU/L. Inhibin B marginally better; anti-müllerian hormone experimental.
- T (ng/dL) : E2 (pg/mL) ratio **<10:1** → aromatase excess.
- Prolactin: only with headache, visual change, erectile dysfunction or low T; labile — repeat if raised. 20–50 µg/L: no further work-up; significant elevation → pituitary MRI.

5 · SEMEN ANALYSIS

- Only direct appraisal of gametes; poor discriminator (all parameters area under the curve ≈0.6). **Only azoospermia is definitive**; fertile and infertile distributions overlap, so normal values never prove fertility.
- ≥2 samples, 2–3 weeks apart, constant abstinence (evidence favours 1–2 days; WHO 2–7). Non-toxic container; special condom if masturbation unacceptable.
- Liquefaction within 30 min; brown semen often SCI; red = blood. pH no longer recommended.

WHO 6th edition (2021)	5th centile	50th centile
Volume (mL)	**1.4**	3.0
Concentration (million/mL)	**16**	66
Total count (million)	39	210
Total motility (%)	42	64
Progressive motility (%)	30	55
Normal forms, strict (%)	**4**	14
Vitality (%)	54	78

Derived from men whose partners conceived within 1 year — a **reference**, not a pass/fail line. Legacy WHO (1st–4th) concentration threshold 20 million/mL. Guzick two-threshold (subfertile/fertile) concentration 13.5/48 million/mL.

- Terms: oligozoospermia (low density, defined in clinical context), cryptozoospermia (too few to count), asthenozoospermia (motility), teratozoospermia (morphology), necrozoospermia (dead sperm).
- Morphology: Kruger strict criteria; isolated teratozoospermia does not predict natural, intrauterine insemination (IUI) or in vitro fertilisation (IVF) outcome. Monomorphic defects matter (below).
- Vitality: with near-total immotility — eosin–nigrosin (kills sperm) vs hypo-osmotic swelling test (live sperm usable for intracytoplasmic sperm injection [ICSI]).
- **Low volume (<1.0 mL) / aspermia**: EDO, CBAVD/seminal vesicle hypoplasia, severe hypoandrogenism, retrograde ejaculation, neuropathy, α-blockers → **postejaculatory urinalysis** (significant if urinary ≈ antegrade sperm) → transrectal ultrasound (TRUS) if no urinary sperm.
- † Sperm DNA fragmentation (e.g. sperm chromatin structure assay, terminal deoxynucleotidyl transferase dUTP nick-end labelling (TUNEL)): not routine; consider in recurrent pregnancy loss or failed assisted reproduction; raised by varicocele, smoking, oxidative stress.

6 · GENETIC TESTING

ALGORITHM
Non-obstructive azoospermia (NOA) or density **<u><5 million/mL</u>** → karyotype + Y-microdeletion (ASRM/AUA 2021) · CBAVD (or idiopathic EDO) → CFTR testing of man **and partner**

Azoospermia factor (AZF) region on Yq	Genes	Surgical sperm retrieval
AZFa	USP9Y, DDX3Y	Complete deletion → essentially nil
AZFb	EIF1AY, KDM5D	Complete deletion → essentially nil
AZFc	DAZ	Sperm often in ejaculate or testis

- Mechanism: Yq amplicons (palindromic repeats) mediate repair but make the region fragile → microdeletions. Y passes father → son: sons of ICSI inherit the deletion.
- AZFc with oligozoospermia → progressive decline → **cryopreserve sperm**.
- CFTR: >1600 mutations; ΔF508 commonest severe. Panels test 25–40 mutations → residual risk if negative. Two severe alleles = cystic fibrosis (CF); mild allele(s) = CBAVD.

7 · IMAGING

- Not for screening — incidental findings (microlithiasis) cause harm. Varicocele is a clinical diagnosis.
- **TRUS** for EDO (azoospermia/oligozoospermia + low volume): seminal vesicle anteroposterior diameter **<u>>1.5 cm</u>** ± midline cyst; low specificity.
- Seminal vesicle aspiration (≥3 sperm/high-power field = positive; useful in partial EDO), vesiculography, chromotubation: more accurate, invasive.

8 · AZOOSPERMIA: OBSTRUCTIVE (OA) VS NON-OBSTRUCTIVE (NOA)

	OA	NOA (spermatogenic dysfunction)
FSH	**<u>≤7.6 IU/L</u>**	>7.6 IU/L
Testis long axis	>4.6 cm	≤4.6 cm
Management	Reconstruction → *Ch 70* or sperm retrieval	Genetic tests; microdissection testicular sperm extraction + ICSI

ALGORITHM
Azoospermia → vasa absent? Yes → CFTR (both partners) · No → volume <1 mL? Yes → postejaculatory urinalysis: sperm → retrograde ejaculation; none → TRUS for EDO · Normal volume → FSH + testis size → OA vs NOA

- NOA histology: hypospermatogenesis; maturation arrest; Sertoli cell–only.
- Consider testis cancer in every azoospermic man.

9 · SPECIFIC CAUSES AND TREATMENT

<u>Klinefelter syndrome (47,XXY)</u>

- **Commonest genetic cause**; 1 in 500–1000; non-mosaic in 80–90%. Small firm testes, ↑ gonadotropins, low T, azoospermia; ↑ mediastinal germ cell tumours.
- Ejaculated sperm ~8%; testicular sperm retrieval ~**<u>50%</u>**.
- **Trap**: exogenous T alone suppresses residual spermatogenesis. Consider sperm extraction early (before T); use aromatase inhibitors/hCG rather than T when fertility is wanted.

<u>Hypogonadotropic hypogonadism — most treatable cause</u>

The testis is normal; replace the missing upstream signal and spermatogenesis follows.

State	Hormones	Treatment
Kallmann (anosmia; KAL1, GNRHR, etc.)/complete HH	↓ LH, ↓ FSH	hCG ± recombinant FSH or human menopausal gonadotropin (hMG)
Isolated LH deficiency ("fertile eunuch")	↓ LH, normal FSH	hCG
Isolated FSH deficiency	↓ FSH	FSH
Hypothalamic gonadotropin-releasing hormone (GnRH) deficiency	↓ GnRH	Pulsatile subcutaneous GnRH pump every 2 h, ≥6 months
Partial HH (common)	LH inappropriately low for T	Selective oestrogen receptor modulator (SERM) or aromatase inhibitor

- **Human chorionic gonadotropin (hCG)** 1500–5000 IU 2–3×/week (max 10,000/week), titrate to T; may initiate spermatogenesis alone. Add hMG/rFSH 75 IU 2–3×/week if needed. After prior androgen abuse, 1–2 years may be needed.
- Prolactinoma: cabergoline (fewer side effects) or bromocriptine.
- CAH: corticosteroids may restore fertility; watch for testicular adrenal rest tumours.

Empirical medical therapy

Drug	Dose	Rationale/notes
Clomiphene (SERM)	25 mg daily or 50 mg alternate days; max 100 mg	Blocks hypothalamic E2 feedback → ↑ LH/FSH → ↑ intratesticular T; off-label; T rises without PSA (prostate-specific antigen) or haematocrit change; target 600–800 ng/dL
Anastrozole / letrozole (aromatase inhibitors)	Anastrozole 1 mg daily	↑ T, ↓ E2; indicated when T:E2 <10:1 (e.g. obesity); caution — long-term bone density

Steroidogenic (primary) failure and androgen resistance

- ↑ LH, ↓ T: no accepted therapy; exogenous T contraindicated. Azoospermia → surgical sperm extraction.
- Androgen receptor defect (Xq11–12, long CAG repeat): ↑ T, ↑ LH, ↑ E2 — very high T with infertility suggests it; ICSI. 5α-reductase type 2 deficiency: ICSI pregnancies reported.

CBAVD

- Often a genital form of CF; nearly all men with CF lack both vasa. Spermatogenesis normal; low-volume azoospermia.
- CF carrier frequency ~1 in 20 (northern Europeans) → test partner; genetic counselling. Treatment: sperm retrieval + ICSI. Urologist may uncover indolent CF.

Varicocele

- ~15–20% of all men vs 35–50% of infertile men — a risk factor, not a sentence.
- Left predominance: left internal spermatic vein enters the left renal vein 8–10 cm higher → taller column → valve incompetence. **Isolated right varicocele → exclude renal/retroperitoneal tumour.**
- Injury: loss of countercurrent heat exchange (main), oxidative stress → DNA fragmentation, hypoxia, apoptosis; progressive testicular atrophy.
- **Repair** palpable varicocele + abnormal semen in an infertile couple: improves semen, pregnancy, live birth and DNA fragmentation; may down-stage assisted reproduction (e.g. IVF → IUI). Never repair subclinical varicocele → *Ch 70*.

Ejaculatory duct obstruction

- <5% of azoospermia; idiopathic EDO carries CFTR mutations in ~half. Causes: cysts, infection, surgery.
- Low volume, ↓ motility, oligo/azoospermia ± painful ejaculation, haematospermia.

- **Transurethral resection of ejaculatory ducts (TURED)** at the verumontanum, depth guided by synchronous TRUS to avoid rectal injury.

Antisperm antibodies and structural defects

- Antisperm antibodies follow barrier breach (vasectomy, torsion, infection, trauma). Washing fails (antibody is bound); steroid trials conflict → **ICSI**.
- Globozoospermia (no acrosome; SPATA16, DPY19L2): ICSI ± oocyte activation. Macrocephalic multiflagellate (AURKC): very high aneuploidy → ICSI not recommended. Primary ciliary dyskinesia (DNAH5, DNAI1): immotile sperm, sinusitis, situs defects → ICSI (testicular sperm better).

10 · LIFESTYLE AND OBESITY

- Obesity: ↑ adipose aromatase → ↑ E2 → ↓ LH and T; ↓ SHBG; ↓ inhibin B; plus hormone-independent effects on epididymis and DNA → correcting hormones alone may not restore fertility.
- Male obesity worsens pregnancy, miscarriage and live birth in assisted reproduction; weight loss advised.
- Stop smoking, cannabis, excess alcohol, anabolic steroids; avoid heat (hot tubs). Offer psychological support — stress may impair semen.

11 · ASSISTED REPRODUCTION

- Sperm preparation: density gradient centrifugation, swim-up; newer selection methods unproven.
- ICSI needs only one sperm per oocyte but bypasses natural selection.
- † Choice by total motile sperm count (TMSC): IUI if processed TMSC ≥5–10 million; IVF below this or after failed IUI; ICSI for severe oligozoospermia, surgically retrieved sperm, antisperm antibodies, prior fertilisation failure.
- † Sperm retrieval: OA — percutaneous/microsurgical epididymal aspiration (PESA/MESA) or testicular aspiration; NOA — microdissection TESE (testicular sperm extraction).

12 · MCQ ONE-LINERS

1. Most important determinant of couple fertility — female partner's age.
2. Commonest identifiable correctable cause — varicocele.
3. Commonest genetic cause — Klinefelter (47,XXY).
4. Optimal abstinence before semen analysis — 1–2 days.
5. Sperm-safe lubricant — Pre-Seed.
6. Exogenous T effect on spermatogenesis — suppression via ↓ LH and intratesticular T.
7. Abnormal caliper testis long axis — ≤4.6 cm.
8. OA vs NOA FSH cut-off — 7.6 IU/L.
9. Adequate total T threshold — 280–300 ng/dL.
0. T:E2 ratio suggesting aromatase excess — <10:1.
1. WHO 6th 5th-centile concentration — 16 million/mL.
2. WHO 6th 5th-centile strict normal forms — 4%.
3. WHO 6th 5th-centile volume — 1.4 mL.
4. Only definitive semen result — azoospermia.
5. Low volume + azoospermia + no urinary sperm — TRUS for EDO.
6. TRUS sign of EDO — seminal vesicle >1.5 cm anteroposterior (AP).
7. Karyotype/Y-microdeletion indication — NOA or <5 million/mL.
18. AZFa or AZFb complete deletion — retrieval essentially nil.
19. Y deletion with retrievable sperm — AZFc.
20. CBAVD — test CFTR in man and partner.
21. Unilateral absent vas — renal ultrasound for agenesis.
22. Kallmann syndrome — HH + anosmia; KAL1.
23. Fertile eunuch — isolated LH deficiency.
24. First-line in HH — hCG ± FSH.
25. Klinefelter testicular sperm retrieval — ~50%.
26. Subclinical varicocele — do not repair.
27. Isolated right varicocele — exclude renal tumour.
28. Main varicocele injury mechanism — heat (lost countercurrent exchange).
29. Orchidopexy timing — by 18 months.
30. Cryptorchidism prognostic factor — laterality.
31. Prolactinoma drug of choice — cabergoline.
32. Globozoospermia — absent acrosome; ICSI.
33. Macrocephalic multiflagellate sperm — AURKC; ICSI not recommended.
34. Vitality test preserving sperm — hypo-osmotic swelling.

70 Surgical Management of Male Infertility

Reading time ≈ 42 min (slow pace)

1 · PRINCIPLES

- In vitro fertilisation (IVF) with intracytoplasmic sperm injection (ICSI) treats even nonobstructive azoospermia (NOA), but is costly, burdens the female partner and bypasses natural barriers (genetic transmission, birth defects).
- Treating the man (microsurgical reconstruction for obstructive azoospermia (OA), varicocelectomy) is safest and most cost-effective, and moves the couple **down the ladder**: donor sperm → IVF/ICSI with testicular sperm → IVF/ICSI with ejaculated sperm or intrauterine insemination (IUI) → natural conception.

2 · SURGICAL ANATOMY

Structure	Arteries (origin)
Testis	**Testicular** (aorta) — main; deferential (internal iliac/superior vesical); cremasteric (inferior epigastric; mainly tunica vaginalis)
Epididymis	Superior/medial epididymal (testicular a.) → caput; inferior epididymal (deferential a.) → cauda; extensive interconnection
Vas	Abdominal end: deferential only. Testicular end: deferential + inferior epididymal. **No supply from cremasteric or cord vessels**

- Testicular artery diameter **exceeds deferential + cremasteric combined**. After vasectomy the vasal artery is presumed lost → the testicular artery must be preserved at any later operation (e.g. varicocelectomy).
- Epididymis may be dissected off the testis to the caput (inferior and medial epididymal arteries ligated) if the **superior epididymal artery is intact** → gains **4–6 cm**.
- A vas obstructed or transected at two sites with vasal vessels interrupted at both → intervening segment fibroses → **never two simultaneous vasovasostomies on one vas**; excise the segment, do one tension-free anastomosis.
- **Excurrent ducts**: rete testis → 7–11 efferent ducts → caput (anastomosing lobules) → **single tubule** from caput–corpus junction → vas. Injury distal to the caput (even **one stitch**) obstructs the whole side. Single puncture is safe only at the most proximal caput.
- **Ejaculatory ducts** enter the prostatic urethra at the level of the utricle; bilateral ejaculatory duct obstruction (EDO) → azoospermia with low volume, acidic pH, low/absent fructose; transrectal ultrasound (TRUS) shows dilated seminal vesicles (SVs)/ducts.

3 · DIAGNOSTIC TESTIS BIOPSY

Indications

- Azoospermia with **normal testis size and consistency, palpable vasa and normal follicle-stimulating hormone (FSH)** → distinguishes OA from NOA.
- **Not needed**: congenital bilateral absence of the vas deferens (CBAVD) with normal FSH (spermatogenesis always present); highly positive antisperm antibodies with prior iatrogenic vasal/epididymal injury (obstruction near-certain).
- Usually bilateral; because ICSI needs only one sperm, biopsy is **diagnostic and therapeutic** — plan as testicular sperm extraction (TESE) with fresh wet examination and **cryopreservation at the same sitting** (andrology laboratory on site).

Open biopsy — gold standard

ALGORITHM
Local with cord block → larger testis (contralateral to a varicocele) → 2 cm scrotal incision → open tunica vaginalis → two 5-0 stay sutures parallel to albugineal vessels → 3–4 mm incision with 15° microknife, bipolar crossing vessels → pea-sized sample with iris scissors (no forceps) → touch prep → wet squash prep → negative: contralateral side → close tunica vaginalis continuous 5-0 (limits scarring); no antibiotics

- Fixative: **Bouin's, Zenker's or buffered glutaraldehyde — never formalin** (distorts histology).
- Microscope at 10–25× → selectively sample larger tubules; non-absorbable suture marks a positive site for future TESE.
- **Percutaneous** 14-G core (blind; avoid in scarred scrotum) or 23-G aspiration (few tubules, poor architecture) — best for fresh retrieval in OA with normal spermatogenesis, poor in NOA.

Complication	Point
Haematoma — most common	Prevent by magnification, bipolar, continuous tunica vaginalis closure; drainage exceedingly rare
Epididymal biopsy — most serious	Sperm in that tubule → obstruction there is now certain; no sperm → no harm

4 · VASOGRAPHY

- **Absolute indications (all three)**: azoospermia + complete spermatogenesis on biopsy + at least one palpable vas. Relative: severe oligospermia with normal biopsy, high sperm-bound antibodies, low volume with poor motility (partial EDO).
- Perform **only at the time of planned reconstruction** (scar makes later surgery harder; poor technique causes stricture). It cannot diagnose epididymal obstruction — the commonest non-vasectomy site.

ALGORITHM
High scrotal incision → vas at straight–convoluted junction → hemivasotomy at 25× → **vasal fluid examined first** → motile sperm: barbotage and cryopreserve **before any dye** → 24-G sheath into abdominal end, 1 mL saline: easy flow = patent → (rarely) dilute indigo carmine, blue urine in catheter confirms → formal contrast study only for obstruction proximal to the internal ring → close with 10-0 mucosa, 8-0 muscularis

Vasal fluid / vasogram	Meaning → action
No sperm, abdominal end patent	Epididymal obstruction → vasoepididymostomy
Copious fluid, many sperm	Vasal obstruction or EDO → locate (vasogram, 2-0 Prolene threaded until it stops)
Obstruction at ejaculatory ducts; both vasa fill from one side	EDO / midline cyst → transurethral resection of ejaculatory ducts (TURED)
Vas ends blindly	Partial vasal agenesis → cystic fibrosis transmembrane conductance regulator (CFTR) testing; retrieve + freeze
Inguinal block	Inguinal or crossed vasovasostomy
Many sperm, normal vasogram	Retrograde ejaculation, emission failure, aperistalsis

- **Indigo carmine, not methylene blue** — methylene blue kills sperm; use it only after sperm are banked.
- Complications: **stricture** (repeated needle punctures, poor closure, non-water-soluble contrast), devascularisation, haematoma, sperm granuloma.
- **Transrectal seminal vesiculography**: dilated SVs/midline cyst → TRUS aspiration; sperm present = ≥1 vas and epididymis patent → contrast + indigo carmine → TURED without scrotal exploration.

No sperm → secondary epididymal obstruction → retrieval + ICSI.

5 · VASOVASOSTOMY (VASECTOMY REVERSAL)

Evaluation

- Prior fertility usually proves spermatogenesis; FSH if testes small/soft (raised = poorer prognosis).
- Examination: small soft testes → poor outcome; indurated epididymis → vasoepididymostomy likely; **sperm granuloma at testicular end → better prognosis** (vents epididymal pressure) regardless of interval; large gap/high vasectomy → inguinal extension; groin scars → iatrogenic obstruction.
- Centrifuged semen pellet with whole sperm → sperm certain in at least one vas (favourable). Low volume → TRUS for EDO.
- General anaesthesia preferred (movement magnified); operate only if able to do vasoepididymostomy (may take ~4 h).

Approach and preparation

- Scrotal for most (length problem is abdominal side → extend toward external ring); inguinal if hernia/orchidopexy obstruction suspected; robotic for transection above the inguinal ligament.
- Preserve perivasal sheath and vessels (**never strip the sheath**); excise granuloma staying on the vas to protect the testicular artery.
- **Length**, escalating: blunt dissection to external then internal ring → free convoluted vas from epididymal tunica (4–6 cm; **do not unravel**) → reroute beneath inguinal floor → dissect epididymis off testis (4–6 cm) → combined, gaps up to **10 cm**.
- Cut transversely through a slotted nerve-holding clamp. Healthy end: white mucosal ring that springs back, soft muscularis, bleeding edge. **Gritty muscularis or poor bleeding → recut.**
- Testicular end: **aspirate, never flush** (pressure damages epididymal tubules); fluid examined. Abdominal end: cannulate, flush saline to prove patency.

Intraoperative fluid grading — the decision

Vasal fluid	Microscopy	Operation
Copious, crystal clear, watery	No sperm	Vasovasostomy
Copious, cloudy, thin	Sperm with tails	Vasovasostomy
Copious, creamy yellow	Many sperm heads	Vasovasostomy
Scant, granuloma present	Barbotage: sperm	Vasovasostomy
Thick white **toothpaste**	No sperm/heads	**Vasoepididymostomy**
Scant white thin, or dry without granuloma	No sperm/heads	**Vasoepididymostomy**
Any	Grape-like clusters of heads	**Vasoepididymostomy**

> **KEY POINT** Why clear watery fluid still gets vasovasostomy: the epididymis is not yet blown out and sperm usually return. Toothpaste fluid means a secondary epididymal blow-out → open tunica vaginalis; a discrete level (dilated above, collapsed below) confirms. When in doubt, or inexperienced, do vasovasostomy.

Anastomotic principles

Principle	Reason
Accurate mucosa-to-mucosa	Testicular lumen 2–5× abdominal; must handle discrepancy
Leakproof	Sperm are antigenic; fluid lacks platelets/clotting factors — seal depends on mucosal sutures
Tension-free	Tension → early sperm then falling counts → azoospermia (thin fibrotic band)
Good blood supply; healthy mucosa/muscularis; atraumatic	Recut until bleeding and soft; redo if back-walled or torn

- **Multilayer microdot**: six dots mapped on each end → **six 10-0 mucosal sutures** (double-armed, inside-out, no back-walling) → 9-0 deep muscularis between → 9-0 adventitia → 8-0 sheath. Two-layer alternative: 10-0 mucosa + 8-0 full-thickness outer layer.
- **Reinforcing vasal (ReVas) technique**: sheath 3 cm either side approximated with 5-0 Prolene into a hairpin loop + 5–10 reinforcing sutures → tension relief, higher patency and pregnancy.
- **Convoluted vas**: perfect transverse cut, do not unravel, small bites, reinforce — better than joining an unhealthy straight segment.
- **Crossed vasovasostomy**: unilateral inguinal obstruction with contralateral atrophic testis, or one side unreconstructable → one good anastomosis beats two poor operations; testicular transposition if vas very short.
- **Varicocele + reversal**: vasal artery/veins likely compromised → microscope and testicular artery preservation mandatory; safer to stage varicocelectomy ≥6 months later.

Post-operative care and outcomes

- Scrotal support and no heavy lifting 4 weeks; ejaculation 2 weeks, intercourse 4 weeks.
- Semen at **6 weeks**, then 6 months if no pregnancy. Poor motility → 4-week prednisone course. Azoospermia at 6 months → redo.
- Complications: small perivasal haematoma (most common); anastomotic granuloma (harbinger of obstruction); **late stricture** — falling motility then count. **Bank sperm as soon as motile sperm appear.**
- Sperm in vasal fluid on ≥1 side → sperm return in virtually all; pregnancy about half, depending on vasectomy interval and female age.

6 · VASOEPIDIDYMOSTOMY

- Epididymis: single tubule ~200 µm, 12–15 feet; wall thinnest at caput → **more distal anastomosis = easier and better pregnancy**. Adaptation may continue up to 2 years.
- **Indications**: vasectomy reversal per fluid criteria; non-vasectomy OA with proven spermatogenesis + no sperm in vasal fluid + patent vas/ejaculatory ducts.

ALGORITHM

Midline raphe → vas hemitransected at straight–convoluted junction, fluid sampled (no sperm), patency confirmed → full transection → open tunica vaginalis → site **above** obstruction/granuloma where tubules are dilated (if unclear: needle-puncture distal to proximal until sperm) → 3–4 mm tunic window → vas through separate tunica vaginalis opening, **fixed with 5-0 Prolene to tunica albuginea (tension relief)** → posterior tunic–vas 8-0 → intussusception sutures → outer tunic–vas adventitia

Technique	For	Against
Longitudinal intussusception (LIVE) — used for all	Two 10-0 longitudinal sutures placed in the **still-distended** tubule; incise between; tubule pulled into vas; bloodless, leakproof	Fluid cannot be checked before sutures are set
End-to-side	Bloodless; fluid checked first	Suturing a collapsed tubule
End-to-end	Rapid level finding; mobilisation bridges gap	Haemostasis hard; inferior epididymal supply lost

- Why intussusception: needles go in before the tubule collapses, and forward epididymal flow pushes the tubule wall against vasal mucosa.
- **No simultaneous varicocelectomy** (raised epididymal venous pressure, bleeding) — stage after 6 months.
- Patency in the majority; late shutdown occurs → **bank sperm intraoperatively and as soon as they appear**. Failure → redo or retrieval + ICSI.

7 · TRANSURETHRAL RESECTION OF THE EJACULATORY DUCTS (TURED)

- EDO: congenital (midline Müllerian cyst, atresia — end of the CBAVD spectrum) or acquired (instrumentation, infection, cyst compression).
- Suspect with azoospermia/severe oligoasthenospermia, ≥1 palpable vas, **low volume, acidic pH, low fructose**, normal FSH. TRUS is key: cyst or dilated ducts/SVs.

ALGORITHM
24-Fr resectoscope; finger in rectum lifts posterior prostate → resect the **verumontanum** to open the dilated orifices/cyst (ducts exit at and lateral to the veru) → spare bladder neck, external sphincter and rectum → **blue efflux confirms**; minimal coagulation → catheter overnight, 7 days antibiotics

- Cold-knife incision alone almost always reobstructs. Avoid confirmatory high-pressure vesiculography (secondary epididymal obstruction).
- Sperm appear in about half of azoospermic men.
- Offer only to azoospermic/severely oligoasthenospermic men after the couple, fully counselled, **declines IVF**.

Complication	Management
Urine reflux into ducts/SVs (anti-reflux valve destroyed) — impairs sperm	Document by voiding cystourethrogram or semen creatinine
Chemical/bacterial epididymitis → epididymal obstruction	Low-dose antibiotic suppression until pregnancy; chronic → vasectomy/epididymectomy
Retrograde ejaculation	Pseudoephedrine 60 mg 90 min before ejaculation; or urinary sperm (alkalinised) for IUI/ICSI

- **Seminal vesiculoscopy** (4–6 Fr via utricle; rinse, laser stones) aims to avoid reflux; developed for refractory haematospermia and duct stones; no comparison with TURED.

8 · SPERM RETRIEVAL

- Candidates: CBAVD, failed/unreconstructable obstruction, men declining reversal, NOA.
- Sperm from chronic obstruction have poor motility → **ICSI essential** regardless of numbers.

Technique	Indication	Yield / features
MESA — microsurgical epididymal sperm aspiration	OA (CBAVD, unreconstructable); not NOA	Highest numbers of motile sperm; multiple vials; low complications; needs microsurgery
PESA — percutaneous epididymal sperm aspiration	OA; not NOA	Local anaesthetic, no microscope; fewer sperm, often too few to freeze; pregnancy ~half of open; may obstruct epididymis
TESA — testicular sperm aspiration	OA (backup to epididymal); poor in NOA	Small numbers, immature/immotile; 10–20 passes
TESE — testicular sperm extraction (open)	OA without epididymal sperm/absent epididymis; NOA	Most consistent, highest yield of open biopsy techniques
Micro-TESE	**NOA — best technique**	Most sperm with least tissue; sperm found in about half of NOA men

MESA technique

- Open tunica vaginalis → 16–25× → incise a dilated tubule → wet prep → no sperm: close (10-0) and move **proximally**, up to efferent ductules → collect with micropipette by **capillary action only** (negative pressure disrupts mucosa) → flush into IVF medium → freeze multiple aliquots.

KEY POINT Paradox: in obstruction the **most motile sperm are proximal** (efferent ductules); distal tubules hold senescent fragments and macrophages. Yet fertilisation is best when the longest epididymis is preserved.

Micro-TESE

- Basis: in NOA spermatogenesis is focal; tubules with sperm are **larger, plumper, opaque/yellow**; Sertoli cell–only tubules are thin and white.

ALGORITHM
General/regional → testis delivered → 10× avascular plane on anterior albuginea → wide equatorial incision → bivalve (clam-shell) → systematic search, excise larger tubules → minced, examined at once by in-theatre andrologist → map positive areas → negative: contralateral → close albuginea 5-0, tunica vaginalis 4-0

Diagnosis	Retrieval
Post-orchidopexy cryptorchidism	~74%
Azoospermia factor c (AZFc) deletion	~70%
Klinefelter (47,XXY)	~68%
Post-chemotherapy	~53%
Maturation arrest	~40%
Sertoli cell–only	~30%
AZFa or AZFb deletion	**0% — do not operate**

- NOA ICSI: high miscarriage (sperm aneuploidy, DNA damage).
- **Postmortem retrieval**: vasal irrigation or epididymal mincing; requires the man's prior consent.

9 · CRYOPRESERVATION

- Freeze at diagnostic biopsy, at every reconstruction, and from the ejaculate as soon as motile sperm return after reversal (late stricture/shutdown).
- Sample before any dye. Frozen sperm do not reduce ICSI success, except in spinal cord injury (SCI), where fresh testicular sperm may be better; epididymal vials allow elective retrieval.

10 · VARICOCELECTOMY

- Most common infertility operation; varicocele in ~15% of men, ~35% of primary infertility.
- Repair halts decline, improves semen, live birth, IVF/ICSI outcomes and testosterone. AUA/ASRM: **consider repair in infertility with abnormal semen**; no definitive evidence for repair before assisted reproductive technology (ART) in NOA.
- Poor results: recurrence, artery ligation, hydrocele; younger age → greater benefit.

Approaches

Approach	Artery preserved	Hydrocele	Recurrence	Note
Retroperitoneal (Palomo)	Often not	Common	15–25%	Misses periarterial and scrotal collaterals
Conventional inguinal	Uncertain	3–15%	5–15%	Misses scrotal and periarterial veins
Laparoscopic	Yes	~12%	3–15%	Serious morbidity; general anaesthetic
Radiographic embolisation	Yes	0	Higher; technical failures	Misses small collaterals

Approach	Artery preserved	Hydrocele	Recurrence	Note
Microsurgical inguinal/subinguinal	**Yes**	**~0**	**<1%**	Preferred

- Subinguinal: no fascia opened → less pain; but more veins, artery often branched and wrapped in venous network. **Open the external oblique (inguinal)** in prepubertal children, solitary testis, tight low ring, short cord, less experienced surgeons or concurrent hernia repair; stay subinguinal after prior inguinal surgery or in obesity.

Microsurgical subinguinal technique

ALGORITHM
Local (liposomal bupivacaine) + sedation → 2–3 cm incision over external ring → cord delivered on Penrose and backboard (exclude ilioinguinal and genital branch of genitofemoral nerves) → **micro-Doppler maps arteries before handling** (vasospasm) → open external then internal spermatic fascia → 10–25×, 1% papaverine → artery freed and looped; temporary occlusion confirms Doppler signal → **all veins except vasal doubly ligated/divided** (clips ≥1 mm, silk <2 mm, bipolar for tiny) → lymphatics preserved → deliver testis: ligate external spermatic and gubernacular veins → re-run cord → impulse test

- **End point**: only testicular arteries, cremasteric arteries, lymphatics and vas with its vessels remain.
- Do not rely on visible pulsation (transmitted); the artery often adheres beneath a large vein.
- Vasal veins kept; ligate only if >2.5 mm, keeping one set — **one intact set of deferential veins guarantees venous return**.
- **Impulse test**: squeezing the plexus above the testis should transmit a thrill to the largest ligated veins on the testicular side; no impulse → a vein was missed → re-run cord.
- Aftercare: intercourse 1 week; all activity 4 weeks (6 weeks if fascia opened).

Complications

Complication	Cause	Prevention / management
Hydrocele (commonest non-microsurgical)	Lymphatic ligation (fluid protein = lymph); impairs heat exchange	Magnification preserves lymphatics; treat as hydrocele
Artery injury	Artery 1–1.5 mm, ~two-thirds of flow, adherent to veins	Microscope + Doppler; atrophy/azoospermia if bilateral
Recurrence	Periarterial, inguinal, retroperitoneal, scrotal/gubernacular collaterals	Microsurgery + testis delivery. Diagnose by upright colour Doppler (reflux); venography + embolisation to treat

> **KEY POINT** Recurrence = a vein missed; hydrocele = a lymphatic tied; atrophy = an artery tied. Microsurgery with testicular delivery addresses all three.

11 · LANDMARK TRIALS

Trial	Design	Result	Take-home
Madgar (1995)	RCT: varicocelectomy vs no surgery in infertile men	1-year pregnancy 44% vs 10%	Repair of clinical varicocele with abnormal semen improves fertility

12 · MCQ ONE-LINERS

1. Vas deferens receives no supply from cremasteric or cord vessels.
2. Never two simultaneous vasovasostomies on one vas — intervening segment necroses.

3. Epididymis off testis is safe if the superior epididymal artery is preserved; gains 4–6 cm.
4. Single puncture safe only at proximal caput; one stitch in corpus/cauda obstructs.
5. Diagnostic biopsy: azoospermia, normal testes, palpable vasa, normal FSH.
6. Biopsy fixative: Bouin's, Zenker's or glutaraldehyde — never formalin.
7. Most common biopsy complication: haematoma; most serious: epididymal biopsy.
8. Methylene blue kills sperm; use indigo carmine for patency.
9. Vasal granuloma at testicular end predicts better reversal outcome.
0. Toothpaste fluid without sperm → vasoepididymostomy.
1. Clear watery fluid without sperm → still vasovasostomy.
2. Grape-like clusters of sperm heads → epididymal obstruction.
13. Microdot method: six mucosal 10-0 sutures.
14. Anastomotic granuloma after vasovasostomy heralds obstruction.
15. LIVE: two longitudinal sutures placed before tubule opened.
16. No varicocelectomy simultaneous with vasoepididymostomy.
17. TURED: resect verumontanum; indigo carmine efflux confirms adequacy.
18. TURED complications: urine reflux, epididymitis, retrograde ejaculation.
19. MESA collection: capillary action, never negative pressure.
20. Best technique in NOA: microdissection TESE.
21. AZFa/AZFb complete deletion: zero retrieval — do not operate.
22. Sperm-containing tubules in NOA: larger, opaque; Sertoli cell–only thin and white.
23. Post-varicocelectomy hydrocele results from lymphatic ligation.

71 Physiology of Penile Erection and Pathophysiology of Erectile Dysfunction

Reading time ≈ 47 min (slow pace)

1 · DEFINITION AND KEY CONCEPTS

- **Erectile dysfunction (ED)**: consistent or recurrent inability to attain and/or maintain an erection sufficient for sexual satisfaction (American Urological Association, AUA, 2018). "ED" replaces the pejorative "impotence".
- Erection = **sinusoidal (trabecular) smooth muscle relaxation + arterial dilation + venous compression (veno-occlusion)**. It is a neurovascular event; rigidity is produced by *trapping* blood, not by inflow alone.
- Historical milestones MCQ: oldest description of ED – Egyptian papyrus c. 2000 BCE; Leonardo da Vinci – erection is due to blood (hanged men); 1980s Doppler (Lue) – increased arterial inflow + decreased venous outflow; 1990s – nitric oxide (NO) as key mediator; 1998 – oral phosphodiesterase type 5 inhibitors (PDE5Is).

2 · FUNCTIONAL ANATOMY

Corpora

- Three cylinders: paired dorsal **corpora cavernosa** and ventral **corpus spongiosum** (around urethra, expands as glans).
- Corpora cavernosa: crura arise separately from the puboischial rami, merge under the pubic arch; in humans the septum is **incomplete** (intercorporeal communication – why one injection fills both corpora).
- Each corpus = interconnected sinusoids (larger centrally) separated by smooth muscle trabeculae with elastic fibres and collagen. Helicine arteries and terminal cavernous nerves are intimately related to the trabecular muscle.
- Flaccid: sinusoidal blood is venous (partial pressure of oxygen, Po_2 **~35 mm Hg**); erect: arterial (**~90 mm Hg**).

Tunica albuginea

- Bilayered; collagen (mostly type I, some III) interwoven with an elastic lattice that permits expansion and recoil.
- **Inner circular layer** – contains the cavernous tissue; gives rise to **intracavernous pillars** (struts), absent between 5 and 7 o'clock. **Outer longitudinal layer** – glans to crura, thicker dorsally, absent in the ventral groove holding the spongiosum.
- Ventral tunica ~0.8 mm vs dorsal ~2.2 mm → ventral (5–7 o'clock) is the zone of weakness: relevant to **penile fracture** and cylinder crossover/perforation in implant surgery.
- **Veno-occlusive mechanism**: emissary veins run obliquely between the inner and outer layers; expanding sinusoids stretch the tunica and compress them, blocking outflow. Cavernous arteries escape compression because they lie in a periarterial soft-tissue sheath.
- Spongiosal tunica: thin, single circular layer, no struts; absent in glans → spongiosum and glans are **low-pressure arteriovenous shunts**; they never become rigid.
- Support: fundiform ligament (from Colles fascia, superficial) and suspensory ligament (from Buck fascia, attaches tunica to pubis). Loose suspensory ligament or large penis → downward-pointing erection.

Arteries

- Internal iliac → **internal pudendal** → common penile artery → three branches:
  - **Cavernous** – main corporal supply; gives **helicine arteries** (tortuous and contracted when flaccid, straight and dilated when erect) opening directly into sinusoids without capillaries.

- **Dorsal** – glans engorgement; perforators often join cavernous artery distally.
- **Bulbourethral** – bulb and spongiosum.

- **Accessory pudendal arteries** (from external iliac, obturator, vesical, femoral) in about one-third; may be dominant or sole supply. Preserving them at radical prostatectomy → faster recovery of erection.

Veins

Sinusoids → venules → **subtunical venous plexus** → **emissary veins** through the tunica.

System	Drains	Destination
Deep dorsal vein (± circumflex, periurethral veins)	Glans + distal two-thirds of corpora	Periprostatic (Santorini) plexus
Cavernous and crural veins	Proximal (infrapubic) penis	Internal pudendal veins
Superficial dorsal vein	Skin, subcutaneous tissue	Saphenous veins

> **KEY POINT** Arterial and venous anatomy vary widely between men — which is why the same "nerve-sparing" or pelvic operation has different erectile outcomes in different patients.

3 · HAEMODYNAMICS: PHASES OF ERECTION AND DETUMESCENCE

Sequence: sexual stimulation → neurotransmitter release from cavernous nerves → smooth muscle relaxation → arteriolar dilation → sinusoids fill and compress subtunical plexus → tunica stretches and occludes emissary veins → intracavernous pressure (ICP) ~100 mm Hg → ischiocavernosus contraction makes it suprasystolic.

Phase	Arterial inflow	Intracavernous pressure
0 Flaccid	Low	Low
1 Latent (filling)	**Peaks sharply**	Barely changes
2 Tumescence	Falls as pressure rises	Rises
3 Full erection	Low (above flaccid)	Plateau **~100 mm Hg**
4 Rigid erection	Near zero / reversed	Suprasystolic (several hundred mm Hg) – pudendal nerve → ischiocavernosus
5 Initial detumescence	—	Transient rise (muscle contracts against closed veins)
6 Slow detumescence	—	Slow fall (veins gradually reopen)
7 Fast detumescence	—	Rapid fall (outflow restored)

- Cavernous (autonomic) nerve drives phases 1–3; pudendal (somatic) nerve drives phase 4.
- **Trap**: arterial inflow is highest in the latent phase, not at full rigidity — the basis of Doppler timing (peak systolic velocity measured early).
- Spongiosum/glans: arterial flow rises but no veno-occlusion → pressure one-third to one-half of corporal pressure; extra engorgement in the rigid phase from bulbocavernosus/ischiocavernosus compression and from dorsal vein compression under Buck fascia.

4 · NEUROANATOMY

Peripheral and spinal

	Sympathetic	Parasympathetic	Somatic
Origin	**T11–L2** (often T10–T12)	**S2–S4** intermediolateral columns	S2–S4; motor from **Onuf nucleus**

	Sympathetic	Parasympathetic	Somatic
Route	Sympathetic chain → superior hypogastric plexus → hypogastric nerves → pelvic plexus	Pelvic nerves (nervi erigentes) → pelvic plexus	Pudendal nerve; afferent via dorsal nerve of penis
Role	**Detumescence**/flaccidity; emission	**Erection** (via cavernous nerves)	Sensation; ischiocavernosus (rigidity), bulbocavernosus (ejaculation)

- **Cavernous nerves** = branches of the pelvic plexus carrying both autonomic fibres; †run posterolateral to the prostate; lateral branch pierces the urogenital diaphragm 4–7 mm lateral to the sphincter. Pelvic ganglion cells are scattered along the bladder–prostate junction and seminal vesicles → vulnerable at radical pelvic surgery.
- Dorsal nerve: mainly somatic sensory but contains neuronal nitric oxide synthase (nNOS)-positive fibres; afferent path: penile receptors → dorsal nerve → pudendal → S2–S4 → spinothalamic tracts → thalamus/cortex.
- **Bulbocavernosus reflex**: glans squeeze → bulbocavernosus contraction; basis of genital neurological exam, value in ED work-up limited.
- Spinal reflex hierarchy: abrupt noxious stimulus → bulbocavernosus reflex; low-intensity continuous → erection (sacral parasympathetic); high-intensity continuous + sympathetic recruitment → ejaculation.

Three types of erection

Type	Trigger	Mechanism	Spinal cord injury (SCI)
Psychogenic	Audiovisual stimuli, fantasy	Brain modulates spinal centres, chiefly by suppressing sympathetic tone	May persist with sacral lesions; absent with lesions above T9
Reflexogenic	Genital touch	Sacral reflex arc → cavernous nerves	Preserved in upper cord lesions
Nocturnal	Rapid eye movement (REM) sleep	Cholinergic pontine tegmentum active; locus coeruleus (noradrenergic) and raphe (serotonergic) silent	—

Supraspinal (central)

- **Medial preoptic area (MPOA)** – partner recognition, integrates hormonal and sensory cues; dopaminergic.
- **Paraventricular nucleus (PVN)** – pro-erectile **oxytocin** neurons projecting to lumbosacral autonomic and somatic nuclei.
- Nucleus paragigantocellularis (brainstem) – **serotonergic brake** on lumbosacral neurons.
- Others: medial amygdala (motivation), periaqueductal grey (relay), locus coeruleus/A5 (noradrenergic), lateral preoptic area (nocturnal erections in rats); pyriform cortex lesions → hypersexuality.
- Arousal involves deactivation of temporal/parietal (self-reflective, inhibitory) cortex; psychogenic ED shows persistent cingulate/frontal activation (normalised by apomorphine).

Central neurotransmitters

Promote erection	Inhibit erection
Dopamine (D2 in PVN → oxytocin → NO; apomorphine = D1/D2 agonist); oxytocin; central NO; central noradrenaline; melanocortins (melanocortin-4 receptor); serotonin (5-hydroxytryptamine, 5-HT) 2C receptor	Serotonin (5-HT_{1A}, 5-HT_2 – but facilitate ejaculation); γ-aminobutyric acid (GABA); opioids (high dose); cannabinoids (CB1); prolactin

KEY POINT nitric oxide synthase (NOS) inhibitors block both apomorphine- and oxytocin-induced erections → NO is the final common central pathway. **Trap**: noradrenaline is pro-sexual centrally but the principal anti-erectile transmitter peripherally.

5 · MOLECULAR MECHANISMS OF SMOOTH MUSCLE TONE

Contraction (flaccid state)

- Corporal smooth muscle is **contracted at rest**. Flaccidity = intrinsic myogenic tone + adrenergic tone + endothelial constrictors.
- **Calcium pathway**: agonist (noradrenaline → α1; endothelin-1 → ET_A; angiotensin II → AT1; prostaglandin F2α, thromboxane A2) → Gq → phospholipase C → inositol trisphosphate (IP_3) → Ca^{2+} release + entry via voltage-dependent channels → Ca^{2+}–calmodulin activates **myosin light chain (MLC) kinase** → phosphorylates MLC20 → cross-bridge cycling.
- **Calcium sensitisation – RhoA/Rho-kinase**: agonist → RhoA → Rho-kinase phosphorylates and inhibits **myosin light chain phosphatase (MLCP)** → MLC stays phosphorylated → sustained (tonic) contraction *without* further Ca^{2+} rise. Latch state (calponin) maintains basal tone cheaply.
- Rho-kinase is upregulated in diabetes, hypertension, hypercholesterolaemia, hypogonadism, ageing and cavernous nerve injury, and drives fibrosis (fibroblast → myofibroblast).

Relaxation (erection)

- **NO–cGMP pathway** (principal): nitric oxide synthase (NOS) converts L-arginine + O_2 → NO + citrulline. **Neuronal NOS (nNOS) initiates** erection; **endothelial NOS (eNOS) sustains** it. NO → soluble guanylyl cyclase (sGC, haem-dependent) → cyclic guanosine monophosphate (cGMP) → **protein kinase G (PKG-I)**.
- PKG effects: opens K^+ channels (hyperpolarisation), closes Ca^{2+} channels, sequesters Ca^{2+}, inhibits IP_3 and Rho-kinase, activates MLCP → ↓ cytosolic Ca^{2+} → relaxation.
- **cAMP pathway**: prostaglandin E1 (PGE1; EP2/EP4 receptors), vasoactive intestinal peptide (VIP), calcitonin gene–related peptide (CGRP) → Gs → adenylyl cyclase → cyclic adenosine monophosphate (cAMP) → protein kinase A (PKA). **Independent of NO** — why intracavernosal alprostadil works when nerves/endothelium have failed.
- **Phosphodiesterases (PDEs)** terminate signalling: **PDE5 (cGMP-specific) is the principal cavernous enzyme**; PDE6 is retinal (visual side effects); cAMP is degraded by PDE2, 3, 4. Papaverine is a nonspecific PDE inhibitor. Raised cGMP also inhibits PDE3 → ↑ cAMP.
- K^+ channels (Ca^{2+}-activated maxi-K, K_{ATP}) → hyperpolarisation. **Gap junctions (connexin-43)** synchronise relaxation across the corpora — why a localised nerve input or injection gives a uniform erection.
- Other gaseous mediators: carbon monoxide (haem oxygenase), hydrogen sulphide (H_2S, cystathionine γ-lyase; acts through cGMP).
- Oxygen is a switch: low flaccid Po_2 suppresses NOS and favours endothelin; erect Po_2 >50–60 mm Hg permits full NOS activity.
- **Adenosine** (A2 receptor → cAMP): excess adenosine with ↑ A2B signalling causes prolonged erection in adenosine deaminase-deficient and sickle cell transgenic mice — the accepted **priapism models**.
- **Sonic hedgehog (SHH)** maintains corporal architecture (stimulates vascular endothelial growth factor, NOS and brain-derived neurotrophic factor); inhibition → rapid corporal atrophy; ↓ after cavernous nerve injury and in diabetes mellitus (DM).
- Androgens, O_2 and repeated intracavernous PGE1 ↑ NOS activity; castration, denervation, hypercholesterolaemia and DM ↓ it.
- Acetylcholine acts indirectly (endothelial NO release, presynaptic adrenergic inhibition); **atropine does not abolish** neurally induced erection MCQ.

- PKG-I knockout mice fail to erect, whereas nNOS or eNOS knockouts often erect (redundancy) → PKG is the bottleneck.

Second messenger	Activators	Kinase	Degraded by	Drug exploiting it
cGMP	NO, CO, H_2S, natriuretic peptides	PKG-I	PDE5	PDE5Is
cAMP	PGE1, VIP, CGRP, adenosine (A2)	PKA	PDE2, 3, 4	Alprostadil (intracavernosal/intraurethral); papaverine (both)

Detumescence

Cessation of NO release + PDE5 hydrolysis of cGMP + sympathetic discharge (noradrenaline – principal transmitter of flaccidity; α1 postjunctional contraction) + endothelin-1 + Rho-kinase → trabecular contraction → venous channels reopen (phases 5–7).

> **KEY POINT** Corporal tone is a tug-of-war over one phosphorylation site (MLC20): Ca^{2+}–calmodulin–MLC kinase and Rho-kinase keep it phosphorylated; NO–cGMP–PKG removes the phosphate. **Trap**: "MLCP" = the phosphatase, not phosphorylated MLC.

6 · CLASSIFICATION OF ED

Approach	Categories
Neurovascular mechanism	Failure to **initiate** (neurogenic, psychogenic, endocrine) · failure to **fill** (arterial) · failure to **store** (veno-occlusive)
Aetiological	Psychogenic · neurogenic · hormonal · arteriogenic · cavernosal (venogenic) · drug-induced · systemic disease · mixed (commonest)

Most organic ED has a psychogenic component; ED rarely has a single cause.

7 · RISK FACTORS AND ENDOTHELIAL DYSFUNCTION

- **Age** is the strongest demographic predictor, but comorbidity (diabetes mellitus, DM; heart disease; hypertension) carries more risk. Massachusetts Male Aging Study (MMAS): incidence rises each decade and is higher with DM, treated heart disease and hypertension.
- **Unifying mechanism – endothelial dysfunction**: oxidative stress oxidises tetrahydrobiopterin (BH4) → **NOS uncoupling** → superoxide + NO → peroxynitrite; arginase depletes L-arginine; asymmetric dimethylarginine (ADMA) inhibits NOS → ↓ NO bioavailability.
- **Artery-size hypothesis**: penile arteries are smaller than coronary arteries, so the same endothelial disease becomes symptomatic in the penis first; ED preceded coronary artery disease (CAD) by a mean of **~2 years**. ED is a **sentinel** for cardiovascular disease and warrants cardiac risk stratification → *Ch 72*.
- Common downstream damage in all vascular/metabolic disease: ↑ Rho-kinase, ↑ transforming growth factor-β (TGF-β) → collagen deposition and ↓ smooth muscle-to-collagen ratio → veno-occlusive failure.

8 · PATHOPHYSIOLOGY BY TYPE

Psychogenic

- Common, possibly ubiquitous as contributor. Mechanisms: (1) direct supraspinal inhibition of spinal erection centres; (2) excess sympathetic outflow/raised catecholamines ↑ smooth muscle tone (men with psychogenic ED have higher serum noradrenaline).
- Contributors: performance anxiety, partner conflict, depression, post-traumatic stress disorder, problematic pornography use.

Neurogenic

- **Brain**: Parkinson disease, stroke, tumours, dementia, temporal lobe epilepsy, multiple system atrophy, traumatic brain injury (dopaminergic/oxytocinergic insufficiency).
- **Spinal cord**: SCI (commonest), spina bifida, disc disease, multiple sclerosis.

Complete lesion	Reflexogenic erection	Psychogenic erection
Upper cord (above sacral centre)	Preserved in **~95%**; brief, need continuous stimulation	Lost
Lower cord (sacral)	Preserved in only **~25%**	May occur (via thoracolumbar pathway); marginal tumescence

- **Peripheral / iatrogenic**: radical prostatectomy, cystoprostatectomy, abdominoperineal resection, pelvic radiation; most men have at least transient decline despite nerve sparing; worse with baseline ED and age >60. Retroperitoneal sympathetic chain injury → loss of emission.
- Cavernous nerve injury → ↓ nNOS, apoptosis, fibrosis, ↓ sonic hedgehog (SHH) → structural (venous leak) ED: the rationale for early penile rehabilitation.
- **Pelvic fracture**: cavernous nerve and/or pudendal artery injury; with posterior urethral injury, early realignment has better potency preservation than delayed anastomosis.
- DM: neuropathy + vasculopathy, both impair NO release. Ageing: ↓ penile sensitivity reduces reflex input.

Hormonal

- **Testosterone (T)** governs **libido** more than mechanics, but maintains NOS, PDE5 and smooth muscle; castration → ↓ arterial inflow, venous leak, apoptosis. Low libido is the commonest sexual symptom of low T.
- AUA: measure **morning, fasting** T in all men with ED (food acutely suppresses T); biochemical low T = **<300 ng/dL**; treat only with ≥2 low levels + attributable symptoms. Nocturnal erections need ~200 ng/dL.
- Nocturnal (REM) erections are androgen-dependent; erections to visual stimuli are largely androgen-independent MCQ. Dihydrotestosterone (DHT) is the key penile androgen.
- Low T is linked to obesity, metabolic syndrome (MetS), insulin resistance — may be a sentinel for diabetes.
- **Hyperprolactinaemia** (prolactinoma, antipsychotics): suppresses gonadotrophin-releasing hormone → low T, ↓ libido and ↓ MPOA dopamine. Low prolactin (<3 ng/mL) is also associated with ED.
- Oestradiol excess/low T:E2 ratio (obesity) ↑ ED; hyperthyroidism (↑ oestrogen, ↑ sympathetic tone) and hypothyroidism (↑ prolactin, ↓ gonadal axis) both cause ED.

Vasculogenic – arteriogenic

- Atherosclerotic or traumatic occlusion of the hypogastric–pudendal–cavernous tree → ↓ perfusion pressure → longer time to maximal erection, ↓ rigidity.
- Risk factors: hypertension, hyperlipidaemia, smoking, DM, obesity, sedentary life, pelvic irradiation, blunt perineal/pelvic trauma (focal pudendal/cavernous stenosis in young men → candidate for revascularisation).
- Chronic hypoxia → ↑ TGF-β1, ↑ collagen I/III, ↓ smooth muscle → secondary veno-occlusive dysfunction. Hence "inflow failure becomes storage failure".
- Hypertension: renin–angiotensin system (RAS) drives structural change; angiotensin receptor blockers (ARBs) and nebivolol protect penile tissue whereas hydralazine/amlodipine lower blood pressure without protection.

Vasculogenic – veno-occlusive (venous leak)

One of the commonest causes of vasculogenic ED. Causes:

- **Tunical**: degenerative (ageing, DM), Peyronie disease, penile fracture, Peyronie surgery disrupting the subtunical layer.

- **Fibroelastic trabecular change**: loss of smooth muscle compliance (ageing, DM, nerve injury, hypercholesterolaemia) — sinusoids cannot expand enough to compress subtunical veins.
- **Insufficient trabecular relaxation**: anxiety, high adrenergic tone, endothelial dysfunction.
- **Acquired shunts**: after priapism shunt surgery (glans–cavernosum, cavernosum–spongiosum); post-priapism fibrosis.
- Congenital: ectopic/aberrant cavernous or crural veins (primary ED).

> **KEY POINT** Veno-occlusion is an emergent property of healthy tissue (elastic tunica + fully relaxed trabeculae), so most "venous leak" is really a cavernosal smooth muscle disease — the reason venous ligation surgery fails and is investigational.

Structure	Change in vasculogenic ED
Cavernous artery	↑ resistance, narrow lumen
Smooth muscle	↑ tone, ↓ content, altered K^+ channels and gap junctions
Cavernous tissue	Fibrosis, elastin loss → impaired veno-occlusion
Endothelium	Impaired endothelium-dependent relaxation
Tunica	Altered elastic/collagen fibres
Nerves	↓ nNOS, ↓ eNOS

Metabolic and systemic disease

- **Diabetes mellitus**: earliest and most severe ED; mechanisms – arterial disease, autonomic neuropathy (selective nitrergic nerve loss), endothelial dysfunction, advanced glycation end products (AGEs), ↑ Rho-kinase/arginase, ↓ smooth muscle, hypogonadism. **Key point**: NO *source* is damaged but response to exogenous NO is preserved → PDE5Is work until fibrosis supervenes.
- **MetS/obesity**: visceral fat is an endocrine organ (cytokines, aromatisation); weight loss, Mediterranean diet, exercise, bariatric surgery improve function.
- **Hyperlipidaemia**: oxidised low-density lipoprotein inhibits NO; statins associated with improved International Index of Erectile Function (IIEF) scores.
- **Chronic kidney disease**: up to **80%** of end-stage renal disease men have ED (hypogonadism, hyperprolactinaemia, atheroma, neuropathy); persists in ~half after transplant.
- Others: HIV, cirrhosis, chronic obstructive pulmonary disease, obstructive sleep apnoea, benign prostatic hyperplasia/lower urinary tract symptoms, hyperuricaemia, hyperhomocysteinaemia, low vitamin D, COVID-19.
- **Ageing**: ↑ latency, ↓ rigidity, ↓ ejaculatory force, longer refractory period, ↓ nocturnal erections.

Drug-induced

Class	Cause ED	Mechanism / note	Alternative
Antihypertensives	**Thiazides**, **nonselective β-blockers** (propranolol), central α2-agonist (clonidine)	Thiazide – unknown; β-blocker – prejunctional α2 effect	Angiotensin-converting enzyme inhibitors (ACEIs), ARBs, nebivolol, α1-blockers (but ejaculatory dysfunction), calcium channel blockers
Antidepressants	Selective serotonin reuptake inhibitors (SSRIs; paroxetine worst), tricyclics	5-HT_2/5-HT_3 spinal inhibition, ↓ dopamine, ↓ NO; post-selective serotonin reuptake inhibitor (SSRI) sexual dysfunction may persist	Bupropion, mirtazapine, nefazodone, buspirone; add phosphodiesterase type 5 inhibitor (PDE5I)
Antipsychotics	Typical > atypical	Dopamine blockade → hyperprolactinaemia; α-block, anticholinergic	Newer agents

Class	Cause ED	Mechanism / note	Alternative
Antiandrogens	Luteinising hormone–releasing hormone (LHRH) agonists/antagonists, androgen receptor (AR) antagonists, cyproterone, **5α-reductase inhibitors (5ARIs)**	5ARI: least effect on serum T yet linked to persistent ED after cessation	None
Recreational	Tobacco (dose–response, ≥20 pack-years), heavy alcohol, opioids (hypogonadism)	Endothelial damage, neuropathy	Cessation/moderation
Others	Cimetidine, spironolactone (antiandrogenic), carvedilol	—	—

Trazodone (metabolite m-chlorophenylpiperazine) → priapism. Anticonvulsants: valproate (↓ desire), carbamazepine/phenytoin worse than lamotrigine.

Primary (lifelong) ED

- Usually psychogenic (performance anxiety, trauma, misinformation), but exclude congenital organic causes: hypoplastic cavernous arteries, aberrant venous drainage (congenital venous leak), congenital corporal absence/atresia, megalourethra.
- Micropenis (stretched length >2.5 standard deviations below mean) and curvature (chordee, Peyronie disease → buckling despite normal rigidity) may present as "ED".

> **KEY POINT** Why ED is a sentinel: nearly every systemic disease reaches the penis through the same three doors — endothelial NO failure, Rho-kinase-driven contraction, and TGF-β-driven fibrosis.

9 · MCQ ONE-LINERS

1. Erection = trabecular relaxation + arterial dilation + veno-occlusion.
2. Arterial inflow peaks in the latent (filling) phase, not at rigidity.
3. Full-erection intracavernous pressure ≈ 100 mm Hg; rigid phase suprasystolic via ischiocavernosus.
4. Emissary veins are compressed between inner circular and outer longitudinal tunica layers.
5. Ventral tunica thinnest (~0.8 mm); intracavernous pillars absent at 5–7 o'clock.
6. Glans and spongiosum lack veno-occlusion; pressure one-third to one-half of corporal.
7. Helicine arteries open directly into sinusoids without capillaries.
8. Accessory pudendal arteries in ~one-third; preserve at prostatectomy.
9. Deep dorsal vein drains into Santorini plexus.
10. Parasympathetic S2–S4 → erection; sympathetic T11–L2 → detumescence.
11. Onuf nucleus (S2–S4) innervates ischiocavernosus and bulbocavernosus via pudendal nerve.
12. Psychogenic erection absent with cord lesions above T9.
13. Complete upper cord lesion: reflexogenic erection preserved in ~95%.
14. PVN oxytocinergic neurons are the central pro-erectile output.
15. Apomorphine is a central D1/D2 agonist acting at PVN.
16. nNOS initiates, eNOS maintains erection.
17. NO → soluble guanylyl cyclase → cGMP → PKG-I → ↓ cytosolic Ca^{2+}.
18. PDE5 is the principal cGMP-hydrolysing enzyme in corpora; PDE6 is retinal.
19. Alprostadil acts via cAMP, independent of NO.
20. Rho-kinase inhibits myosin light chain phosphatase → sustained contraction.
21. Noradrenaline (α1) is the principal peripheral transmitter of flaccidity.
22. Atropine does not abolish neurally induced erection.
23. Gap junction protein connexin-43 synchronises corporal smooth muscle.
24. NOS needs cavernous Po_2 above ~50–60 mm Hg.
25. Prostacyclin (PGI2) contracts, PGE1 relaxes corporal tissue.
26. ED precedes coronary artery disease by ~2 years (artery-size hypothesis).
27. Nocturnal erections are androgen-dependent; visually evoked erections largely androgen-independent.
28. AUA low testosterone: <300 ng/dL on morning sample.
29. Commonest sexual symptom of low testosterone: reduced libido.

30. Thiazides and nonselective β-blockers carry highest antihypertensive ED risk.
31. Nebivolol and ARBs spare or improve erectile function.
32. 5α-reductase inhibitors: persistent ED despite minimal serum testosterone change.
33. Diabetes damages NO source but preserves response to exogenous NO.
34. Up to 80% of end-stage renal disease men have ED.
35. Venous leak after priapism shunt = persistent glans–cavernosum or cavernosum–spongiosum communication.

72 Evaluation and Management of Erectile Dysfunction

Reading time ≈ 44 min (slow pace)

1 · DEFINITION AND SIGNIFICANCE

- **Erectile dysfunction (ED)** (American Urological Association, AUA): consistent or recurrent inability to attain and/or maintain an erection sufficient for sexual satisfaction; minimum duration **3 months** (except trauma/surgery-induced ED).
- Prevalence ~**20%** of adult men, rising with age (40–49 y up to 15%; 60–69 y up to 40%). One in four men presenting with new ED is <40 y, half of them severe.
- Comorbid associations: type 2 diabetes mellitus (DM), obesity, cardiovascular disease (CVD), hypertension, dyslipidaemia, depression, benign prostatic hyperplasia; opioid use doubles risk (screen for it); prior COVID-19 infection ~sixfold risk.
- **ED predicts CVD**: shared endothelial dysfunction; ED is a sentinel of cardiac events comparable to smoking or family history of myocardial infarction (MI), and is associated with CVD and all-cause mortality. Younger age at onset → greater cardiovascular (CV) risk.

Strength of independent association (multivariate)	Risk factors
Strongest	Antidepressant use, then antihypertensive use
Strong	DM (strongest disease); obstructive urinary symptoms
Moderate	Hypertension, prostate enlargement, smoking, body mass index (BMI) ≥30, inactivity
Weak/null after adjustment	CVD, hypercholesterolaemia

- **Drug-induced ED** ≈ up to 25% of presentations: thiazides, nonselective β-blockers, selective serotonin reuptake inhibitors (SSRIs), tricyclics, phenothiazines, antiandrogens, cimetidine, cytotoxics, opiates → *Ch 71*.

2 · MANAGEMENT PRINCIPLES

- Diagnosis rests on the patient's report; extensive testing is usually unnecessary; first-line therapy is simple — but evaluation must still be structured and aimed at the **disease behind the ED**.
- **Shared decision making** (AUA 2018 – state of the art): all non-contraindicated options (phosphodiesterase type 5 inhibitors (PDE5Is), vacuum erection device (VED), intraurethral (IU) alprostadil, intracavernosal injection (ICI), penile prosthesis) are presented **in parallel, not ranked**; patient may choose any first, or defer. Counsel partner, lifestyle change and psychosocial support.
- The older **step-care / Process of Care Model (1999)** ranked therapy first → second → third line by ease, reversibility, invasiveness and cost (oral → local/VED → prosthesis); still used informally.
- Testosterone (T) deficiency (symptoms + total T <300 ng/dL): restoring T likely improves efficacy of other ED treatments.
- **Partner interview** alters diagnosis/treatment in a large proportion (over half in some series); partner attitude drives adherence.
- **Follow-up**: discontinuation is high without reassessment; review efficacy, dose titration, adverse effects (AEs), interactions, comorbidity.
- Direct-to-consumer platforms: questionnaire-only screening, unproven therapies, frequently non-physicians → reinforce in-person/telemedicine urological evaluation.

Cardiac risk assessment – Princeton Consensus

Every man with ED, even if asymptomatic, is considered at potential CV risk. Sexual activity ≈ moderate exertion.

Risk	Features	Action
High	Unstable/refractory angina, recent MI, certain arrhythmias, uncontrolled hypertension	**Defer sexual activity and ED treatment** until stabilised; cardiology referral, stress testing
Indeterminate	Not clearly high or low	CV assessment (e.g. †exercise stress test) and restratify
Low	Controlled risk factors	Initiate/resume sexual activity and ED therapy

All men: lifestyle counselling, risk-factor control; Framingham or global risk score is a useful first step.

Specialist referral

Failure of initial therapy; young man with pelvic/perineal trauma (vascular work-up); significant penile deformity (Peyronie disease, chordee); medicolegal; complicated endocrinopathy (pituitary adenoma) → endocrinologist; refractory psychiatric/psychosexual disorder → psychiatrist; vascular/neurosurgical lesion.

3 · BASIC EVALUATION

Sexual, medical and psychosocial history

- Private, nonjudgemental, culturally competent; do not assume a heterosexual monogamous relationship.
- Define onset, duration, situational vs global, morning/nocturnal and masturbatory erections, libido, ejaculation, orgasm, penile curvature/pain, partner factors, prior treatment. Determine whether ED is primary or secondary to another phase (desire, ejaculation) — e.g. premature ejaculation causing secondary ED.
- Medical: DM, CVD, neurological disease, pelvic surgery/radiation, trauma, drugs, recreational substances, depression (bidirectional). Whooley two-question depression screen (low mood; anhedonia in past month).

Psychogenic	Organic
Sudden onset	Gradual onset
Complete, immediate loss	Incremental progression
Situational	Global
Waking/nocturnal erections present	Poor or absent

> **KEY POINT** Preserved nocturnal/morning erections prove an intact neurovascular axis — the single strongest historical pointer to psychogenic ED.

Questionnaires

- **International Index of Erectile Function (IIEF)**: 15 items, 5 domains (erectile function, orgasm, desire, intercourse satisfaction, overall satisfaction).
- **IIEF-5 / Sexual Health Inventory for Men (SHIM)**: 5 items, each scored 1–5 (total 5–25), over the past 6 months †.

SHIM score	5–7	8–11	12–16	17–21	22–25
Severity	Severe	Moderate	Mild–moderate	Mild	No ED

Questionnaires document presence, severity and treatment response but do **not** identify cause. Others: Erectile Dysfunction Inventory of Treatment Satisfaction (EDITS), Male Sexual Health Questionnaire.

Physical examination

Finding	Implies
Small/soft testes, gynaecomastia, poor virilisation, Klinefelter/Kallmann habitus	Hypogonadism
Obesity, raised blood pressure (BP), weak femoral/pedal pulses	Vascular
Abnormal genital/perineal sensation, absent bulbocavernosus reflex (glans squeeze → anal contraction)	Neurogenic (neuropathy, DM)
Peyronie plaque, micropenis, chordee	Structural

Include waist circumference, BP, and digital rectal examination where prostate disease is relevant.

Laboratory tests

- **Recommended**: fasting glucose or glycated haemoglobin (HbA1c), lipid profile, serum chemistry, full blood count, and **morning (7–11 am) total testosterone**.
- Low T → repeat (≈one-third of mildly low values normalise) with luteinising hormone (LH) and prolactin; free/calculated bioavailable T if sex hormone–binding globulin (SHBG) altered or borderline T; thyroid tests at discretion; prostate-specific antigen (PSA) if prostate disease suspected or before T therapy.

> **KEY POINT** The basic work-up is designed to find the disease behind the ED — undiagnosed DM, dyslipidaemia, hypogonadism, CV risk — not to prove ED.

4 · HORMONAL EVALUATION

- T fractions: free 0.5–3%, SHBG-bound ~30%, albumin-bound ~67%; **bioavailable T = free + albumin-bound**. Commercial free-T assays are unreliable; use calculated bioavailable T.
- SHBG ↓ (bioavailable T ↑): obesity, nephrotic syndrome, hypothyroidism, glucocorticoids. SHBG ↑ (bioavailable T ↓): ageing, cirrhosis, hyperthyroidism, HIV, anticonvulsants, oestrogens. In DM use bioavailable T.
- AUA threshold **300 ng/dL (10.4 nmol/L)**; sexual function correlates weakly with T above ~230 ng/dL.
- Primary hypogonadism: low T, **high** LH/FSH (follicle-stimulating hormone) → testicular cause. Secondary: low T, low/inappropriately normal LH → prolactin; pituitary MRI if T **<150 ng/dL**, persistent hyperprolactinaemia or mass effect.
- Prolactin: hyperprolactinaemia suppresses gonadotrophin-releasing hormone (GnRH) pulsatility → low T and ↓ libido. Causes: antipsychotics, tricyclics, opiates, prolactinoma, hypothyroidism, renal failure, cirrhosis.
- Oestradiol if gynaecomastia/breast symptoms. Hyperthyroidism: high T4/T3, low thyroid-stimulating hormone (TSH).

5 · SPECIALISED TESTING

Indications: complicated/atypical cases — young patient with trauma, primary (lifelong) ED, suspected surgical candidate (revascularisation, venous surgery), medicolegal, failure of therapy, Peyronie disease planning.

Vascular testing ladder

Test	Line	Answers	Indication
Combined intracavernosal injection and stimulation (CIS)	1st	Can vessels produce a rigid erection?	Screening vascular integrity
Penile duplex Doppler ultrasonography	2nd	Arterial vs veno-occlusive; asymmetry	Most reliable, least invasive
Dynamic infusion cavernosometry and cavernosography (DICC)	3rd	Is there a venous leak, and where?	Trauma, lifelong ED, before venous surgery

Test	Line	Answers	Indication
Selective internal pudendal arteriography	3rd	Site of arterial lesion; inferior epigastric donor suitability	Young trauma patient before revascularisation

CIS test

- Alprostadil 10–20 µg, or Bimix/Trimix 0.3 mL, with a 27–29 G needle into the lateral base of the corpus cavernosum; compress 5 min; stimulation; assessor rates rigidity.
- Sustained rigid erection = normal haemodynamics → psychogenic/neurogenic/endocrine cause. False negatives: anxiety, needle phobia, low dose (redose). False normal in up to a fifth with borderline arterial inflow.
- Must be flaccid before discharge; if no detumescence by ~1 h → intracavernosal phenylephrine (diluted 500 µg/mL) every 3–5 min.

Penile duplex Doppler

- 7.5–12 MHz linear probe; measure cavernous artery peak systolic velocity (PSV) and end-diastolic velocity (EDV) at the penoscrotal junction/base, every 5 min up to 20 min after ICI; redose if poor erection (anxiety).
- Waveform evolves: flaccid high resistance → filling (high systolic + diastolic flow) → tumescence (diastolic falls) → rigid (diastolic zero or reversed).

Parameter	Arterial insufficiency	Veno-occlusive dysfunction
PSV	**<25 cm/s** (25–35 borderline; >35 normal)	Persistently >25 cm/s
EDV	—	**>5 cm/s**
Resistive index (RI) = (PSV − EDV)/PSV	—	**<0.75** (normal >0.9)
Other	Acceleration time >122 ms; side-to-side PSV difference >10 cm/s; cavernous artery dilation <75%	Rapid detumescence
When assessed	Within 5 min	15–20 min (needs full smooth muscle relaxation)

> **KEY POINT** Venous leak can be diagnosed only when arterial inflow is adequate and smooth muscle is fully relaxed — an anxious patient with incomplete relaxation will mimic veno-occlusive dysfunction, so redose before labelling a leak. Anatomical variants (dorsal–cavernous communications, early branching) can falsely suggest arteriogenic ED.

DICC and arteriography

- DICC: intracavernosal vasodilator → saline infusion with pressure monitoring → flow-to-maintain, pressure decay, cavernous artery systolic occlusion pressure (CASOP; gradient vs brachial systolic abnormal) → contrast cavernosography localises leaks (glans, spongiosum, dorsal, cavernous and crural veins). Veno-occlusive dysfunction = inability to reach mean systolic pressure or rapid pressure fall after infusion stops.
- Arteriography (after ICI vasodilation): reserved for young men with traumatic arterial lesions being considered for revascularisation; also assesses inferior epigastric donor.
- Obsolete: penile–brachial index, plethysmography (flaccid penis — wrong physiological state).

Nocturnal penile tumescence and rigidity (NPTR)

- Sleep-laboratory NPTR is the standard: tumescence, rigidity, duration plus polysomnography (documents REM and sleep apnoea).
- **RigiScan** (home): two loops (base and coronal sulcus) record tumescence and **radial** rigidity; normal = 4–5 erections/night, ≥ †60% base-tip rigidity lasting ≥10 min. Cannot verify REM; radial ≠ axial rigidity.
- Use: obscure aetiology, medicolegal, suspected psychogenic ED. False positive (normal) in multiple sclerosis; false negative in ageing, depression, anxiety.

- Audiovisual sexual stimulation (AVSS) — cheaper, more physiological alternative for organic vs psychogenic.

Neurological tests

Limited routine role (research/medicolegal). Bulbocavernosus reflex latency (grade B), dorsal nerve conduction velocity (neurogenic ED in DM), biothesiometry (poor), genitocerebral evoked potentials, sympathetic skin response, corpus cavernosum electromyography (questioned). Somatic tests are reproducible; autonomic tests are not. Penile neuropathy in DM can exist without foot neuropathy.

6 · TREATMENT

ALGORITHM
Evaluate and treat reversible causes (lifestyle, drugs, hormones, psychosocial) + CV risk stratification → shared decision: phosphodiesterase type 5 inhibitor (PDE5I) | VED | IU alprostadil | ICI | (low-intensity extracorporeal shockwave therapy (Li-ESWT) investigational) → inadequate response: optimise dose/instructions, combine, switch → penile prosthesis → *Ch 75*

Lifestyle and risk-factor modification

- Smoking cessation, exercise, weight loss, Mediterranean diet improve erectile function (randomised trial: obese men improved IIEF with exercise + weight control). Control DM, lipids, BP; no-nose saddle for cyclists.
- **Medication change**: thiazides/β-blockers → calcium channel blockers, angiotensin-converting enzyme inhibitors (ACEIs), angiotensin receptor blockers (ARBs); SSRI → bupropion, mirtazapine, buspirone, drug holiday, or add PDE5I.
- **Psychosexual therapy**: sensate focus, cognitive-behavioural therapy, couples therapy; best integrated with medical therapy.

Hormonal therapy

- T therapy for confirmed deficiency: ~3-month trial, judged clinically; improves libido more than erection; synergistic with PDE5Is.
- Monitoring: baseline digital rectal examination, PSA, haematocrit, liver function, lipids → 3–6 months → yearly. AEs: erythrocytosis, sleep apnoea, acne, gynaecomastia, lower urinary tract symptoms (LUTS). Meta-analysis of placebo-controlled RCTs: no causal link with CV events.
- Preparations: gels (daily; risk of transfer to partner/children), patches, intramuscular enanthate/cypionate (200–250 mg every 2–3 weeks; supraphysiological peaks, polycythaemia), undecanoate depot (750 mg at 0, 4 weeks, then every 10 weeks), pellets, nasal gel (no transfer), oral undecanoate (raises BP). Avoid 17α-alkylated methyltestosterone (hepatotoxic). Prefer short-acting initially.
- Hyperprolactinaemia: T replacement is not corrective — stop offending drug; dopamine agonist (bromocriptine) for prolactinoma; surgery for failure or visual compromise.

PDE5 inhibitors (first-line oral)

Mechanism: block cGMP (cyclic guanosine monophosphate) breakdown in cavernous smooth muscle → **enhance, not initiate** erection; sexual stimulation (nitric oxide release) is obligatory.

	Sildenafil	Vardenafil	Tadalafil	Avanafil
Doses (mg)	25, 50, 100	5, 10, 20	5, 10, 20 (2.5/5 daily)	50, 100, 200
T_{max} (h)	0.8 (take ~1 h before)	0.7–0.9 (~1 h)	**2** (~2 h)	**0.3–0.5** (~30 min)
Onset (min)	15–60	15–60	15–120	15–60
Half-life (h)	3–5	4–5	**17.5**	3–5
Fatty food	↓ absorption	↓ absorption	**No effect**	↓ absorption

	Sildenafil	Vardenafil	Tadalafil	Avanafil
Characteristic adverse effect (AE)	Visual (blue tinge; PDE6), flushing	Visual, flushing; **QT prolongation – avoid with class 1A/3 antiarrhythmics**	**Back pain/myalgia** (PDE11)	Fewest visual effects
Nitrates	**Contraindicated (all)**			

- Common AEs: headache (commonest), dyspepsia, flushing, nasal congestion. Efficacy equivalent across agents (~70% successful intercourse; 40–50% in DM and post-prostatectomy).
- **Nitrates** in any form (glyceryl trinitrate, isosorbide, amyl nitrite "poppers") → profound hypotension; nitrate use >2 weeks earlier is not a contraindication. If angina occurs: no nitrates for **24 h** (sildenafil, vardenafil) or **48 h** (tadalafil). Hypotension: Trendelenburg → IV fluids → phenylephrine → intra-aortic balloon pump; no antidote.
- Caution: α-blockers (additive hypotension — start low dose, stabilise first), aortic stenosis/left ventricular outflow obstruction, hypovolaemia. CYP3A4 inhibitors (ketoconazole, ritonavir) → reduce dose; inducers (rifampicin) → higher dose.
- Not studied: MI/stroke within 6 months, hypotension <90/50 or hypertension >170/100, retinitis pigmentosa, Child-Pugh C, dialysis.
- Nonarteritic anterior ischaemic optic neuropathy (NAION): no proven increase, but stop and seek care for sudden visual loss.
- **Before labelling PDE5I failure**: confirm adequate stimulation, correct timing/food, maximum dose, **up to 9–10 attempts**, correct hypogonadism, glycaemia and lipids; consider switching agent or daily tadalafil.
- **Penile rehabilitation** after radical prostatectomy (scheduled PDE5I): nightly sildenafil trial positive, but REINVENT (vardenafil) and nightly vs on-demand trials negative — role unproven.

Other oral/central agents

Yohimbine (α2 antagonist) – psychogenic only; apomorphine sublingual (D1/D2 agonist, nausea; not US-approved); trazodone – no better than placebo; phentolamine – minimal ED only.

Vacuum erection device

- Cylinder + pump creates negative pressure → corporal engorgement; constriction ring at base retains blood; **remove ring within 30 min**. Erection is largely venous blood, hinged at base (pivoting), cool and cyanotic.
- Satisfactory erection up to 90%, but satisfaction 30–70%; better accepted by older men in stable relationships. Engorges glans → useful in soft glans syndrome.
- AEs: pain, numbness, trapped ejaculate, petechiae; rarely skin necrosis, Fournier gangrene. Caution with anticoagulants/bleeding disorders. Use prescription devices; no metal rings.
- Also: post-prostatectomy rehabilitation, preserving length after explantation/priapism/Peyronie surgery, salvage of failing prosthesis.

Intraurethral alprostadil (MUSE)

- Prostaglandin E1 (PGE1) pellet 125–1000 µg placed 3 cm into urethra after voiding; massage, stay upright; absorbed via spongiosum into corpora.
- Responders ~50%; niche second line — e.g. PDE5I failure after nerve injury, combination with PDE5I, soft glans after prosthesis. Constriction band (ACTIS) improves response.
- AEs: urethral/penile pain (~one-third), minor bleeding, hypotension, dizziness; priapism rare. Condom if partner pregnant.

Intracavernosal injection

- Agents act directly on smooth muscle, **independent of nerves and NO** → effective when PDE5Is fail (neurogenic, post-prostatectomy, DM).

Agent	Mechanism	Notes
Alprostadil (PGE1; Caverject, Edex) 5–40 µg	Receptor → adenylate cyclase → cAMP	Only FDA-approved ICI; ~70% efficacy; locally metabolised (96% within 60 min); **penile pain** commonest AE; least priapism and fibrosis
Papaverine	Nonspecific PDE inhibitor; blocks Ca^{2+} channels	Cheap, stable; hepatic enzyme rise, priapism, fibrosis → not monotherapy
Phentolamine	α-adrenergic antagonist	Poor alone; adjunct; hypotension, reflex tachycardia
Trimix (alprostadil 10 µg + papaverine 30 mg + phentolamine 1 mg per mL)	Synergistic	≈90% efficacy; rescues monotherapy failures; less pain
Bimix	Papaverine + phentolamine, or alprostadil + phentolamine	≈90%

- Technique: in-office titration from low dose (especially neurogenic/psychogenic ED, which are very sensitive); inject lateral shaft (2 or 10 o'clock †) avoiding dorsal neurovascular bundle and urethra; compress; aim for erection lasting ≤1 h.
- **Priapism**: highest with papaverine-containing mixes and in neurogenic/young patients. Instruct to seek care if erection persists >†4 h → aspiration ± intracavernosal phenylephrine.
- Other AEs: pain, haematoma, fibrosis/curvature with long-term use (examine regularly).
- Contraindications: psychological instability, priapism risk (sickle cell disease), severe coagulopathy, unstable CVD, poor dexterity (partner may inject), monoamine oxidase inhibitors (α-agonist reversal could cause hypertensive crisis).

Topical agents

Alprostadil cream with permeation enhancer (Vitaros) – modest benefit; MD3000 (Eroxon) – first over-the-counter FDA-cleared glans gel (evaporative cooling/warming).

Restorative and investigational

Therapy	Proposed mechanism	Evidence/status
Low-intensity extracorporeal shock wave therapy (Li-ESWT)	Angiogenesis; nerve, endothelial and smooth muscle regeneration	Meta-analysis of RCTs: ↑ IIEF erectile function vs sham in **mild–moderate vasculogenic** ED — best evidence of the group; still investigational (SMSNA: research protocols only)
Platelet-rich plasma	Growth factors	Conflicting RCTs
Stem cells	Tissue regeneration	Uncontrolled series only
Botulinum toxin ICI	Blocks sympathetic noradrenaline release	Short-term benefit in refractory ED; no long-term data

Combination therapy

Monotherapy nonresponse approaches 40%. Effective pairs: daily tadalafil 5 mg + on-demand sildenafil; PDE5I + T; PDE5I + IU/ICI/VED; ICI + VED; prosthesis + IU (soft glans). Combining PDE5I with ICI is off-label — test in office first.

Surgery

- **Penile prosthesis** (malleable or inflatable): when other therapy is contraindicated, ineffective or unwanted; also corporal fibrosis after ischaemic priapism or infection, Peyronie disease → *Ch 75*.
- **Arterial revascularisation** (inferior epigastric artery → dorsal artery or deep dorsal vein): strictly for **age <55, nonsmoker, nondiabetic, no venous leak, isolated internal pudendal stenosis** on arteriography — best in young men after perineal/pelvic trauma. Commonest complication: glans hyperaemia.

- **Venous ligation** surgery: investigational (leak is usually a smooth muscle disorder); complications – glans hypoaesthesia, shortening.

7 · LANDMARK TRIALS

Trial	Design	Result	Take-home
Prostate Cancer Prevention Trial (placebo arm)	Prospective cohort, men ≥55	ED → 45% higher cardiac events at 5 years	ED is a CV sentinel; stratify risk
REINVENT	RCT: nightly vs on-demand vardenafil vs placebo after nerve-sparing prostatectomy	No difference in spontaneous recovery	Scheduled "rehabilitation" PDE5I unproven

8 · MCQ ONE-LINERS

1. Minimum duration for ED diagnosis: 3 months (except trauma/surgery).
2. ED precedes and predicts CV events; evaluate CV risk in all.
3. Strongest adjusted ED risk factor: antidepressant use; strongest disease: diabetes.
4. Princeton high-risk patient: defer sexual activity and ED therapy until cardiac stabilisation.
5. SHIM 5–7 severe, 8–11 moderate, 12–16 mild–moderate, 17–21 mild, 22–25 normal.
6. IIEF has 15 items across 5 domains.
7. Preserved morning erections suggest psychogenic ED.
8. Testosterone drawn 7–11 am; AUA cut-off 300 ng/dL.
9. About one-third of mildly low testosterone values normalise on repeat.
0. Pituitary MRI if testosterone <150 ng/dL with low/normal LH.
1. Bioavailable T = free + albumin-bound testosterone.
2. Ageing, cirrhosis, hyperthyroidism raise SHBG; obesity lowers it.
3. Duplex: PSV <25 cm/s = arterial insufficiency.
4. Duplex: EDV >5 cm/s with RI <0.75 = veno-occlusive dysfunction.
5. Side-to-side PSV difference >10 cm/s suggests unilateral arterial lesion.
6. Prolonged CIS/ICI erection reversed with diluted phenylephrine every 3–5 min.
7. RigiScan measures radial, not axial, rigidity.
18. NPTR false-negative in depression and ageing; false-positive in multiple sclerosis.
19. PDE5Is enhance but do not initiate erection.
20. Tadalafil: half-life 17.5 h, food-independent, back pain.
21. Avanafil: fastest Tmax (~30 min), least visual effects.
22. Vardenafil: avoid with class 1A/3 antiarrhythmics and congenital long QT.
23. Sildenafil blue vision due to PDE6 inhibition.
24. No nitrates for 24 h after sildenafil/vardenafil, 48 h after tadalafil.
25. Declare PDE5I failure only after up to 9–10 adequate attempts.
26. Alprostadil is the only FDA-approved intracavernosal agent; pain is commonest AE.
27. Papaverine: highest priapism and fibrosis risk; raises liver enzymes.
28. ICI contraindicated with monoamine oxidase (MAO) inhibitors (phenylephrine hypertensive crisis).
29. MUSE pellet inserted 3 cm into urethra; urethral pain in one-third.
30. VED constriction ring removed within 30 minutes.
31. Li-ESWT best evidence in mild–moderate vasculogenic ED; still investigational.
32. Revascularisation ideal: young nonsmoker, nondiabetic, isolated post-traumatic pudendal stenosis.
33. Commonest revascularisation complication: glans hyperaemia.

73 Priapism

Reading time ≈ 27 min (slow pace)

1 · DEFINITION AND CLASSIFICATION

- **Priapism**: full or partial erection lasting **≥4 h** beyond sexual stimulation/orgasm, or unrelated to stimulation. A urological emergency because of the risk of fibrosis, shortening and permanent erectile dysfunction (ED).
- It is classified by **oxygenation, not duration**. Failed venous outflow causes ischaemia (a time-dependent emergency); excess arterial inflow gives a well-oxygenated, painless erection that can wait.
- Guidelines: American Urological Association/Sexual Medicine Society of North America (AUA/SMSNA) 2021–22, European Association of Urology (EAU), British Association of Urological Surgeons (BAUS).

Feature	Ischaemic (low-flow, veno-occlusive)	Stuttering (recurrent ischaemic)	Non-ischaemic (high-flow, arterial)
Share	**95%**	Subset of ischaemic	<5%
Lesion	Failed outflow, compartment-syndrome-like	Repeated self-limiting ischaemic episodes, often nocturnal	Cavernosal artery–sinusoid fistula
Typical cause	Drugs, sickle cell disease (SCD), haematological	SCD	Blunt perineal/penile trauma, needle injury
Pain / rigidity	Painful (within 6–8 h); fully rigid, tender	Painful; rigid during episode	Painless; partially rigid
Blood gas	Hypoxic, hypercarbic, acidotic	Ischaemic per episode	Arterial (normal)
Doppler	Absent/low cavernosal flow	As ischaemic during episode	High flow; fistula/blush
Management	Emergency decompression	Treat episode as ischaemic + prophylaxis	Observe → embolise
Fibrosis/ED	Common if untreated	Risk with each episode	Usually absent

> **KEY POINT** Trap: stuttering priapism is **ischaemic**, not a benign separate entity — episodes ≥4 h need the same first-line treatment.

2 · AETIOLOGY

Ischaemic

- **Intracavernosal injection (ICI)** for ED — papaverine, phentolamine, prostaglandin E1 (PGE1): wrong dosing, escalation or recreational use (younger men, late presentation, often HIV-positive).
- **Drugs** (take a full drug history every time): α-blockers (doxazosin, prazosin, tamsulosin, terazosin); antidepressants/antipsychotics with α-antagonist activity (**trazodone**, chlorpromazine, clozapine, olanzapine, risperidone…); antihypertensives (hydralazine, propranolol); oral phosphodiesterase type 5 inhibitors (PDE5Is); heparin, warfarin; attention-deficit/hyperactivity disorder agents (methylphenidate, atomoxetine); gonadotropin-releasing hormone (GnRH), testosterone; hydroxyzine; cocaine, alcohol, marijuana. Most either block α-mediated detumescence or relax smooth muscle directly.
- **Haematological/prothrombotic**: SCD, thalassaemia, leukaemia (1–5% of leukaemic males; high white cell count — sludging, splenic venous congestion, sacral/central nervous system infiltration; treat with cytotoxics + leukapheresis alongside penile treatment), myeloma, asplenia, erythropoietin, warfarin cessation, haemodialysis with heparin.
- **Metastases** (testis, kidney, lung) — usually venous obstruction. **Toxins**: scorpion (*Tityus*), Brazilian banana spider (*Phoneutria*).

- **Recurrence after a shunt** may be shunt thrombosis (tunical collagen exposure → tissue factor/VIIa cascade). Regimen: **aspirin 325 mg + heparin 5000 U subcutaneously** pre-op → aspirin 81 mg daily × 2 weeks to keep the shunt patent.

Sickle cell disease

- Haemoglobin S (HbS): valine replaces glutamic acid in β-globin. Deoxygenation → HbS polymerisation → sickling, haemolysis, endothelial dysfunction, low nitric oxide (NO) → venous outflow obstruction.
- Lifetime priapism risk **29–42%** of males with SCD; causes 23% of adult and **63% of paediatric** ischaemic priapism. Onset ~15 years; stuttering commoner than major episodes.
- Precipitants: sexual arousal, sleep, fever, cold, dehydration. Sickle trait: low risk.

Non-ischaemic

- Straddle/coital injury, pelvic fracture, needle injury (including after treating ischaemic priapism), neoplastic erosion. Men >55 years: think malignancy.
- Onset often **delayed ≥1 day**: the clot over the arterial tear disperses during a nocturnal erection.
- Turbulent high-velocity inflow → shear stress → endothelial NO → dilatation; rigidity incomplete.
- Mimic: transient **post-ischaemic hyperaemia** after treatment — colour Doppler distinguishes.

3 · MOLECULAR PATHOPHYSIOLOGY

- Priapism is a disorder of **failed detumescence**. Chronically low basal NO from dysregulated endothelial nitric oxide synthase (eNOS) reduces cyclic guanosine monophosphate (cGMP)-dependent feedback → **PDE5 down-regulated**. A neuronal NO surge then produces a cGMP surge that cannot be degraded → unopposed relaxation.
- Also ↓Rho-kinase (loses contractile tone), ↑adenosine (HIF-1α dependent). Every change removes a brake on relaxation.
- Prolonged hypoxia → smooth-muscle apoptosis and transforming growth factor-β (TGF-β)-driven fibrosis → ED.
- This explains the paradox: **daily PDE5I restores PDE5 expression and prevents** recurrent episodes.

4 · EVALUATION

- **History**: duration, pain, prior episodes, drugs/ICI, trauma, SCD/malignancy, baseline erectile function (needed for prosthesis counselling).
- **Examination**: vitals before sympathomimetics; corpora rigid and tender (ischaemic) vs partial and non-tender (non-ischaemic). **Glans and spongiosum are soft in both** — involvement suggests other pathology.
- **Corporal aspiration + blood gas** — the definitive test (may skip after known recent ICI). Colour first: dark vs bright red.

Parameter	Ischaemic	Flaccid (mixed venous)	Non-ischaemic (arterial)
pO_2 (mm Hg)	**<30**	40	>90
pCO_2 (mm Hg)	**>60**	50	<40
pH	**<7.25**	7.35	7.4

- Adjuncts in parallel (never delaying treatment): complete blood count, reticulocytes, haemoglobin electrophoresis, coagulation, toxicology.
- **Penile duplex ultrasound (PDUS)**: for equivocal gas, especially recurrence after treatment. Ischaemic: absent flow or peak systolic velocity (PSV) <50 cm/s without diastolic flow; non-ischaemic: PSV >50 cm/s, may show fistula. Magnetic resonance imaging: viability (non-acute). Angiography only for embolisation.

ALGORITHM
History + examination + gas → ischaemic → acute ischaemic algorithm | indeterminate → penile/perineal PDUS | non-ischaemic → expectant algorithm

5 · MANAGEMENT OF ACUTE ISCHAEMIC PRIAPISM (AUA/SMSNA)

- Irreversible damage begins 4–12 h after presentation. Counsel on ED risk first. Cold packs, exercise, ejaculation, oral agents, intravenous (IV) fluids: **adjunct only — never delay first-line**. Oral terbutaline/pseudoephedrine: conflicting trials, not recommended.

Duration	ED rate (SCD series)
≤12 h	0%
12–24 h	22%
24–36 h	56%
≥36 h	**100%**

ALGORITHM
<36 h: aspiration ± irrigation + intracavernosal phenylephrine (blood pressure and cardiac monitoring) → distal shunt ± tunnelling → tunnelling if not done → proximal shunt / penoscrotal decompression → prosthesis if refractory
>36 h: counsel high ED risk + analgesia/follow-up, OR early penile prosthesis

Aspiration / irrigation

- Penile block → **19-G needle** at the penoscrotal junction, **3 or 9 o'clock** (avoids dorsal neurovascular bundle). †One side usually suffices (septum fenestrated). Aspirate while compressing the shaft; refill, repeat; irrigate saline through the same needle.
- Aspiration alone works in <1 in 4. Cold (10 °C) saline irrigation gave 96% vs 60% detumescence in a randomised controlled trial (RCT) — but **avoid chilled saline in SCD** (cold precipitates sickling).

Phenylephrine

- Relatively selective α1-agonist (minimal β → least cardiac effect). Aspiration empties the compartment; the α-agonist contracts smooth muscle so it stays empty.
- Dilute to **100–500 µg/mL**; **1-mL aliquots every 3–5 min**, continue up to **1 h** before declaring failure; †maximum ≈1 mg in 1 h (EAU).
- Toxicity: hypertension, headache, arrhythmia, **reflex bradycardia** — monitor all, especially cardiovascular disease or monoamine oxidase inhibitors.

Distal (corporoglanular) shunts

All create a cavernosum–spongiosum fistula that drains to spongiosal veins and closes spontaneously after detumescence. No head-to-head comparisons exist.

Shunt	Technique	Note
Winter	Tru-Cut biopsy needle through glans into corpus; several passes	Bedside; least invasive; lowest success
Ebbehoj	No. 11 scalpel stab through glans into corpus	Bedside
T-shunt	No. 10 blade vertical through glans lateral to meatus → rotate 90° laterally; bilateral if needed	Larger window; ± tunnelling
Al-Ghorab	Open: transverse incision 1 cm distal to dorsal corona; excise corporal tips	Sedation, spinal or general anaesthesia; durable
Corporal snake	7/8 Hegar dilator through the distal corporotomy towards the crura	Breaks proximal thrombus; rescues prolonged/failed cases

Proximal shunts and decompression

- **Quackels** (unilateral) / **Sacher** (bilateral): transperineal/scrotal; 1-cm tunical ellipses on corpus cavernosum and bulb anastomosed (5-0 polydioxanone); stagger bilateral shunts ≥1 cm to reduce urethral stricture.
- **Grayhack**: saphenous vein (mobilised ~10 cm from the saphenofemoral junction) to corpus. **Barry**: deep dorsal vein to corpus.
- **Penoscrotal decompression (Morey)**: proximal corporotomy, paediatric Yankauer suction of clot bilaterally; glans-sparing, permits concurrent prosthesis.
- AUA: proximal shunts only after failed distal shunt with tunnelling. Complications: oedema, haematoma, urethral injury/fistula, necrosis, infection, pulmonary embolism.

Early penile prosthesis

- Indicated for priapism **>36 h** or refractory to shunt ± tunnelling. Oedematous corpora dilate easily; delayed implantation meets dense fibrosis.
- Early vs delayed: less shortening (3% vs 40%), fewer revisions and infections, satisfaction 96% vs 60%; rigidity similar.
- Prior distal shunts raise distal perforation risk — if recovery is unlikely, weigh shunting against a future implant. → *Ch 75*

6 · NON-ISCHAEMIC PRIAPISM

- **Not an emergency**: ~60% resolve with observation, compression, ice.

ALGORITHM
Observe, reassess at 4 weeks → persists → PDUS (fistula site, size) → counsel → selective internal pudendal/cavernosal artery embolisation → recurrence → repeat embolisation with non-resorbable agent → surgical ligation only after repeated failure

Agent	Type	Feature
Autologous clot	Temporary	Children; early recurrence
Gelatin sponge	Temporary (2–6 weeks)	Reabsorbed
Microcoils	Permanent	Theoretical ED risk
Polyvinyl alcohol particles	Permanent	**Preferred for repeat** (AUA)
N-butyl cyanoacrylate	Permanent	Second line for recurrence

- Embolisation success ≥75%; recurrence 30–40%. Open transcorporal ligation (intra-op Doppler localisation) carries ~50% ED, gangrene, abscess.

7 · STUTTERING PRIAPISM — PREVENTION

- Each episode ≥4 h is treated as acute ischaemic priapism. Short episodes: oral fluids, walking, warm shower, analgesia.
- Prophylaxis targets the trigger (androgen-driven nocturnal erections) or the molecular imbalance (α-tone, PDE5). No agent has robust evidence.

Agent	Dose	Note
PDE5I (sildenafil, tadalafil)	Sildenafil 25–50 mg daily; tadalafil 5–10 mg	Most promising; †start when flaccid; only RCT missed primary endpoint
Pseudoephedrine	60–120 mg at night	Off-label first-line α-agonist
Etilefrine	50–100 mg daily	α1-agonist; Cochrane: insufficient evidence
GnRH agonists, antiandrogens (bicalutamide, flutamide, cyproterone)	Depot / oral	Effective but suppress libido and erections; †avoid before puberty completes
Finasteride	5 → 3 → 1 mg taper	One SCD series

Agent	Dose	Note
Digoxin	0.25–0.5 mg	Na^+/K^+-ATPase inhibition
Baclofen	Oral 40 mg nocte; intrathecal	Neurogenic/reflex erections

- Self-injection of intracavernosal phenylephrine at home for episodes that begin despite prophylaxis.

SCD specifics

- Treat with the same first-line aspiration/phenylephrine; IV hydration and analgesia run **concurrently but never instead**. †Exchange transfusion is not a primary treatment (AUA) — involve haematology.
- Avoid chilled irrigation. Labs: low haemoglobin, high reticulocytes, lactate dehydrogenase, bilirubin.

8 · LANDMARK TRIALS

Trial	Design	Result	Take-home
Ateyah 2005	RCT, ICI-induced priapism, irrigation temperature	10 °C saline 96% vs 60% at 37 °C	Cold irrigation (not in SCD)
Burnett 2014	RCT, SCD, sildenafil vs placebo	Primary endpoint missed; open-label benefit	PDE5I prophylaxis plausible, unproven
Zacharakis 2014	Early vs delayed prosthesis	Shortening 3% vs 40%; satisfaction 96% vs 60%	Implant early when ED is inevitable

9 · MCQ ONE-LINERS

1. Priapism definition: erection ≥4 h unrelated to or beyond sexual stimulation.
2. Commonest type: ischaemic (95%).
3. Ischaemic gas: pO_2 <30, pCO_2 >60, pH <7.25.
4. Non-ischaemic gas resembles arterial blood: pO_2 >90, pCO_2 <40.
5. Glans and spongiosum remain flaccid in priapism.
6. Commonest cause in children: SCD (63%).
7. Classic antidepressant causing priapism: trazodone.
8. First-line ischaemic treatment: aspiration ± irrigation + intracavernosal phenylephrine.
9. Phenylephrine: 100–500 µg/mL, 1-mL aliquots every 3–5 min, up to 1 h.
0. Phenylephrine's key cardiac adverse effect: reflex bradycardia.
1. Aspiration site: 3 or 9 o'clock, avoiding dorsal neurovascular bundle.
2. Avoid chilled saline irrigation in SCD.
3. Winter shunt uses a Tru-Cut biopsy needle.
4. T-shunt: blade rotated 90° laterally after glans insertion.
5. Al-Ghorab: open excision of corporal tips via dorsal glans incision.
6. Corporal snake uses a 7/8 Hegar dilator.
17. Quackels shunt: cavernosum to bulbar spongiosum.
18. Grayhack shunt: saphenous vein to corpus cavernosum.
19. Bilateral proximal shunts are staggered ≥1 cm to avoid urethral stricture.
20. Priapism ≥36 h: ED essentially 100% — consider early prosthesis.
21. Early prosthesis: less shortening and higher satisfaction than delayed.
22. Non-ischaemic priapism onset is often delayed after trauma.
23. Non-ischaemic first-line: observation; ~60% resolve.
24. Failed resorbable embolisation → repeat with non-resorbable polyvinyl alcohol, not surgery.
25. Stuttering priapism was first described in SCD.
26. Priapism is a low-NO, low-PDE5 state.
27. PDE5Is cause priapism yet prevent stuttering episodes.
28. Penile duplex: ischaemic shows absent cavernosal flow.
29. Post-shunt recurrence may be thrombotic — aspirin + heparin prophylaxis.
30. Leukaemic priapism: leukapheresis plus cytotoxics.
31. First case report of priapism: 1824.

74 Disorders of Male Orgasm and Ejaculation

Reading time ≈ 33 min (slow pace)

1 · SPECTRUM

- Ejaculatory dysfunction (EjD) is the **most common male sexual disorder**: premature ejaculation (PE) → delayed ejaculation (DE) → anejaculation; plus retrograde ejaculation (RE), painful ejaculation, ejaculatory anhedonia, postorgasmic illness syndrome (POIS).
- Orgasm is a distinct **cortical** event — perception of pelvic striated muscle contraction and semen expulsion; ejaculation and orgasm can therefore dissociate.

2 · PHYSIOLOGY OF EJACULATION

- Reflex arc: genital sensory afferents → brain → spinal motor centres → efferents. Five groups of thoracolumbar and lumbosacral nuclei are coordinated by lumbar interneurons at **L3–L5** — the **spinal ejaculation generator**, modulated by genital and supraspinal input.
- Brain nuclei active at ejaculation: bed nucleus of stria terminalis, medial amygdala, posterodorsal preoptic nucleus, parvicellular subparafascicular thalamus; medial preoptic area and paraventricular nucleus also involved.

Phase	Events	Mediation
Emission	Vas, seminal vesicles, prostate contract → semen into posterior urethra	Autonomic (sympathetic + parasympathetic), **T10–L2**
Ejection (expulsion)	Bulbospongiosus, ischiocavernosus, levator ani contract rhythmically; **bladder neck closes**; external sphincter relaxes	Somatic **S2–S4 (pudendal)** + sympathetic spinal reflex
Orgasm	Cerebral processing of pudendal input (raised posterior urethral pressure, verumontanum, bulb contraction)	Cortical

- Key transmitters: **dopamine promotes** emission/ejaculation (D2 receptors); **serotonin (5-hydroxytryptamine, 5-HT) inhibits** (raphe nuclei). Also noradrenaline, acetylcholine, oxytocin, gamma-aminobutyric acid (GABA), nitric oxide (NO).
- 5-HT2C stimulation delays ejaculation; 5-HT1A stimulation shortens it → lifelong PE hypothesis: **5-HT2C hyposensitivity and/or 5-HT1A hypersensitivity**.

> **KEY POINT** Serotonin brakes, dopamine accelerates: PE drugs raise synaptic 5-HT; DE drugs are dopaminergic, noradrenergic or anti-serotonergic.

3 · PREMATURE EJACULATION — DEFINITION AND CLASSIFICATION

	Lifelong	Acquired	Variable	Subjective
Onset	First encounters, every partner	After previously normal latency	—	—
Latency	≤ ~1 min (mostly 30–60 s)	Markedly reduced, often ≤3 min	Occasionally short	Normal or long
Basis	Neurobiological, genetic	Medical/psychological	Normal variation	Psychological/cultural
Disorder?	Yes	Yes	No	No

- Waldinger–Schweitzer subtypes. Variable and subjective dominate the community; lifelong and acquired dominate the clinic. Acquired PE sufferers are more likely to seek treatment.

International Society for Sexual Medicine (ISSM) definition (2008 lifelong; unified 2014)

- All three required: (1) ejaculation always/nearly always before or within **about 1 min** of vaginal penetration (lifelong) **or** a clinically significant, bothersome reduction, often to **≤ about 3 min** (acquired); (2) inability to delay on all/nearly all penetrations; (3) **negative personal consequences** — distress, bother, frustration, avoidance of intimacy.
- First **evidence-based** definition; limited to vaginal intercourse. "About 1 min" allows up to ~90 s. **Latency alone never suffices** — short latency with control and no distress is not PE.

	ISSM	Diagnostic and Statistical Manual of Mental Disorders (DSM-5) (2013)	AUA/SMSNA (2020)
Lifelong latency	≤ ~1 min	≈1 min	≈**2 min** since sexual debut
Acquired	Bothersome fall, often ≤3 min	Specifier only	Marked fall (guide 2–3 min or ≥50%)
Duration/frequency	Always/nearly always	≥6 months, 75–100% of occasions	—
Severity	—	Mild 30–60 s; moderate 15–30 s; severe <15 s	—
Basis	Evidence	Same data	Expert opinion

- DSM-5 = Diagnostic and Statistical Manual of Mental Disorders, 5th edition; AUA/SMSNA = American Urological Association/Sexual Medicine Society of North America. The AUA's 2-min line trades specificity for sensitivity.
- MCQ International Classification of Diseases (ICD-10) is the only older definition with a number (≤15 s); Masters and Johnson defined PE by partner satisfaction in ≥50% of coital episodes.
- Community stopwatch **intravaginal ejaculatory latency time (IELT)**: positively skewed, median **5.4 min**; ≤1 min is statistically abnormal.

4 · PREMATURE EJACULATION — AETIOLOGY

Lifelong PE	Acquired PE
Serotonergic receptor sensitivity, genetic; younger, fewer comorbidities	Secondary: performance anxiety, relationship problems, **erectile dysfunction (ED)**, chronic prostatitis, **hyperthyroidism**, drug withdrawal; older, metabolic syndrome, diabetes

- **PE with ED** (up to half of ED men): need for more stimulation and "rushing" to ejaculate before a partial erection is lost, plus performance anxiety. Pitfall: potent PE men confuse losing erection after ejaculation with ED → false-positive Sexual Health Inventory for Men (SHIM).
- **Chronic prostatitis/chronic pelvic pain syndrome (CP/CPPS)**: PE is the main sexual symptom; antibiotics for proven bacterial prostatitis lengthen IELT.
- **Hyperthyroidism**: PE correlates with suppressed thyroid-stimulating hormone (TSH); normalising thyroid function reduces PE — but routine TSH screening is not indicated.
- Anxiety → sympathetic drive → earlier emission (hypothesis). Circumcision: no effect on PE.

5 · PREMATURE EJACULATION — EVALUATION

- **History-based diagnosis**; three recommended questions mirror ISSM: time to ejaculation, ability to delay, bother. Then lifelong vs acquired, erectile function, relationship impact, prior treatment.
- Self-estimated IELT in practice (correlates with stopwatch); stopwatch for trials.
- Questionnaires: Premature Ejaculation Profile (PEP), Index of Premature Ejaculation (IPE), Premature Ejaculation Diagnostic Tool (PEDT) — adjuncts.
- Assess ED with International Index of Erectile Function (IIEF): normal erectile domain ≥26; IIEF-5 ≥22.

- Examination: desirable in lifelong, **mandatory in acquired** (find cause); digital rectal examination (digital rectal examination (DRE)) for prostatitis. Tests only as history dictates.

6 · PREMATURE EJACULATION — TREATMENT

ALGORITHM
PE confirmed → secondary to ED/other dysfunction? **Yes** → treat primary cause → **No**: lifelong → selective serotonin reuptake inhibitor (SSRI) ± behavioural therapy (patient preference); acquired → treat cause ± behavioural/SSRI, attempt withdrawal after 6–8 weeks; variable/subjective → reassurance, education, psychotherapy

- **Psychosexual education** for all (normative latency data, dispel myths, reduce anxiety). Cognitive behavioural therapy (CBT) has a limited first-line role in lifelong PE but an important adjunctive role in acquired PE. Lifestyle: exercise, weight, alcohol.

Drugs

Drug	Dose / timing	Notes
Dapoxetine	30–60 mg, 1–3 h before	**Only licensed oral PE drug** (>50 countries, not US Food and Drug Administration (FDA)); short half-life; effective from first dose; no interaction with phosphodiesterase type 5 inhibitor (PDE5I); high discontinuation
Paroxetine (daily)	10–40 mg	**Strongest delay**; off-label
Sertraline / fluoxetine / citalopram	50–200 / 20–40 / 20–40 mg daily	Off-label
Clomipramine (TCA)	12.5–50 mg daily or 3–4 h before	Off-label
Topical lidocaine/prilocaine	20–30 min before (spray ≥5 min)	Condom advised; level 1a evidence
PDE5 inhibitors	30–50 min before	Best evidence with comorbid ED; add-on to SSRI
Tramadol	25–50 mg, 3–4 h before	Low evidence; addiction risk

- FDA = US Food and Drug Administration; TCA = tricyclic antidepressant; PDE5I = phosphodiesterase type 5 inhibitor.
- **Daily SSRIs**: block 5-HT reuptake → postsynaptic receptor stimulation. Onset 5–10 days, full effect **2–3 weeks**. Side effects (week 1, fade): fatigue, yawning, nausea, diarrhoea, sweating. Cautions: bipolar history (hypomania), bleeding with aspirin/NSAIDs (non-steroidal anti-inflammatory drugs), under-18s/depression (suicidality), sperm DNA fragmentation, weight gain. **Never stop abruptly** (withdrawal syndrome).
- On-demand off-label SSRIs (3–6 h before) delay less than daily dosing.
- **Topical anaesthetics** reduce glans sensitivity; risk penile hypoaesthesia and partner vaginal numbness/anorgasmia unless condom used. Metered lidocaine–prilocaine spray (Fortacin) penetrates glans mucosa but not keratinised shaft skin → minimal transfer.
- **PDE5Is**: rationale via NO/cyclic guanosine monophosphate (cGMP) nitrergic pathway; meta-analyses show PDE5I > placebo and PDE5I + SSRI > SSRI → guidelines allow use with or without ED.

On-demand dapoxetine	Daily off-label SSRI
Licensed; fewer side effects; little withdrawal; suits infrequent intercourse	Greater delay; full effect 2–3 weeks; must taper; suits established relationships

KEY POINT The real limiter in PE is adherence, not efficacy — most patients stop SSRIs and topicals (cost, loss of spontaneity, "antidepressant" stigma).

7 · DELAYED EJACULATION, ANEJACULATION AND ANORGASMIA

- DE: marked delay or absence of ejaculation despite adequate stimulation and desire, ~75–100% of occasions, with distress (DSM-5). **No validated latency cut-off**; most men ejaculate at 4–10 min, so >25–30 min with distress, or stopping from exhaustion, suggests DE.
- **Anorgasmia** = absent perceived orgasm, independent of ejaculation. Lifelong ~25%, acquired ~75% of DE; rises with age.
- Hallmark: normal (even prolonged) erections with **low subjective arousal**.

Causes (same reflex arc gives DE, anejaculation, RE, anorgasmia)

Category	Causes
Afferent	Ageing (penile afferents, Pacinian corpuscles)
Psychogenic	Inhibited ejaculation; performance anxiety; religious orthodoxy; **idiosyncratic vigorous masturbation**
Endocrine	Hypogonadism, **hypothyroidism** (DE; hyperthyroidism → PE), hyperprolactinaemia
Neurogenic disease	Spinal cord injury (SCI) > diabetic autonomic neuropathy > multiple sclerosis
Neurogenic surgery	Radical prostatectomy, rectal surgery, sympathectomy, aortic aneurysm repair, retroperitoneal lymph node dissection (RPLND)
Anatomical (→ RE)	Transurethral resection of prostate (TURP), bladder neck incision
Congenital / infective	Müllerian cyst, Wolffian anomaly, prune belly; urethritis, tuberculosis, schistosomiasis
Drugs	**SSRIs** (~7-fold DE risk), tricyclic antidepressants (TCAs), antipsychotics, α-methyldopa, thiazides, alcohol

- Antipsychotics: DE (dopamine block, prolactin) and RE (α-block at bladder neck).
- **Superior hypogastric plexus** (sympathetic, anterior to lower aorta) mediates emission → injury at RPLND gives dry ejaculation. **Nerve-sparing RPLND** preserves antegrade ejaculation in most.
- After radical prostatectomy: no ejaculation, orgasm preserved; climacturia (leak at orgasm). After radiotherapy: reduced volume, weaker orgasm. Prior TURP explains RE after radiotherapy.
- Rectal cancer: total mesorectal excision with autonomic nerve preservation gives good ejaculatory function.

Evaluation

- History: exclude RE, obstruction, anorgasmia, ED, low libido, partner factors. Ask why he stops, masturbatory ejaculation (separates coital-specific from global), and "ever feel you ejaculated but no semen?" (screens RE/obstruction).
- Examine testes, epididymides, vasa; DRE (anal tone, bulbocavernosus reflex). Morning testosterone; HbA1c (glycated haemoglobin), lipids; evoked potentials or sympathetic skin response if neuropathy suspected.

ALGORITHM
Orgasm? **Never/sometimes** → inhibited orgasm (psychosexual therapy) or ageing (reassure, change technique) → **Always** → ejaculate? No → **sperm in post-orgasm urine?** Yes = RE; No = aspermia → failure of emission (neurogenic, drug, metabolic) or ejaculatory duct obstruction (EDO)

	Bilateral EDO	RE	Failure of emission
Antegrade volume	Very low	Low/absent	Absent
pH, fructose	Low (no seminal vesicle fluid, ~80% of volume)	—	—
Post-orgasm urine	No sperm	**Sperm + fructose**	No sperm

Treatment

- Aetiology-specific; address fertility; counsel before pelvic surgery about anejaculation and sperm retrieval. Refer to a psychosexual therapist once organic/drug causes excluded (AUA). Normalise testosterone if low. Change offending drug.
- Behavioural: more stimulation, change masturbatory style, less alcohol, sex when not tired.
- Drugs — all off-label, modest (~40%) success; poor in psychogenic and neuropathic DE:

Drug	Action	Dose
Cabergoline	Central D2 agonist	0.5–2 mg every 3 days; **annual echocardiography** (valvular fibrosis)
Bupropion	Dopaminergic	150 mg once/twice daily
Pseudoephedrine	α1 agonist	120 mg 1–2 h before
Reboxetine	Noradrenaline reuptake inhibitor	4–8 mg daily
Cyproheptadine	5-HT antagonist	4–12 mg 3–4 h before (SSRI-induced); sedation
Amantadine, buspirone	—	Anecdotal
Intranasal oxytocin	Oxytocinergic	Both RCTs negative

- Neuropathic DE is usually irreversible → counsel alternative routes to satisfaction.

8 · RETROGRADE EJACULATION

- Failure of **bladder neck closure**. Commonest causes: **TURP and diabetic autonomic neuropathy**. Others: SCI, multiple sclerosis, RPLND, drugs (α-blockers, antipsychotics, antidepressants, thiazides).
- TURP (RE 42–100%) > transurethral incision of prostate (TUIP; 5–45% — less if one incision not extended to verumontanum). Ejaculation-sparing: modified TURP/photovaporisation, prostatic urethral lift and Aquablation (superior to TURP in RCTs).
- **Diagnosis**: post-masturbation urine; **>5–10 sperm/high-power field** confirms RE; more sperm in urine than ejaculate = significant RE.
- **Treatment**: sympathomimetics (pseudoephedrine, ephedrine, midodrine, phenylpropanolamine) close the internal sphincter. **Pseudoephedrine 120 mg 6-hourly the day before and twice on the day**; side effects dry mouth, hypertension. Imipramine 25 mg twice daily; antimuscarinic + sympathomimetic better than either. Bladder neck reconstruction — poor results.
- Fertility if drugs fail: **urinary sperm retrieval** (alkalinised urine, empty bladder) → intrauterine insemination (IUI), in vitro fertilisation (IVF) or intracytoplasmic sperm injection (ICSI).

9 · SPERM RETRIEVAL IN SCI

- Erection and ejaculation move oppositely with level: reflexogenic erection needs the intact sacral arc (upper motor neuron (UMN) lesions); ejaculation is rare with complete UMN (<5%) and needs intact thoracolumbar outflow (better with incomplete, lower lesions). RE common; orgasm may be absent.

ALGORITHM

Penile vibratory stimulation (PVS) — first line, success up to ~70% (needs intact T12–S1-level reflex) → **electroejaculation (EEJ)** — rectal probe stimulating hypogastric sympathetic efferents, effective and safe → prostate massage → surgical retrieval (TESE/TESA/MESA/PESA; testicular sperm extraction succeeds in most)

- Prepare bladder (empty, instil sperm-wash medium) because of RE.
- **Autonomic dysreflexia** risk (lesions above T6) with PVS and EEJ → pretreat with **nifedipine**, monitor blood pressure.
- Semen: low motility initially (infection, urine, stasis, scrotal heat) — improves with repeated ejaculation. Fresh testicular sperm may outperform frozen in SCI.

10 · OTHER DISORDERS

Disorder	Key points
Painful ejaculation (odynorgasmia)	Pain in penis, scrotum, perineum during/after ejaculation. Causes: prostatitis/CPPS, urethritis, BPH (benign prostatic hyperplasia), seminal vesicle calculi, EDO, α-blockers (less with alfuzosin); often none. Treat cause; surgery (TURP, transurethral resection of the ejaculatory ducts (TURED), pudendal neurolysis) weakly supported
Haematospermia	Alarming but usually benign, self-limiting. Associations: schistosomiasis, prostatitis; seminal vesiculoscopy for refractory cases → *Ch 70*. †Evaluate men >40 y or persistent cases: DRE, prostate-specific antigen (PSA), urinalysis/culture, transrectal ultrasound (TRUS); red flags — haematuria, weight loss, bleeding disorder
Climacturia	Leak at orgasm after radical prostatectomy (arousal incontinence = leak in foreplay). Void first → condom → pelvic floor training → constriction ring → male sling
Ejaculatory anhedonia	Orgasm without pleasure; depression, hypogonadism, hyperprolactinaemia, SSRIs
Post-SSRI sexual dysfunction	Genital anaesthesia, low desire, ED, anhedonia persisting months–years after SSRI (vs transient withdrawal syndrome)
POIS	Myalgia, fatigue, flu-like state within 30 min of orgasm
Orgasmic headache	Dull bilateral, rising with excitement, suddenly intense at orgasm

- PSA = prostate-specific antigen; TRUS = transrectal ultrasound; TURED = transurethral resection of ejaculatory ducts.

11 · MCQ ONE-LINERS

1. Most common male sexual disorder: ejaculatory dysfunction.
2. Spinal ejaculation generator: lumbar interneurons L3–L5.
3. Emission: sympathetic T10–L2; expulsion: somatic S2–S4 pudendal.
4. Bladder neck closes during expulsion, preventing retrograde flow.
5. Serotonin inhibits; dopamine (D2) promotes ejaculation.
6. Lifelong PE: 5-HT2C hyposensitivity / 5-HT1A hypersensitivity.
7. ISSM lifelong PE: IELT within about 1 min.
8. ISSM acquired PE: bothersome reduction, often ≤3 min.
9. ISSM three criteria: latency, inability to delay, negative consequences.
0. AUA/SMSNA lifelong PE latency: about 2 min.
11. DSM-5 severe PE: <15 s or before penetration.
12. Median community IELT: 5.4 min.
13. Only licensed oral PE drug: dapoxetine (not FDA).
14. Strongest ejaculatory delay among SSRIs: paroxetine.
15. Daily SSRI full effect: 2–3 weeks.
16. Hyperthyroidism → PE; hypothyroidism → DE.
17. Main sexual symptom in CP/CPPS: PE.
18. Normal IIEF-5 score: ≥22.
19. DE: ~25% lifelong, ~75% acquired.
20. Cabergoline for DE needs annual echocardiography.
21. Intranasal oxytocin for DE: RCTs negative.
22. Commonest causes of RE: TURP and diabetic neuropathy.
23. RE confirmed: >5–10 sperm/high-power field (HPF) in post-ejaculate urine.
24. Pseudoephedrine treats RE by closing bladder neck.
25. EDO semen: low volume, acidic, fructose-negative.

26. Dry ejaculation after RPLND: superior hypogastric plexus injury.
27. SCI first-line sperm retrieval: penile vibratory stimulation.
28. PVS/EEJ hazard: autonomic dysreflexia — nifedipine pretreatment.
29. Climacturia: urinary leakage at orgasm after radical prostatectomy.
30. Painful ejaculation with α-blockers: least with alfuzosin.

75 Surgery for Erectile Dysfunction

Reading time ≈ 27 min (slow pace)

1 · PLACE OF SURGERY

- Penile prosthesis is the **most effective but last-line** treatment for erectile dysfunction (ED): for motivated men who fail, cannot use or reject oral phosphodiesterase type 5 inhibitors (PDE5Is), intracavernosal injection (ICI), intraurethral pellets or vacuum erection devices (VEDs).
- **Goal-directed approach** (Lue): offer all options, trial as the patient wishes, settle on the acceptable one. Extensive aetiological testing does not change outcome.
- Satisfaction 80–90%; repair needed in ~15% at 5 years and ~30% at 10 years.
- Firsts: three-piece inflatable (Scott, 1972); semirigid rod (Small–Carrion, 1978).

2 · PROSTHESIS TYPES

Feature	Malleable (semirigid rod)	Two-piece inflatable	Three-piece inflatable (IPP)
Components	Two bendable rods	Cylinders + scrotal pump-reservoir	Two cylinders + scrotal pump + abdominal reservoir
Flaccidity	Permanently firm; concealment harder	Partial	**True flaccidity** (fluid fully evacuated)
Rigidity	Intrinsic rod + less dead space	Moderate	Best
Mechanical failure	Exceedingly rare	—	Commonest complication of implant surgery
Best for	Poor manual dexterity/cognition; where hydraulic repair undesirable	Never popular; last model (Ambicor) discontinued	Most patients — most popular device

- **Rods**: Tactra (nitinol core), Genesis (braided silver), Rigi 10 (stainless steel); fixed girths; sized 0.5–1 cm shorter than corporal length for easier bending.
- **Inflatables**: AMS 700 (CX girth; LGX girth + length up to 25%; CXR narrow), Coloplast Titan (Titan Narrow), Rigicon Infla 10 (AX length-expanding; NB narrow).
- **Material**: Coloplast uses polyurethane (Bioflex) — cautery **<35 W**; Boston/Rigicon silicone–fabric–silicone trilaminate tolerates cautery.
- Narrow cylinders suit small or scarred corpora; length-expanding cylinders correct curvature poorly.

3 · SELECTION AND COUNSELLING

- **Consent is expectation management**: the device restores rigidity, not the former erection. Commonest disappointment — **shorter erection**, because the pseudocapsule around the device limits expansion (worse in Peyronie disease).
- Key points: penis may be shorter/thinner; goal is a firm penis for intercourse; sensation and ejaculation may diminish; pain; mechanical revision; infection/erosion may need reoperation; alternatives exist; device options. Patients may be litigious — document.
- Limited dexterity → malleable rod.

Pre-operative preparation (infection prevention by source control)

- Treat skin lesions, fungal infection; **phimosis** → circumcise and implant ~6 weeks later. Wanted circumcision with redundant foreskin → simultaneous (implant first, close, then circumcise).
- Sterile urine; stop anticoagulants including aspirin; antibiotic allergy history.
- Shave in theatre (nicks colonise if done earlier). Chlorhexidine–alcohol skin preparation (≈40% fewer surgical site infections than povidone-iodine).

- Diabetes: **blood glucose <200 mg/dL on the day** matters more than glycated haemoglobin (HbA1c); if raised, intravenous insulin.
- Nasal methicillin-resistant *Staphylococcus aureus* (MRSA) carriers: chlorhexidine bathing + mupirocin 2% × 5 days.

4 · SURGICAL APPROACHES

Approach	Advantages	Cautions
Penoscrotal (transverse upper scrotal or raphe)	Majority choice; secure subdartos pump; good crural exposure; cosmetic	Urethra at risk — Foley first; reservoir via inguinal canal
Infrapubic (original Scott)	Reservoir under vision	Dorsal neurovascular bundle (sensory to glans) — stay sutures at 10 and 2 o'clock; patient pulls pump down daily
Subcoronal	Rods; allows straightening	Circumcise uncircumcised men simultaneously (else lymphoedema/maceration)
Ventral penile	Rods without bending; foreskin kept	—

5 · OPERATIVE TECHNIQUE (THREE-PIECE)

ALGORITHM
Catheter → expose tunica → two 2-0 polydioxanone stay sutures per corpus → longitudinal corporotomy → dilate proximal then distal (Metzenbaum, Hegar/Brooks 7→13 mm) → goalpost test → measure (Furlow) → cylinders → test inflation 1 → close corporotomies → test inflation 2 → reservoir → pump → connect → test inflation 3 → irrigate, close

- **Dilation**: instruments always point **dorsolaterally**, parallel to the shaft, away from the urethra and thin septum. Proximally to the ischial tuberosity gently (crural perforation).
- **Goalpost test**: dilators in both corpora at equal height and symmetry — detects crossover.
- **Sizing**: inflatable cylinder equal to or 1 cm shorter than measured length — oversizing causes buckling and S-shaped deformity; rear tip extenders adjust. Minimal-dilation technique in unscarred primary implants preserves length.
- Close corporotomy protecting the cylinder; input tubing at the proximal end.
- **Pump**: subdartos pouch, most dependent scrotum, anterolateral to testis. Trim tubing; straight connection with shodded clamps.
- **Malleable girth**: two parallel Hegar dilators in the corpora — slight separation = ideal.

Reservoir placement

- **Space of Retzius**: via linea alba (infrapubic) or through the external ring/inguinal canal piercing transversalis fascia above the pubic tubercle (penoscrotal), bladder empty, Trendelenburg. Iliac vessels lie ~1 inch lateral.
- Hostile pelvis (radiation, cystectomy, robotic prostatectomy — bowel adherent at internal ring, renal transplant) → **ectopic site**, made safe by the **lockout valve** preventing autoinflation.
- **Submuscular suprafascial** (behind rectus, above transversalis) — commonest ectopic; fill nearly to capacity (underfilling → migration); confirm not intraperitoneal. Others: lateral retroperitoneal (after diversion), epigastric, subcutaneous.
- Large reservoirs can compress the iliac vein → leg oedema.

6 · INTRA-OPERATIVE PROBLEMS

- **Fibrotic corpora** (post-priapism, post-explant; pelvic radiation does *not* add difficulty): midshaft corporotomy, opposite incision to previous, spread not cut, then cavernotomes (Rossello, Uramix), Otis urethrotome, excision of scar. Use **narrow cylinders**; always two cylinders (single cylinder →

lopsided, poor rigidity). Last resort: incise scar and cover with graft (biologic — oversize, absorbable suture; synthetic — size to defect).

- **Crossover**: both cylinders on one side, lopsided erection, unequal measurements → remove, leave large dilator in the common cavity, redilate opposite side laterally, replace. Confirm Foley midline between cylinders.
- **Proximal (crural) perforation**: dilator deeper than mate, lengths differ >1 cm → **suture sling** (2-0 Prolene through rear tip tied over the corporotomy). Windsock abandoned.
- **Urethral perforation**: at corporotomy → close and proceed (abort if doubt). Distal (fossa navicularis; blood/irrigant at meatus) → **abandon that cylinder**, place contralateral, no catheter, replace at 5–6 months. Never leave a cylinder beside an unclosed urethral injury.
- **Hypermobile glans** (ventral = **supersonic transporter (SST) deformity**, like the drooping Concorde nose): glans fixation sutures (3-0 non-absorbable) to distal tunica via hemicircumcising incision. Distinguish from short cylinders (→ longer cylinder/rear tip extenders).
- **Residual curvature** (Peyronie): high-rigidity cylinders (AMS 700 CX, Titan, Infla 10 X), then **modelling** — clamp tubing, bend 90° away from curve 90 s, repeat; then plication or incision ± graft. <20° acceptable. → *Ch 76*

> **KEY POINT** Why no catheter after distal urethral perforation? Distal defects are not closed (poor exposure); the urinary stream draws tissue fluid into the urethra (Bernoulli effect), so urine does not enter the defect. Why wait 5–6 months? Solid scar must obliterate the false tract, or the dilator re-enters it. Almost every intra-operative error is **prevented by instrument direction** and **detected by symmetry** (goalpost test, midline Foley).

> **KEY POINT** After robotic prostatectomy the peritoneal reflection is not replaced; bowel becomes fixed near the internal ring, so a blind inguinal Retzius reservoir can compress bowel on filling — choose an ectopic site or place under vision.

7 · COMPLICATIONS

Infection (1–3%)

- **Risk factors**: acute hyperglycaemia, revision surgery (disturbed biofilm), skin or urinary sources, MRSA carriage, post-priapism/scarred corpora.
- **Prophylaxis** (American Urological Association, AUA): intravenous (IV) vancomycin or 3rd-generation cephalosporin + gentamicin 1 h before incision, ≤24 h. Extended oral courses have no proven benefit. High risk: add antifungal ± piperacillin–tazobactam (watch nephrotoxicity).
- Antibiotic coatings (InhibiZone rifampin–minocycline on AMS 700; hydrophilic Titan/Infla dipped in surgeon's solution) halved infections. **No-touch technique** → <0.5%: organisms enter via the wound, not blood.
- Diagnosis: fever, erythema (suggestive); pus, exposed device, progressive pain, **pump fixed to scrotal wall** (definitive).
- Treatment: **remove ALL components** (retained parts harbour biofilm).

Option	Technique	Caveats
Explant, delayed replacement	Drains with antibiotic irrigation × 3 days; reimplant at 2–6 months	Fibrosis → difficult reinsertion; loss of several cm length; vacuum erection device (VED) meanwhile
Mulcahy salvage	Remove all → antiseptic washes → change drapes → immediate new device	~84% success; fails with onset <2 months, aggressive organisms, purulence with cellulitis
Delayed salvage	IV vancomycin + gentamicin, salvage at 3 days	For aggressive infection
Malleable salvage	Rods instead	Quick; convert later

- Current washes: dilute povidone-iodine (0.35–3.5%) 3 min, then saline/antibiotic; **drop hydrogen peroxide** (cytotoxic, no gain).
- **Revision washout**: revision of an uninfected device has ~10% infection (disturbed biofilm); antiseptic washout of all cavities lowers it to ~3% — use in every revision.

Mechanical failure

- **Commonest complication**: fluid leak, pump malfunction, autoinflation, tube kinking, cylinder aneurysm (AMS fabric shreds; Titan polyurethane wears).
- Device >3 years → replace **all** components. If parts retained, verify integrity — **ohmmeter test** (saline conducts through a leak).
- Rod loss of rigidity → core fracture → replace.

Extrusion and erosion

Feature	Extrusion	Erosion
Position	Tip through tunica, still under skin/glans	Through skin or into urethra
Cause	Forceful dilation, oversizing, aggressive use	Infection, crossover pressure, catheterisation; rods in insensate (spinal cord injury) men
Management	**Distal corporoplasty**: new dorsomedial plane behind pseudocapsule	Remove all, washout, contralateral spacer, replace at 5–6 months

- Use inflatables (not rods) in insensate or chronically catheterised men.
- **Reservoir erosion**: bladder (sterile) → exchange reservoir and repair; bowel → remove whole device, general surgery.
- **Penile necrosis**: diabetes, vascular disease, smoking, prior implant, radiation; avoid circumferential incisions/aggressive reconstruction, wrap with glans visible; if necrosis → remove cylinders promptly.
- Early bulge at base = corporotomy disruption (repair); late = cylinder aneurysm (replace).

8 · SPECIAL SITUATIONS

- **Post-priapism fibrosis**: priapism >36 h → **immediate implant** preserves length (oedematous, unscarred corpora) and relieves pain; delayed → dense fibrosis, shortening. Document necrosis (biopsy); close distal shunt defects; avoid distally expanding cylinders (LGX, Infla 10 AX). → *Ch 73*
- **Peyronie disease**: more length loss; non-expanding high-rigidity cylinders + modelling.
- Transplant recipients and HIV-positive men on therapy with adequate CD4 counts: no excess infection.
- Lax suspensory ligament (buckling despite good sizing) → suture tunica to symphysis periosteum.
- Neophallus: single cylinder anchored to pubis; high erosion (no tunica).
- Long-distance cycling: noseless saddle.

9 · POST-OPERATIVE CARE

- Day surgery; cylinders **partially inflated** so the pseudocapsule forms large; mummy wrap; scrotal drain <24 h; penis upward 4–6 weeks.
- Analgesia: dorsal nerve block (liposomal bupivacaine), short opioid course, then non-steroidal anti-inflammatory drugs.
- Activation at **4–6 weeks**, partner present; daily cycling; intercourse thereafter.
- Satisfaction high; partners slightly less; dissatisfaction = length and rigidity.

10 · PENILE REVASCULARISATION †

- †Candidate: **young man (<55)** with ED after pelvic/perineal trauma, no vascular risk factors or veno-occlusive dysfunction, and a **focal** common penile/internal pudendal artery occlusion on selective pudendal arteriography.

- †Operation: microvascular **inferior epigastric artery → dorsal penile artery** (or deep dorsal vein arterialisation). Generalised atherosclerosis and venous-leak surgery are not recommended.

11 · MCQ ONE-LINERS

1. First three-piece inflatable prosthesis: Scott, 1972.
2. Commonest complication of IPP: mechanical failure.
3. Commonest patient complaint after implant: shorter erection.
4. Shortening cause: pseudocapsule limiting cylinder expansion.
5. Poor dexterity → malleable rod.
6. Tactra core: nitinol; Genesis: silver; Rigi 10: stainless steel.
7. Titan cylinders are polyurethane (Bioflex); keep cautery <35 W.
8. Length-expanding cylinders: AMS 700 LGX and Infla 10 AX.
9. Best cylinders for curvature: AMS 700 CX, Titan, Infla 10 X.
0. Infrapubic approach risks the dorsal nerve; penoscrotal risks the urethra.
1. Day-of-surgery glucose <200 mg/dL matters more than HbA1c.
2. Prophylaxis: vancomycin or cephalosporin + gentamicin, ≤24 h.
3. InhibiZone coating: rifampin + minocycline.
4. Infected implant: remove all components.
15. Mulcahy salvage: explant, antiseptic washout, immediate replacement.
16. Salvage failure predictors: early onset, aggressive organism, cellulitis.
17. Hydrogen peroxide is dropped from salvage washes.
18. Revision washout lowers infection from ~10% to ~3%.
19. Lockout valve prevents autoinflation and allows ectopic reservoirs.
20. Commonest ectopic reservoir site: submuscular suprafascial.
21. Crural perforation repair: suture sling.
22. Distal urethral perforation: abandon ipsilateral cylinder; replace at 5–6 months.
23. SST deformity: ventral glans hypermobility; treat by glans fixation.
24. Extrusion: distal corporoplasty; erosion: explant.
25. Rods erode in insensate men — prefer inflatables.
26. Bladder reservoir erosion: exchange reservoir; bowel: remove entire device.
27. Pelvic radiation does not increase implant difficulty.
28. Post-priapism: early implant preserves length.
29. Activation 4–6 weeks after surgery.
30. Ohmmeter test detects leaks in retained parts.

76 Diagnosis and Management of Peyronie Disease

Reading time ≈ 27 min (slow pace)

1 · DEFINITION

- **Peyronie disease (PD)** (American Urological Association, AUA 2015): acquired penile abnormality with **fibrosis of the tunica albuginea**, which may be accompanied by pain, deformity, erectile dysfunction (ED) and/or distress. Old name: induratio penis plastica; described by de la Peyronie (1743).
- An acquired **wound-healing disorder**: the inelastic scar ("plaque") never remodels, so scar and deformity persist.

> **KEY POINT** Misconceptions corrected: PD is **not rare** (3–20%; higher in diabetes and ED); **spontaneous resolution is rare** — up to half worsen untreated, so "wait 6–12 months" is a myth; occurs from teens to the 70s and in all races; flaccid trauma can trigger it; **calcification is not a marker of chronic disease**.

2 · ANATOMY AND PATHOPHYSIOLOGY

- Tunica albuginea: mainly type I collagen + elastin; **inner circular and outer longitudinal layers**, 1.5–3 mm thick; septum anchored to the inner layer. The septum is essential — a single corporal body would rupture at full erection.
- Longitudinal layer is thinnest at 3 and 9 o'clock and **absent between 5 and 7 o'clock** → dorsal buckling; **60–70% of plaques are dorsal** with septal involvement.
- **Microtrauma**: coital bending (often partner-on-top, missed thrust) raises intracorporal pressure beyond tunical elasticity → **delamination at septal insertions** → microhaemorrhage traps the healing process within the tunica.
- Cascade: **transforming growth factor-β1 (TGF-β1)** (central), platelet-derived and fibroblast growth factors, reactive oxygen species, inducible nitric oxide synthase, plasminogen activator inhibitor → disorganised collagen, reduced elastin → asymmetric expansion with **curvature toward the plaque**.
- Trauma alone is insufficient (penile fracture rarely causes PD) — needs a **susceptible host** (genetics, aberrant healing; associated with Dupuytren and Ledderhose disease). Only 16–40% recall trauma.
- Medicolegal: do not attribute PD to intracavernosal injection (ICI), catheterisation or endoscopy — these are opportunities, not causes.

3 · CLINICAL FEATURES AND PHASES

- Presents with pain, deformity (curvature, indentation, **hourglass**, hinge, shortening), palpable plaque, ED.
- Circumferential/transverse scar → hourglass or hinge → **unstable penis** that cannot bear axial load. Distal softness beyond the plaque may occur.
- Curvature severity (Kelami): ≤30° mild, 31–60° moderate, >60° severe. Many manage intercourse up to 60° (especially dorsal); ventral and lateral curves are harder.
- Pain on erection usually settles once stable; persistent "torque" pain on strong erection is mechanical, not inflammatory.

Feature	Active (acute) phase	Stable (chronic) phase
Pain	Present (flaccid tenderness, erection, intercourse)	Usually resolved
Deformity	Progressive scar growth	No further progression
Typical aim	Prevent progression; non-steroidal anti-inflammatory drugs (NSAIDs) for pain; traction	Correct deformity (injection, surgery)

Feature	Active (acute) phase	Stable (chronic) phase
Surgery	Not appropriate	When **≥12 months from onset and stable ≥6 months**

- Differential: congenital curvature, chordee without hypospadias, epispadias; post-traumatic fibrosis; tumours (epithelioid sarcoma mimics PD).

4 · EVALUATION

- **History**: onset, trauma, progression, pain, intromission ability, vascular risk factors, Dupuytren/Ledderhose.
- **Key predictive question**: "If your penis were straight with its current rigidity, would it be adequate for penetration?" — **No** → prosthesis with straightening. This predicts post-operative erectile function.
- **Peyronie's Disease Questionnaire (PDQ)**: 15 items, 3 domains (psychosexual symptoms, pain, bother); no total score.
- **Examination**: plaque palpated on stretch; plaque size is unreliable and its change does not track deformity. **Stretched penile length (SPL)**, pubis to corona dorsally, stretched at 90° — essential baseline because patients overestimate loss.
- **Calcification**: ultrasound (hyperechoic, shadowing). Extensive calcification → poor response to intralesional agents and more likely surgery/prosthesis.

Levine calcification grade	Size	Surgery likelihood
1	<0.3 cm	Not increased
2	0.3–1.5 cm	Not increased
3	>1.5 cm, or multiple ≥1 cm	Increased

- **Intracavernosal injection (ICI) ± penile duplex Doppler ultrasonography (PDDUS)**: AUA — required **before any invasive treatment**. Gives objective curvature (goniometer), indentation, hinge, calcification and vascular status (peak systolic velocity, end-diastolic velocity, resistive index). Patient-estimated curvature is unreliable.
- Ventral curvature is most associated with veno-occlusive dysfunction. Home photographs: a rough guide only.
- Morning testosterone only if ED coexists; biothesiometry if sensory change.

5 · NON-SURGICAL MANAGEMENT

- No pain, no penetration difficulty → **reassurance** (not cancer, not life-threatening).
- Goal: minimum — prevent progression in the active phase; ultimate — reduce deformity and restore function.
- **Oral agents**: AUA "should not offer" any except **non-steroidal anti-inflammatory drugs (NSAIDs) for pain**. Vitamin E, tamoxifen, colchicine, carnitine — no benefit; Potaba shrinks plaque without straightening; pentoxifylline — uncertain.
- Phosphodiesterase type 5 inhibitors (PDE5Is): treat associated ED; not PD itself.
- Extracorporeal shock wave therapy (ESWT): helps pain/function only — should not be used for curvature. Radiotherapy: no benefit, real harm.

Collagenase *Clostridium histolyticum* (CCH, Xiaflex)

- First and only FDA-approved PD drug (2013). Degrades collagen **types I and III** (even with elevated tissue inhibitors of metalloproteinases), sparing elastin, vascular smooth muscle and nerves.
- **Protocol (IMPRESS)**: cycle = 2 injections of **0.58 mg** 24–72 h apart → **modelling** 24–72 h after the second; up to **4 cycles, 6 weeks apart** (max 8 injections). Modelling is essential — CCH without it is inferior. Most gain in the first 3 cycles.

- **Criteria (AUA)**: stable disease, curvature **>30° and <90°**, intact erectile function (± medication). IMPRESS excluded ventral plaques, extensive calcification, disease <12 months.
- Poor response: calcification, curvature ≥60°, severe indentation, hinge. Better response: International Index of Erectile Function (IIEF-5) ≥17.
- Adverse: bruising, swelling, pain; serious — **corporal rupture/fracture** (0–4.9%) and haematoma (tunica weakened around the plaque) — advise no intercourse until after modelling period.
- Later plication/grafting is not made harder. Ventral use not routine (spongiosum/urethra contain the same collagens).

Other intralesional agents

Agent	Mechanism	Evidence	AUA
Verapamil	Calcium channel blocker: ↓fibroblast proliferation and extracellular matrix, ↑collagenase	Uncontrolled benefit; RCTs conflicting; stabilises; inferior to CCH head-to-head; cheap	May offer
Interferon α-2b	↓fibroblast proliferation and collagen, ↑collagenase	randomised controlled trial (RCT) vs saline: curvature ↓27% vs 9%; flu-like symptoms (pre-treat with NSAIDs)	May administer
Nicardipine	Calcium channel blocker	↓pain and plaque; no curvature benefit	—

- Verapamil: 10 mg diluted (10–20 mL); poor candidates — calcification, curvature >90°, ventral curves.

Mechanical therapy

- **Penile traction therapy (PTT)**: mechanotransduction — sustained stretch reorients collagen, ↓myofibroblasts, ↑matrix metalloproteinases. Acute phase: 6–9 h/day (remove 30 min every 2 h) → less curvature and pain, fewer needing surgery, grafting downgraded to plication. Gains length; benefit needs ≥3 h/day.
- **RestoreX** (counter-bending, 30–90 min/day): RCT — better length, curvature, IIEF-5.
- Vacuum device: modest; AUA no recommendation for traction, vacuum or combinations.

6 · SURGERY

Indications

- Gold standard for rapid, reliable correction. **Stable disease ≥12 months from onset, deformity stable ≥6 months** (expert consensus).
- Deformity and/or inadequate rigidity preventing satisfactory intercourse; failed conservative therapy; wish for most reliable result.

Consent

- Goal: **functionally straight** (≤20°; EAU ≤15°), not the pre-PD penis. Residual/recurrent curvature up to 16%.
- **Length loss** with every procedure (plication > grafting); 70–80% already lost length — document SPL.
- **New ED** (grafting > plication), ~30% if pre-op rigidity is suboptimal. Pre-operative erection quality is the strongest predictor. Sensory reduction ~20%, usually transient.

Choice of procedure — algorithm

ALGORITHM

Erection adequate for penetration (± phosphodiesterase type 5 inhibitor (PDE5I))? **No** → inflatable penile prosthesis + modelling → if curve persists, incision ± graft (graft if defect >2 cm)

Yes → PDDUS → simple curve <60–70°, no hourglass/hinge, acceptable length loss → **plication** (if length loss unacceptable → grafting)

Yes → complex/multiplanar >60–70°, severe indentation/hourglass, hinge, short penis (<9 cm stretched) → **plaque incision/partial excision + graft** if ED risk accepted (else prosthesis)

Tunical plication (shortening) procedures

- Shortens the **convex (long) side opposite the plaque**. Simple, quick, minimal rigidity effect, least new ED. Does not correct hourglass/hinge or existing shortening; knot pain, granulomas.
- Most common PD operation (~half of surgery). Nesbit failure predictors: poor erections, shortening >2 cm, curvature >30°.

Technique	Method	Point
Nesbit (1965)	Full-thickness transverse elliptical tunical excision opposite the curve, transverse closure	Original; ventral curve → lift Buck fascia, dorsal wedges
Yachia	Vertical 0.5–1.5 cm incision closed transversely (Heineke–Mikulicz)	No tunica removed; long incisions narrow shaft
Essed–Schroeder	Non-absorbable figure-of-eight, buried knots	First plication without incision
16-dot	Permanent sutures, extended Lembert, 4 dots per plication	Popular; no incision
Tunica albuginea plication (Levine)	Partial-thickness paired transverse incisions through outer longitudinal layer only; buried permanent suture + 3-0 polydioxanone (PDS)	Cavernosal tissue undisturbed

Plaque incision/partial excision and grafting (lengthening)

- For complex deformity with **strong erections** and no veno-occlusive dysfunction. Restores calibre, regains 0.5–3 cm length.
- ED predictors: age >55, veno-occlusive dysfunction (resistive index <0.80), large graft, ventral curve, curvature >60°. Total plaque excision abandoned (ED).
- **Steps**: artificial erection → circumcising (degloving) incision → elevate Buck fascia with neurovascular bundle (dorsal: incisions beside urethra) → mark point of maximum deformity → H/double-Y incision or rectangular partial excision (must **cross the septum**), dart corners → graft (pericardium ≤10% oversize; porcine small intestinal submucosa 25%) with running 4-0 polydioxanone (PDS) → re-test, plicate residual curve (up to 25%).
- Grafts: pericardium and small intestinal submucosa commonest; collagen fleece (TachoSil) sutureless; synthetics not recommended; autologous out of favour (donor site). Outcome depends on selection and technique, not graft.
- Rehabilitation: massage/stretch from 2 weeks; nightly PDE5I (graft take, preserves cavernosal tissue); traction ≥3 h/day from 3–4 weeks for 3 months.

Penile prosthesis with straightening

- PD + ED refractory to PDE5I: one operation treats both. **Inflatable** preferred (better modelling and girth).
- **Manual modelling** (Wilson–Delk): inflate fully, clamp tubing to protect pump, bend opposite to curve 60–90 s; hand on shaft not glans (urethral perforation), other hand supports corporotomies. Straightens 86–100%.
- Residual >30° or buckling → relaxing tunical incision (cylinders deflated, cautery <30 W with Titan); graft if defect >2 cm (pericardium/submucosa, never local dermis — infection).

- Residual <45°: home modelling from 4 weeks. Length loss commonest complaint; pre- and post-implant traction/prolonged inflation preserve length. → *Ch 75*

7 · LANDMARK TRIALS

Trial	Design	Result	Take-home
IMPRESS I and II	Phase III RCT, CCH vs placebo 2:1, curvature 30–90°	Curvature ↓34% (17°) vs 18% (9.3°); bother improved	Basis of US Food and Drug Administration (FDA) approval; modelling essential
Interferon α-2b multicentre RCT	Intralesional interferon (IFN) vs saline	↓27% vs 9% curvature	First RCT of intralesional therapy; saline inert

8 · MCQ ONE-LINERS

1. PD = fibrosis of the tunica albuginea.
2. Commonest plaque site: dorsal.
3. Longitudinal tunical layer absent between 5 and 7 o'clock.
4. Key cytokine: TGF-β1.
5. Associated: Dupuytren and Ledderhose disease.
6. Calcification does not indicate chronic disease.
7. Up to 50% worsen untreated; spontaneous resolution rare.
8. Surgery when ≥12 months from onset and stable ≥6 months.
9. Objective curvature: after ICI, goniometer, with PDDUS.
0. ICI erection test required before any invasive therapy (AUA).
1. Only AUA-endorsed oral therapy: NSAIDs for pain.
2. Vitamin E, tamoxifen, colchicine, carnitine: no benefit.
3. ESWT treats pain, not curvature.
4. Only FDA-approved PD drug: collagenase *Clostridium histolyticum*.
5. CCH dose: 0.58 mg, two injections per cycle, up to four cycles.
16. CCH indication: stable curvature 30–90°, intact erections.
17. Most serious CCH complication: corporal rupture.
18. CCH degrades collagen types I and III.
19. Modelling is essential after CCH.
20. Verapamil: calcium channel blocker; AUA "may offer".
21. Traction acts by mechanotransduction; needs ≥3 h/day.
22. Plication shortens the convex side.
23. 16-dot plication: four dots per stitch, extended Lembert pattern.
24. First incisionless plication: Essed–Schroeder.
25. Grafting for curves >60–70°, hourglass, hinge.
26. Strongest predictor of post-operative ED: pre-operative erection quality.
27. Ventral curves do poorly with grafting (veno-occlusive dysfunction).
28. Pericardium graft oversize ≤10%; small intestinal submucosa 25%.
29. PD + refractory ED → inflatable prosthesis + modelling.
30. Graft during prosthesis if tunical defect >2 cm.
31. Functionally straight: residual curvature ≤20°.

★ Putting it together: andrology and sexual medicine decisions

Reading time ≈ 10 min (slow pace) · Integrates Ch 66–76 · use these as the skeleton of long-essay answers

A · INFERTILE MAN: THE WORK-UP

ALGORITHM

1. History and examination of both partners: duration, toxins, drugs (exogenous testosterone), mumps, cryptorchidism; testis volume, vasa (palpable?), varicocele, secondary sex characteristics (→ *Ch 69*).

2. Two semen analyses (abstinence 2–7 days). Low volume with acid pH → think ejaculatory duct obstruction or congenital bilateral absence of the vas deferens (CBAVD); check a post-ejaculate urine for retrograde ejaculation.

3. Hormones if the count is low or sexual function is impaired: follicle-stimulating hormone (FSH), testosterone, then luteinising hormone (LH) and prolactin if testosterone is low.

4. Genetics for azoospermia or severe oligozoospermia: karyotype and Y-chromosome microdeletions; cystic fibrosis transmembrane conductance regulator (CFTR) testing in CBAVD (test the partner too).

B · AZOOSPERMIA: OBSTRUCTIVE OR NOT?

	Obstructive (OA)	**Non-obstructive (NOA)**
Testis	Normal volume	Small, soft
FSH	Normal (≤7.6 IU/L)	Raised (>7.6 IU/L)
Clues	Absent vasa, low-volume ejaculate, prior vasectomy or hernia repair	Klinefelter (47,XXY), Y microdeletion, cryptorchidism, chemotherapy
Treatment	Reconstruction (vasovasostomy or vasoepididymostomy), transurethral resection of the ejaculatory ducts for obstruction there, or sperm retrieval (epididymal or testicular aspiration) for intracytoplasmic sperm injection (ICSI) (→ *Ch 70*)	Microdissection testicular sperm extraction (micro-TESE) + ICSI; correct a clinical varicocele first; no retrieval with complete azoospermia factor a or b (AZFa/AZFb) deletion

KEY POINT † Vasectomy reversal or ICSI? Reconstruction suits a younger partner and the wish for more than one child; ICSI suits an older partner or a female factor. At reversal, inspect the fluid from the testicular vas: copious fluid — even clear and sperm-free — or sperm/sperm parts → vasovasostomy; thick toothpaste-like fluid or none → vasoepididymostomy (→ *Ch 70*).

C · ERECTILE DYSFUNCTION: THE LADDER

ALGORITHM

1. Assess: history and a validated score (International Index of Erectile Function, IIEF-5, or Sexual Health Inventory for Men, SHIM), examination, glucose/glycated haemoglobin (HbA1c), lipids, morning testosterone. Erectile dysfunction (ED) is a cardiovascular sentinel — stratify cardiac risk before restarting sexual activity (→ *Ch 72*).

2. Modify: lifestyle, risk factors, offending drugs; treat hypogonadism (testosterone may rescue phosphodiesterase type 5 inhibitor [PDE5I] non-responders).

3. Shared decision on any of: PDE5I (never with nitrates; retry after correct-use counselling before calling it failure), vacuum erection device, intraurethral alprostadil, intracavernosal injection (alprostadil or trimix).

4. Penile prosthesis (inflatable, three-piece preferred) when the others fail or are unwanted; highest satisfaction. Infection: remove all components ± immediate salvage (→ *Ch 75*).

Young man, focal arterial occlusion after pelvic or perineal trauma → consider penile revascularisation.

D · PRIAPISM: ISCHAEMIC VERSUS NON-ISCHAEMIC

	Ischaemic (emergency)	Non-ischaemic (high-flow)
Clinical	Rigid, painful	Partially rigid, painless; often after perineal trauma
Cavernous blood gas	pO_2 <30, pCO_2 >60 mm Hg, pH <7.25	Like arterial blood: pO_2 >90, pCO_2 <40
Doppler	Little or no flow	High flow, fistula or pseudoaneurysm
Treatment	Aspiration ± saline irrigation → intracavernosal phenylephrine (100–500 µg every 3–5 min, up to 1 h) → distal shunt ± tunnelling → proximal shunt; >36 h or refractory → consider early prosthesis	Not an emergency: observe (many resolve); selective arterial embolisation if persistent

KEY POINT Sickle cell disease: treat the priapism exactly as above (no delay for transfusion), avoid chilled irrigation, and involve haematology. Stuttering priapism: daily PDE5I (paradoxical), pseudoephedrine at night, or hormonal suppression after puberty (→ *Ch 73*).

E · PEYRONIE DISEASE: WHICH TREATMENT?

ALGORITHM

Active phase (pain, changing curve): oral analgesia, penile traction; collagenase *Clostridium histolyticum* (CCH) for stable dorsal/lateral curves 30–90° with intact erections; intralesional verapamil or interferon α-2b as alternatives.

Surgery only when stable — ≥12 months from onset and deformity unchanged ≥6 months. Document stretched penile length and erectile function first (→ *Ch 76*).

Erections adequate + simple curve curvature <60°, no hourglasslt;60–70°, no hourglass or hinge, adequate length → tunical plication (shortens). **Erections adequate** + complex curve, hourglass or hinge, or short penis → plaque incision or partial excision + graft (more new ED). **Erections inadequate** for penetration despite PDE5I → inflatable prosthesis ± modelling.

F · PREMATURE EJACULATION (PE)

KEY POINT Diagnose by latency (lifelong: about 1 min or less), inability to delay and distress — latency alone never suffices. Treat co-existing ED first. Options: behavioural (stop–start, squeeze), topical lidocaine–prilocaine spray, on-demand dapoxetine (only licensed oral drug outside the US), daily selective serotonin reuptake inhibitor (paroxetine delays most), tramadol (off-label, addiction risk) (→ *Ch 74*).